The proof of the quotient rule is very similar to the proof of the product rule, so it is omitted here. Instead, we apply this new rule for finding derivatives in the next example.

Example 3.3.9: Applying the Quotient Rule

Use the quotient rule to find the derivative of $q(x) = \dfrac{5x^2}{4x+3}$.

Solution

Let $f(x) = 5x^2$ and $g(x) = 4x + 3$. Thus, $f'(x) = 10x$ and $g'(x) = 4$.

Substituting into the quotient rule, we have

$$q'(x) = \frac{f'(x)g(x) - g'(x)f(x)}{(g(x))^2} = \frac{10x(4x+3) - 4(5x^2)}{(4x+3)^2}.$$

Simplifying, we obtain

$$q'(x) = \frac{20x^2 + 30x}{(4x+3)^2}$$

Exercise 3.3.7

Find the derivative of $h(x) = \dfrac{3x+1}{4x-3}$.

Hint

Apply the quotient rule with $f(x) = 3x + 1$ and $g(x) = 4x - 3$.

Answer

$h'(x) = -\dfrac{13}{(4x-3)^2}$.

It is now possible to use the quotient rule to extend the power rule to find derivatives of functions of the form x^k where k is a negative integer.

Extended Power Rule

If k is a negative integer, then

$$\frac{d}{dx}(x^k) = kx^{k-1}.$$

Proof

If k is a negative integer, we may set $n = -k$, so that n is a positive integer with $k = -n$. Since for each positive integer n,$x^{-n} = \dfrac{1}{x^n}$, we may now apply the quotient rule by setting $f(x) = 1$ and $g(x) = x^n$. In this case, $f'(x) = 0$ and $g'(x) = nx^{n-1}$. Thus,

$$\frac{d}{dx}(x^{-n}) = \frac{0(x^n) - 1(nx^{n-1})}{(x^n)^2}.$$

Simplifying, we see that

$$\begin{aligned}\frac{d}{dx}(x^{-n}) &= \frac{-nx^{n-1}}{x^{2n}} \\ &= -nx^{(n-1)-2n} \\ &= -nx^{-n-1}.\end{aligned}$$

Finally, observe that since $k = -n$, by substituting we have

$$\frac{d}{dx}(x^k) = kx^{k-1}.$$

□

Example 3.3.10: Using the Extended Power Rule

Find $\frac{d}{dx}(x^{-4})$.

Solution

By applying the extended power rule with $k = -4$, we obtain

$$\frac{d}{dx}(x^{-4}) = -4x^{-4-1} = -4x^{-5}.$$

Example 3.3.11: Using the Extended Power Rule and the Constant Multiple Rule

Use the extended power rule and the constant multiple rule to find $f(x) = \frac{6}{x^2}$.

Solution

It may seem tempting to use the quotient rule to find this derivative, and it would certainly not be incorrect to do so. However, it is far easier to differentiate this function by first rewriting it as $f(x) = 6x^{-2}$.

$$\begin{aligned} f'(x) = \frac{d}{dx}\left(\frac{6}{x^2}\right) = \frac{d}{dx}\left(6x^{-2}\right) & \quad \text{Rewrite } \frac{6}{x^2} \text{ as } 6x^{-2}. \\ = 6\frac{d}{dx}\left(x^{-2}\right) & \quad \text{Apply the constant multiple rule.} \\ = 6(-2x^{-3}) & \quad \text{Use the extended power rule to differentiate } x^{-2}. \\ = -12x^{-3} & \quad \text{Simplify.} \end{aligned}$$

Exercise 3.3.8

Find the derivative of $g(x) = \frac{1}{x^7}$ using the extended power rule.

Hint

Rewrite $g(x) = \frac{1}{x^7} = x^{-7}$. Use the extended power rule with $k = -7$.

Answer

$g'(x) = -7x^{-8}$.

Combining Differentiation Rules

As we have seen throughout the examples in this section, it seldom happens that we are called on to apply just one differentiation rule to find the derivative of a given function. At this point, by combining the differentiation rules, we may find the derivatives of any polynomial or rational function. Later on we will encounter more complex combinations of differentiation rules. A good rule of thumb to use when applying several rules is to apply the rules in reverse of the order in which we would evaluate the function.

Example 3.3.12: Combining Differentiation Rules

For $k(x) = 3h(x) + x^2g(x)$, find $k'(x)$.

Solution: Finding this derivative requires the sum rule, the constant multiple rule, and the product rule.

$k'(x) = \frac{d}{dx}\big(3h(x) + x^2 g(x)\big) = \frac{d}{dx}\big(3h(x)\big) + \frac{d}{dx}\big(x^2 g(x)\big)$	Apply the sum rule.
$= 3\frac{d}{dx}\big(h(x)\big) + \Big(\frac{d}{dx}(x^2)g(x) + \frac{d}{dx}(g(x))x^2\Big)$	Apply the constant multiple rule to differentiate $3h(x)$ and the product rule to differentiate $x^2 g(x)$.
$= 3h'(x) + 2xg(x) + g'(x)x^2$	

Example 3.3.13: Extending the Product Rule

For $k(x) = f(x)g(x)h(x)$, express $k'(x)$ in terms of $f(x), g(x), h(x)$, and their derivatives.

Solution

We can think of the function $k(x)$ as the product of the function $f(x)g(x)$ and the function $h(x)$. That is, $k(x) = (f(x)g(x)) \cdot h(x)$. Thus,

$$k'(x) = \frac{d}{dx}\big(f(x)g(x)\big) \cdot h(x) + \frac{d}{dx}\big(h(x)\big) \cdot \big(f(x)g(x)\big)$$ Apply the product rule to the product of $f(x)g(x)$ and $h(x)$.

$$= \big(f'(x)g(x) + g'(x)f(x)\big)h(x) + h'(x)f(x)g(x)$$ Apply the product rule to $f(x)g(x)$

$$= f'(x)g(x)h(x) + f(x)g'(x)h(x) + f(x)g(x)h'(x)$$ Simplify.

Example 3.3.14: Combining the Quotient Rule and the Product Rule

For $h(x) = \frac{2x^3 k(x)}{3x+2}$, find $h'(x)$.

Solution

This procedure is typical for finding the derivative of a rational function.

$$h'(x) = \frac{\frac{d}{dx}(2x^3 k(x)) \cdot (3x+2) - \frac{d}{dx}(3x+2) \cdot (2x^3 k(x))}{(3x+2)^2}$$ Apply the quotient rule.

$$= \frac{(6x^2 k(x) + k'(x) \cdot 2x^3)(3x+2) - 3(2x^3 k(x))}{(3x+2)^2}$$ Apply the product rule to find $\frac{d}{dx}(2x^3 k(x))$. Use $\frac{d}{dx}(3x+2) = 3$.

$$= \frac{-6x^3 k(x) + 18x^3 k(x) + 12x^2 k(x) + 6x^4 k'(x) + 4x^3 k'(x)}{(3x+2)^2}$$ Simplify

Exercise 3.3.9

Find $\frac{d}{dx}(3f(x) - 2g(x))$.

Hint

Apply the difference rule and the constant multiple rule.

Answer

$3f'(x) - 2g'(x)$.

Example 3.3.15: Determining Where a Function Has a Horizontal Tangent

Determine the values of x for which $f(x) = x^3 - 7x^2 + 8x + 1$ has a horizontal tangent line.

Solution

To find the values of x for which $f(x)$ has a horizontal tangent line, we must solve $f'(x) = 0$.

Since $f'(x) = 3x^2 - 14x + 8 = (3x-2)(x-4)$,

we must solve $(3x-2)(x-4)=0$. Thus we see that the function has horizontal tangent lines at $x=\frac{2}{3}$ and $x=4$ as shown in the following graph.

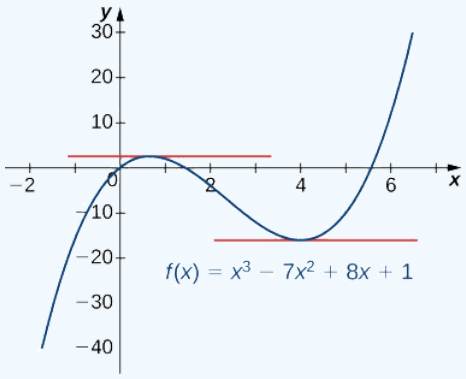


Figure 3.3.2: This function has horizontal tangent lines at $x=2/3$ and $x=4$.

Example 3.3.16: Finding a Velocity

The position of an object on a coordinate axis at time t is given by $s(t)=\frac{t}{t^2+1}$. What is the initial velocity of the object?

Solution

Since the initial velocity is $v(0)=s'(0)$, begin by finding $s'(t)$ by applying the quotient rule:

$$s'(t)=\frac{1(t^2+1)-2t(t)}{(t^2+1)^2}=\frac{1-t^2}{(t^2+1)^2}.$$

After evaluating, we see that $v(0)=1$.

Exercise 3.3.10

Find the values of x for which the line tangent to the graph of $f(x)=4x^2-3x+2$ has a tangent line parallel to the line $y=2x+3$.

Hint

Solve $f'(x)=2$.

Answer

$\frac{5}{8}$

Formula One Grandstands

Formula One car races can be very exciting to watch and attract a lot of spectators. Formula One track designers have to ensure sufficient grandstand space is available around the track to accommodate these viewers. However, car racing can be dangerous, and safety considerations are paramount. The grandstands must be placed where spectators will not be in danger should a driver lose control of a car (Figure 3.3.3).



Figure 3.3.3: The grandstand next to a straightaway of the Circuit de Barcelona-Catalunya race track, located where the spectators are not in danger.

Safety is especially a concern on turns. If a driver does not slow down enough before entering the turn, the car may slide off the racetrack. Normally, this just results in a wider turn, which slows the driver down. But if the driver loses control completely, the car may fly off the track entirely, on a path tangent to the curve of the racetrack.

Suppose you are designing a new Formula One track. One section of the track can be modeled by the function $f(x) = x^3 + 3x^2 + x$ (Figure 3.3.4). The current plan calls for grandstands to be built along the first straightaway and around a portion of the first curve. The plans call for the front corner of the grandstand to be located at the point $(-1.9, 2.8)$. We want to determine whether this location puts the spectators in danger if a driver loses control of the car.

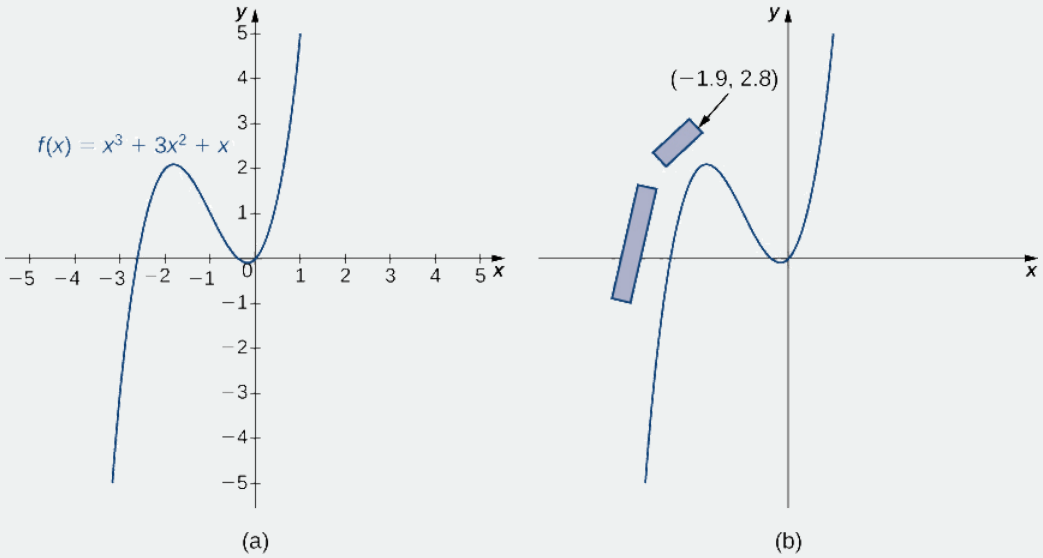


Figure 3.3.4: (a) One section of the racetrack can be modeled by the function $f(x) = x^3 + 3x^2 + x$. (b) The front corner of the grandstand is located at $(-1.9, 2.8)$.

1. Physicists have determined that drivers are most likely to lose control of their cars as they are coming into a turn, at the point where the slope of the tangent line is 1. Find the (x, y) coordinates of this point near the turn.
2. Find the equation of the tangent line to the curve at this point.
3. To determine whether the spectators are in danger in this scenario, find the x-coordinate of the point where the tangent line crosses the line $y = 2.8$. Is this point safely to the right of the grandstand? Or are the spectators in danger?
4. What if a driver loses control earlier than the physicists project? Suppose a driver loses control at the point $(-2.5, 0.625)$. What is the slope of the tangent line at this point?
5. If a driver loses control as described in part 4, are the spectators safe?
6. Should you proceed with the current design for the grandstand, or should the grandstands be moved?

Key Concepts

- The derivative of a constant function is zero.
- The derivative of a power function is a function in which the power on x becomes the coefficient of the term and the power on x in the derivative decreases by 1.
- The derivative of a constant c multiplied by a function f is the same as the constant multiplied by the derivative.
- The derivative of the sum of a function f and a function g is the same as the sum of the derivative of f and the derivative of g.
- The derivative of the difference of a function f and a function g is the same as the difference of the derivative of f and the derivative of g.
- The derivative of a product of two functions is the derivative of the first function times the second function plus the derivative of the second function times the first function.
- The derivative of the quotient of two functions is the derivative of the first function times the second function minus the derivative of the second function times the first function, all divided by the square of the second function.
- We used the limit definition of the derivative to develop formulas that allow us to find derivatives without resorting to the definition of the derivative. These formulas can be used singly or in combination with each other.

Glossary

constant multiple rule

the derivative of a constant c multiplied by a function f is the same as the constant multiplied by the derivative: $\frac{d}{dx}\big(cf(x)\big) = cf'(x)$

constant rule

the derivative of a constant function is zero: $\frac{d}{dx}(c) = 0$, where c is a constant

difference rule

the derivative of the difference of a function f and a function g is the same as the difference of the derivative of f and the derivative of g: $\frac{d}{dx}\big(f(x) - g(x)\big) = f'(x) - g'(x)$

power rule

the derivative of a power function is a function in which the power on x becomes the coefficient of the term and the power on x in the derivative decreases by 1: If n is an integer, then $\frac{d}{dx}(x^n) = nx^{n-1}$

product rule

the derivative of a product of two functions is the derivative of the first function times the second function plus the derivative of the second function times the first function: $\frac{d}{dx}\big(f(x)g(x)\big) = f'(x)g(x) + g'(x)f(x)$

quotient rule

the derivative of the quotient of two functions is the derivative of the first function times the second function minus the derivative of the second function times the first function, all divided by the square of the second function: $\frac{d}{dx}\left(\frac{f(x)}{g(x)}\right) = \frac{f'(x)g(x) - g'(x)f(x)}{\big(g(x)\big)^2}$

sum rule

the derivative of the sum of a function f and a function g is the same as the sum of the derivative of f and the derivative of g: $\frac{d}{dx}\big(f(x) + g(x)\big) = f'(x) + g'(x)$

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3.3E: EXERCISES FOR SECTION 3.3

In exercises 1 - 12, find $f'(x)$ for each function.

1) $f(x) = x^7 + 10$

2) $f(x) = 5x^3 - x + 1$

Answer

$f'(x) = 15x^2 - 1$

3) $f(x) = 4x^2 - 7x$

4) $f(x) = 8x^4 + 9x^2 - 1$

Answer

$f'(x) = 32x^3 + 18x$

5) $f(x) = x^4 + 2x$

6) $f(x) = 3x\left(18x^4 + \dfrac{13}{x+1}\right)$

Answer

$f'(x) = 270x^4 + \dfrac{39}{(x+1)^2}$

7) $f(x) = (x+2)(2x^2 - 3)$

8) $f(x) = x^2\left(\dfrac{2}{x^2} + \dfrac{5}{x^3}\right)$

Answer

$f'(x) = \dfrac{-5}{x^2}$

9) $f(x) = \dfrac{x^3 + 2x^2 - 4}{3}$

10) $f(x) = \dfrac{4x^3 - 2x + 1}{x^2}$

Answer

$f'(x) = \dfrac{4x^4 + 2x^2 - 2x}{x^4}$

11) $f(x) = \dfrac{x^2 + 4}{x^2 - 4}$

12) $f(x) = \dfrac{x + 9}{x^2 - 7x + 1}$

Answer

$f'(x) = \dfrac{-x^2 - 18x + 64}{(x^2 - 7x + 1)^2}$

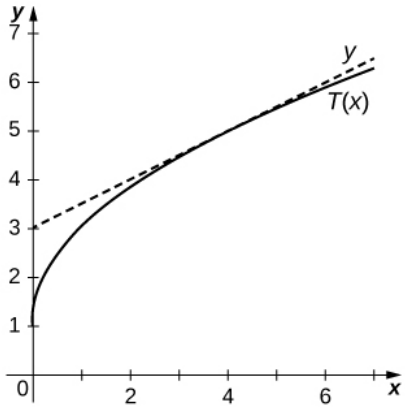
In exercises 13 - 16, find the equation of the tangent line $T(x)$ to the graph of the given function at the indicated point. Use a graphing calculator to graph the function and the tangent line.

13) [T] $y = 3x^2 + 4x + 1$ at $(0, 1)$

14) [T] $y = 2\sqrt{x} + 1$ at $(4, 5)$

Answer

$T(x) = \frac{1}{2}x + 3$

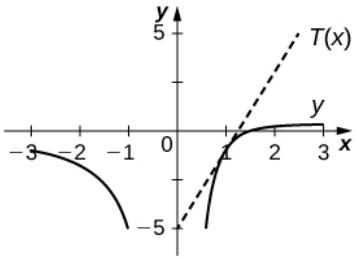


15) [T] $y = \dfrac{2x}{x-1}$ at $(-1, 1)$

16) [T] $y = \dfrac{2}{x} - \dfrac{3}{x^2}$ at $(1, -1)$

Answer

$T(x) = 4x - 5$



In exercises 17 - 20, assume that $f(x)$ and $g(x)$ are both differentiable functions for all x. Find the derivative of each of the functions $h(x)$.

17) $h(x) = 4f(x) + \dfrac{g(x)}{7}$

18) $h(x) = x^3 f(x)$

Answer

$h'(x) = 3x^2 f(x) + x^3 f'(x)$

19) $h(x) = \dfrac{f(x)g(x)}{2}$

20) $h(x) = \dfrac{3f(x)}{g(x)+2}$

Answer

$h'(x) = \dfrac{3f'(x)(g(x)+2) - 3f(x)g'(x)}{(g(x)+2)^2}$

For exercises 21 - 24, assume that $f(x)$ and $g(x)$ are both differentiable functions with values as given in the following table. Use the following table to calculate the following derivatives.

x	1	2	3	4
$f(x)$	3	5	−2	0
$g(x)$	2	3	−4	6
$f'(x)$	−1	7	8	−3
$g'(x)$	4	1	2	9

21) Find $h'(1)$ if $h(x) = xf(x) + 4g(x)$.

22) Find $h'(2)$ if $h(x) = \dfrac{f(x)}{g(x)}$.

Answer

$h'(2) = \frac{16}{9}$

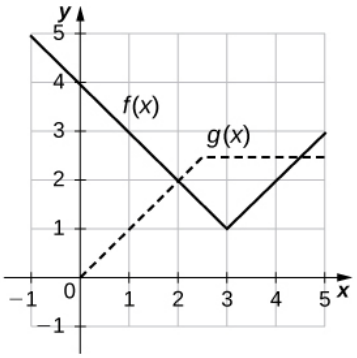
23) Find $h'(3)$ if $h(x) = 2x + f(x)g(x)$.

24) Find $h'(4)$ if $h(x) = \dfrac{1}{x} + \dfrac{g(x)}{f(x)}$.

Answer

$h'(4)$ is undefined.

In exercises 25 - 27, use the following figure to find the indicated derivatives, if they exist.



25) Let $h(x) = f(x) + g(x)$. Find

a) $h'(1)$,

b) $h'(3)$, and

c) $h'(4)$.

26) Let $h(x) = f(x)g(x)$. Find

a) $h'(1)$,

b) $h'(3)$, and

c) $h'(4)$.

Answer

a. $h'(1) = 2$,
b. $h'(3)$ does not exist,
c. $h'(4) = 2.5$

27) Let $h(x) = \dfrac{f(x)}{g(x)}$. Find

a) $h'(1)$,

b) $h'(3)$, and

c) $h'(4)$.

In exercises 28 - 31,

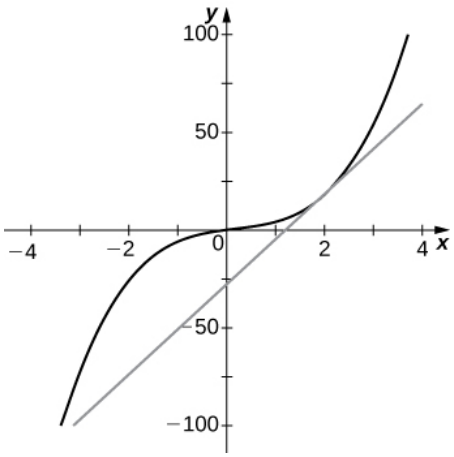
a) evaluate $f'(a)$, and

b) graph the function $f(x)$ and the tangent line at $x = a$.

28) [T] $f(x) = 2x^3 + 3x - x^2, \quad a = 2$

Answer

a. 23
b. $y = 23x - 28$



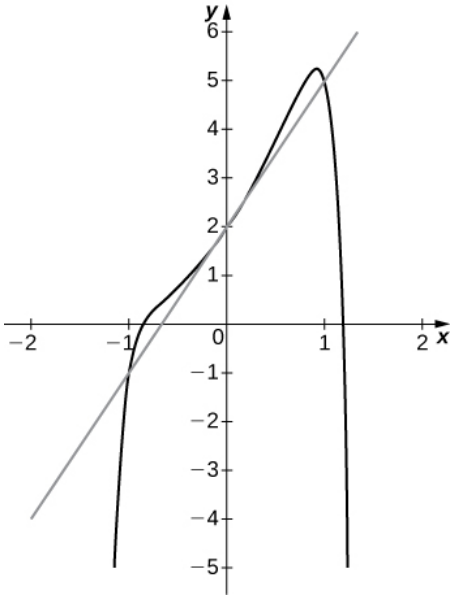
29) [T] $f(x) = \frac{1}{x} - x^2, \quad a = 1$

30) [T] $f(x) = x^2 - x^{12} + 3x + 2, \quad a = 0$

Answer

a. 3

b. $y = 3x + 2$



31) [T] $f(x) = \frac{1}{x} - x^{2/3}, \quad a = -1$

32) Find the equation of the tangent line to the graph of $f(x) = 2x^3 + 4x^2 - 5x - 3$ at $x = -1$.

Answer

$y = -7x - 3$

33) Find the equation of the tangent line to the graph of $f(x) = x^2 + \frac{4}{x} - 10$ at $x = 8$.

34) Find the equation of the tangent line to the graph of $f(x) = (3x - x^2)(3 - x - x^2)$ at $x = 1$.

Answer

$y = -5x + 7$

35) Find the point on the graph of $f(x) = x^3$ such that the tangent line at that point has an x-intercept of $(6, 0)$.

36) Find the equation of the line passing through the point $P(3,3)$ and tangent to the graph of $f(x)=\dfrac{6}{x-1}$.

Answer

$y=-\frac{3}{2}x+\frac{15}{2}$

37) Determine all points on the graph of $f(x)=x^3+x^2-x-1$ for which the slope of the tangent line is

a. horizontal

b. −1.

38) Find a quadratic polynomial such that $f(1)=5,\ f'(1)=3$ and $f''(1)=-6$.

Answer

$y=-3x^2+9x-1$

39) A car driving along a freeway with traffic has traveled $s(t)=t^3-6t^2+9t$ meters in t seconds.

a. Determine the time in seconds when the velocity of the car is 0.

b. Determine the acceleration of the car when the velocity is 0.

40) [T] A herring swimming along a straight line has traveled $s(t)=\dfrac{t^2}{t^2+2}$ feet in t seconds. Determine the velocity of the herring when it has traveled 3 seconds.

Answer

$\frac{12}{121}$ or 0.0992 ft/s

41) The population in millions of arctic flounder in the Atlantic Ocean is modeled by the function $P(t)=\dfrac{8t+3}{0.2t^2+1}$, where t is measured in years.

a. Determine the initial flounder population.

b. Determine $P'(10)$ and briefly interpret the result.

42) [T] The concentration of antibiotic in the bloodstream t hours after being injected is given by the function $C(t)=\dfrac{2t^2+t}{t^3+50}$, where C is measured in milligrams per liter of blood.

a. Find the rate of change of $C(t)$.

b. Determine the rate of change for $t=8,12,24$,and 36.

c. Briefly describe what seems to be occurring as the number of hours increases.

Answer

a. $\dfrac{-2t^4-2t^3+200t+50}{(t^3+50)^2}$

b. -0.02395 mg/L-hr, -0.01344 mg/L-hr, -0.003566 mg/L-hr, -0.001579 mg/L-hr

c. The rate at which the concentration of drug in the bloodstream decreases is slowing to 0 as time increases.

43) A book publisher has a cost function given by $C(x)=\dfrac{x^3+2x+3}{x^2}$, where x is the number of copies of a book in thousands and C is the cost, per book, measured in dollars. Evaluate $C'(2)$and explain its meaning.

44) [T] According to Newton's law of universal gravitation, the force F between two bodies of constant mass m_1 and m_2 is given by the formula $F=\dfrac{Gm_1m_2}{d^2}$, where G is the gravitational constant and d is the distance between the bodies.

a. Suppose that G, m_1, and m_2 are constants. Find the rate of change of force F with respect to distance d.

b. Find the rate of change of force F with gravitational constant $G=6.67\times10^{-11}\text{Nm}^2/\text{kg}^2$, on two bodies 10 meters apart, each with a mass of 1000 kilograms.

Answer

a. $F'(d)=\dfrac{-2Gm_1m_2}{d_3}$

b. -1.33×10^{-7} N/m

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3.4: Derivatives as Rates of Change

Learning Objectives

- Determine a new value of a quantity from the old value and the amount of change.
- Calculate the average rate of change and explain how it differs from the instantaneous rate of change.
- Apply rates of change to displacement, velocity, and acceleration of an object moving along a straight line.
- Predict the future population from the present value and the population growth rate.
- Use derivatives to calculate marginal cost and revenue in a business situation.

In this section we look at some applications of the derivative by focusing on the interpretation of the derivative as the rate of change of a function. These applications include **acceleration** and velocity in physics, **population growth rates** in biology, and marginal functions in economics.

Amount of Change Formula

One application for derivatives is to estimate an unknown value of a function at a point by using a known value of a function at some given point together with its rate of change at the given point. If $f(x)$ is a function defined on an interval $[a, a+h]$, then the **amount of change** of $f(x)$ over the interval is the change in the y values of the function over that interval and is given by

$$f(a+h) - f(a).$$

The **average rate of change** of the function f over that same interval is the ratio of the amount of change over that interval to the corresponding change in the x values. It is given by

$$\frac{f(a+h) - f(a)}{h}.$$

As we already know, the instantaneous rate of change of $f(x)$ at a is its derivative

$$f'(a) = \lim_{h \to 0} \frac{f(a+h) - f(a)}{h}.$$

For small enough values of h, $f'(a) \approx \frac{f(a+h)-f(a)}{h}$. We can then solve for $f(a+h)$ to get the amount of change formula:

$$f(a+h) \approx f(a) + f'(a)h. \tag{3.4.1}$$

We can use this formula if we know only $f(a)$and $f'(a)$ and wish to estimate the value of $f(a+h)$. For example, we may use the current population of a city and the rate at which it is growing to estimate its population in the near future. As we can see in Figure 3.4.1, we are approximating $f(a+h)$ by the y coordinate at a+h on the line tangent to $f(x)$ at $x = a$. Observe that the accuracy of this estimate depends on the value of h as well as the value of $f'(a)$.

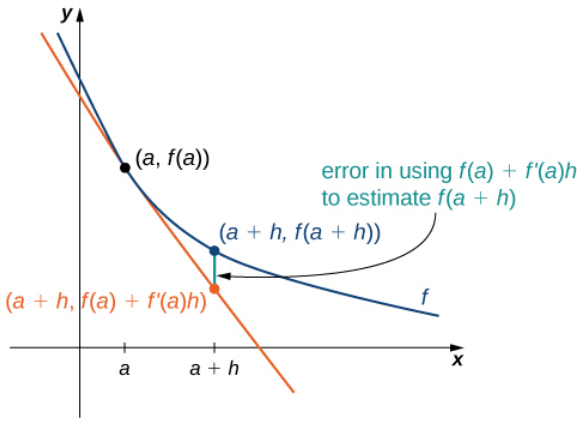


Figure 3.4.1: The new value of a changed quantity equals the original value plus the rate of change times the interval of change: $f(a+h) \approx f(a) + f'(a)h$.

Example 3.4.1: Estimating the Value of a Function

If $f(3) = 2$ and $f'(3) = 5$, estimate $f(3.2)$.

Solution

Begin by finding h. We have $h = 3.2 - 3 = 0.2$. Thus,

$$f(3.2) = f(3 + 0.2) \approx f(3) + (0.2)f'(3) = 2 + 0.2(5) = 3.$$

Exercise 3.4.1

Given $f(10) = -5$ and $f'(10) = 6$, estimate $f(10.1)$.

Hint

Use the same process as in the preceding example.

Answer

-4.4

Motion along a Line

Another use for the derivative is to analyze motion along a line. We have described velocity as the rate of change of position. If we take the derivative of the velocity, we can find the acceleration, or the rate of change of velocity. It is also important to introduce the idea of **speed**, which is the magnitude of velocity. Thus, we can state the following mathematical definitions.

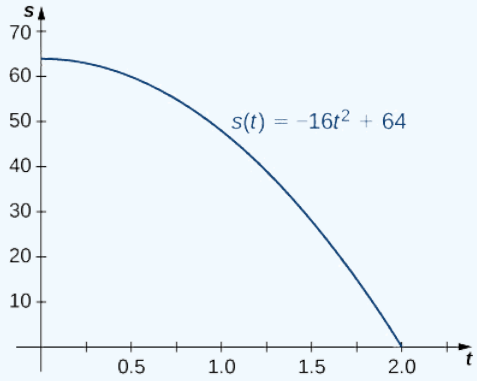
Definition

Let $s(t)$ be a function giving the position of an object at time t.

- The velocity of the object at time t is given by $v(t) = s'(t)$.
- The speed of the object at time t is given by $|v(t)|$.
- The acceleration of the object at t is given by $a(t) = v'(t) = s''(t)$.

Example 3.4.2: Comparing Instantaneous Velocity and Average Velocity

A ball is dropped from a height of 64 feet. Its height above ground (in feet) t seconds later is given by $s(t) = -16t^2 + 64$.



a. What is the instantaneous velocity of the ball when it hits the ground?
b. What is the average velocity during its fall?

Solution

The first thing to do is determine how long it takes the ball to reach the ground. To do this, set $s(t) = 0$. Solving $-16t^2 + 64 = 0$, we get $t = 2$, so it takes 2 seconds for the ball to reach the ground.

a. The instantaneous velocity of the ball as it strikes the ground is $v(2)$. Since $v(t) = s'(t) = -32t$, we obtain $v(t) = -64$ ft/s.
b. The average velocity of the ball during its fall is

$$v_{ave} = \frac{s(2)-s(0)}{2-0} = \frac{0-64}{2} = -32 \text{ ft/s}.$$

Example 3.4.3: Interpreting the Relationship between $v(t)$ and $a(t)$

A particle moves along a coordinate axis in the positive direction to the right. Its position at time t is given by $s(t) = t^3 - 4t + 2$. Find $v(1)$ and $a(1)$ and use these values to answer the following questions.

a. Is the particle moving from left to right or from right to left at time $t = 1$?
b. Is the particle speeding up or slowing down at time $t = 1$?

Solution

Begin by finding $v(t)$ and $a(t)$.

$v(t) = s'(t) = 3t^2 - 4$ and $a(t) = v'(t) = s''(t) = 6t$.

Evaluating these functions at $t = 1$, we obtain $v(1) = -1$ and $a(1) = 6$.

a. Because $v(1) < 0$, the particle is moving from right to left.
b. Because $v(1) < 0$ and $a(1) > 0$, velocity and acceleration are acting in opposite directions. In other words, the particle is being accelerated in the direction opposite the direction in which it is traveling, causing $|v(t)|$ to decrease. The particle is slowing down.

Example 3.4.4: Position and Velocity

The position of a particle moving along a coordinate axis is given by $s(t) = t^3 - 9t^2 + 24t + 4,\ t \geq 0$.

a. Find $v(t)$.
b. At what time(s) is the particle at rest?
c. On what time intervals is the particle moving from left to right? From right to left?
d. Use the information obtained to sketch the path of the particle along a coordinate axis.

Solution

a. The velocity is the derivative of the position function:

$$v(t) = s'(t) = 3t^2 - 18t + 24.$$

b. The particle is at rest when $v(t) = 0$, so set $3t^2 - 18t + 24 = 0$. Factoring the left-hand side of the equation produces $3(t - 2)(t - 4) = 0$. Solving, we find that the particle is at rest at $t = 2$ and $t = 4$.

c. The particle is moving from left to right when $v(t) > 0$ and from right to left when $v(t) < 0$. Figure 3.4.2 gives the analysis of the sign of $v(t)$ for $t \geq 0$, but it does not represent the axis along which the particle is moving.



Figure 3.4.2:The sign of $v(t)$ determines the direction of the particle.

- Since $3t^2 - 18t + 24 > 0$ on $[0, 2) \cup (4, +\infty)$, the particle is moving from left to right on these intervals.
- Since $3t^2 - 18t + 24 < 0$ on $(2, 4)$, the particle is moving from right to left on this interval.

d. Before we can sketch the graph of the particle, we need to know its position at the time it starts moving $(t = 0)$ and at the times that it changes direction $(t = 2, 4)$. We have $s(0) = 4$, $s(2) = 24$, and $s(4) = 20$. This means that the particle begins on the coordinate axis at 4 and changes direction at 24 and 20 on the coordinate axis. The path of the particle is shown on a coordinate axis in Figure 3.4.3.

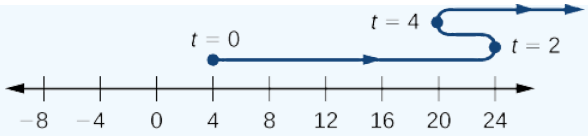


Figure 3.4.3: The path of the particle can be determined by analyzing $v(t)$.

Exercise 3.4.2

A particle moves along a coordinate axis. Its position at time t is given by $s(t) = t^2 - 5t + 1$. Is the particle moving from right to left or from left to right at time $t = 3$?

Hint

Find $v(3)$ and look at the sign.

Answer

left to right

Population Change

In addition to analyzing velocity, speed, acceleration, and position, we can use derivatives to analyze various types of populations, including those as diverse as bacteria colonies and cities. We can use a current population, together with a growth rate, to estimate the size of a population in the future. The population growth rate is the rate of change of a population and consequently can be represented by the derivative of the size of the population.

Definition

If $P(t)$ is the number of entities present in a population, then the population growth rate of $P(t)$ is defined to be $P'(t)$.

Example 3.4.5: Estimating a Population

The population of a city is tripling every 5 years. If its current population is 10,000, what will be its approximate population 2 years from now?

Solution

Let $P(t)$ be the population (in thousands) t years from now. Thus, we know that $P(0) = 10$ and based on the information, we anticipate $P(5) = 30$. Now estimate $P'(0)$, the current growth rate, using

$$P'(0) \approx \frac{P(5)-P(0)}{5-0} = \frac{30-10}{5} = 4.$$

By applying Equation 3.4.1 to $P(t)$, we can estimate the population 2 years from now by writing

$$P(2) \approx P(0) + (2)P'(0) \approx 10 + 2(4) = 18;$$

thus, in 2 years the population will be 18,000.

Exercise 3.4.3

The current population of a mosquito colony is known to be 3,000; that is, $P(0) = 3,000$. If $P'(0) = 100$, estimate the size of the population in 3 days, where t is measured in days.

Hint

Use $P(3) \approx P(0) + 3P'(0)$

Answer

3,300

Changes in Cost and Revenue

In addition to analyzing motion along a line and population growth, derivatives are useful in analyzing changes in cost, revenue, and profit. The concept of a marginal function is common in the fields of business and economics and implies the use of derivatives. The marginal cost is the derivative of the cost function. The marginal revenue is the derivative of the revenue function. The marginal profit is the derivative of the profit function, which is based on the cost function and the revenue function.

Definition

- If $C(x)$ is the cost of producing x items, then the **marginal cost** $MC(x)$ is $MC(x) = C'(x)$.
- If $R(x)$ is the revenue obtained from selling x items, then the marginal revenue $MR(x)$ is $MR(x) = R'(x)$.
- If $P(x) = R(x) - C(x)$ is the profit obtained from selling x items, then the **marginal profit** $MP(x)$ is defined to be $MP(x) = P'(x) = MR(x) - MC(x) = R'(x) - C'(x)$.

We can roughly approximate

$$MC(x) = C'(x) = \lim_{h \to 0} \frac{C(x+h) - C(x)}{h}$$

by choosing an appropriate value for h. Since x represents objects, a reasonable and small value for h is 1. Thus, by substituting $h = 1$, we get the approximation $MC(x) = C'(x) \approx C(x+1) - C(x)$. Consequently, $C'(x)$ for a given value of x can be thought of as the change in cost associated with producing one additional item. In a similar way, $MR(x) = R'(x)$ approximates the revenue obtained by selling one additional item, and $MP(x) = P'(x)$ approximates the profit obtained by producing and selling one additional item.

Example 3.4.6: Applying Marginal Revenue

Assume that the number of barbeque dinners that can be sold, x, can be related to the price charged, p, by the equation $p(x) = 9 - 0.03x, 0 \leq x \leq 300$.

In this case, the revenue in dollars obtained by selling x barbeque dinners is given by

$$R(x) = xp(x) = x(9 - 0.03x) = -0.03x^2 + 9x \quad \text{for } 0 \leq x \leq 300.$$

Use the marginal revenue function to estimate the revenue obtained from selling the 101st barbeque dinner. Compare this to the actual revenue obtained from the sale of this dinner.

Solution

First, find the marginal revenue function: $MR(x) = R'(x) = -0.06x + 9$.

Next, use $R'(100)$ to approximate $R(101) - R(100)$, the revenue obtained from the sale of the 101st dinner. Since $R'(100) = 3$, the revenue obtained from the sale of the 101st dinner is approximately \$3.

The actual revenue obtained from the sale of the 101st dinner is

$$R(101) - R(100) = 602.97 - 600 = 2.97, \text{ or } \$2.97.$$

The marginal revenue is a fairly good estimate in this case and has the advantage of being easy to compute.

Exercise 3.4.4

Suppose that the profit obtained from the sale of x fish-fry dinners is given by $P(x) = -0.03x^2 + 8x - 50$. Use the marginal profit function to estimate the profit from the sale of the 101st fish-fry dinner.

Hint

Use $P'(100)$ to approximate $P(101) - P(100)$.

Answer

$2

Key Concepts

- Using $f(a+h) \approx f(a) + f'(a)h$, it is possible to estimate $f(a+h)$ given $f'(a)$ and $f(a)$.
- The rate of change of position is velocity, and the rate of change of velocity is acceleration. Speed is the absolute value, or magnitude, of velocity.
- The population growth rate and the present population can be used to predict the size of a future population.
- Marginal cost, marginal revenue, and marginal profit functions can be used to predict, respectively, the cost of producing one more item, the revenue obtained by selling one more item, and the profit obtained by producing and selling one more item.

Glossary

acceleration
is the rate of change of the velocity, that is, the derivative of velocity

amount of change
the amount of a function $f(x)$ over an interval $[x, x+h] is f(x+h) - f(x)$

average rate of change
is a function $f(x)$ over an interval $[x, x+h]$ is $\frac{f(x+h)-f(a)}{b-a}$

marginal cost
is the derivative of the cost function, or the approximate cost of producing one more item

marginal revenue
is the derivative of the revenue function, or the approximate revenue obtained by selling one more item

marginal profit
is the derivative of the profit function, or the approximate profit obtained by producing and selling one more item

population growth rate
is the derivative of the population with respect to time

speed
is the absolute value of velocity, that is, $|v(t)|$ is the speed of an object at time t whose velocity is given by $v(t)$

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3.4E: EXERCISES FOR SECTION 3.4

In exercises 1 - 3, the given functions represent the position of a particle traveling along a horizontal line.

a. Find the velocity and acceleration functions.

b. Determine the time intervals when the object is slowing down or speeding up.

1) $s(t) = 2t^3 - 3t^2 - 12t + 8$

2) $s(t) = 2t^3 - 15t^2 + 36t - 10$

Answer

a. $v(t) = 6t^2 - 30t + 36, \quad a(t) = 12t - 30$;
b. speeds up for $(2, 2.5) \cup (3, \infty)$, slows down for $(0, 2) \cup (2.5, 3)$

3) $s(t) = \dfrac{t}{1 + t^2}$

4) A rocket is fired vertically upward from the ground. The distance s in feet that the rocket travels from the ground after t seconds is given by $s(t) = -16t^2 + 560t$.

a. Find the velocity of the rocket 3 seconds after being fired.

b. Find the acceleration of the rocket 3 seconds after being fired.

Answer

a. 464 ft/s^2
b. -32 ft/s^2

5) A ball is thrown downward with a speed of 8 ft/s from the top of a 64-foot-tall building. After t seconds, its height above the ground is given by $s(t) = -16t^2 - 8t + 64$.

a. Determine how long it takes for the ball to hit the ground.

b. Determine the velocity of the ball when it hits the ground.

6) The position function $s(t) = t^2 - 3t - 4$ represents the position of the back of a car backing out of a driveway and then driving in a straight line, where s is in feet and t is in seconds. In this case, $s(t) = 0$ represents the time at which the back of the car is at the garage door, so $s(0) = -4$ is the starting position of the car, 4 feet inside the garage.

a. Determine the velocity of the car when $s(t) = 0$.

b. Determine the velocity of the car when $s(t) = 14$.

Answer

a. 5 ft/s
b. 9 ft/s

7) The position of a hummingbird flying along a straight line in t seconds is given by $s(t) = 3t^3 - 7t$ meters.

a. Determine the velocity of the bird at $t = 1$ sec.

b. Determine the acceleration of the bird at $t = 1$ sec.

c. Determine the acceleration of the bird when the velocity equals 0.

8) A potato is launched vertically upward with an initial velocity of 100 ft/s from a potato gun at the top of an 85-foot-tall building. The distance in feet that the potato travels from the ground after t seconds is given by $s(t) = -16t^2 + 100t + 85$.

a. Find the velocity of the potato after 0.5 s and 5.75 s.

b. Find the speed of the potato at 0.5 s and 5.75 s.

c. Determine when the potato reaches its maximum height.

d. Find the acceleration of the potato at 0.5 s and 1.5 s.

e. Determine how long the potato is in the air.

f. Determine the velocity of the potato upon hitting the ground.

Answer

a. 84 ft/s, −84 ft/s
b. 84 ft/s
c. $\frac{25}{8}$ s
d. $-32\ \text{ft/s}^2$ in both cases
e. $\frac{1}{8}(25+\sqrt{965})$ s
f. $-4\sqrt{965}$ ft/s

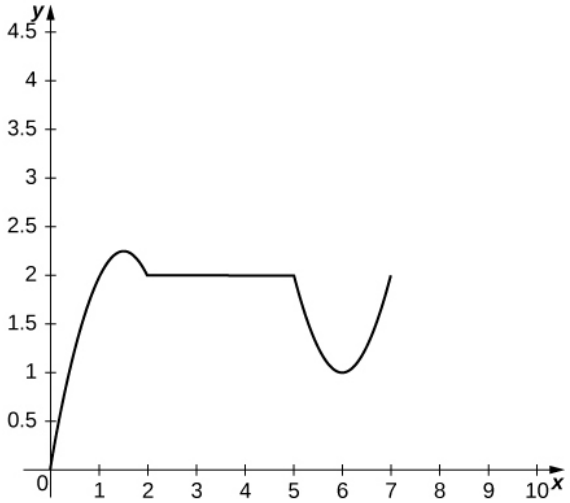
9) The position function $s(t)=t^3-8t$ gives the position in miles of a freight train where east is the positive direction and t is measured in hours.

a. Determine the direction the train is traveling when $s(t)=0$.

b. Determine the direction the train is traveling when $a(t)=0$.

c. Determine the time intervals when the train is slowing down or speeding up.

10) The following graph shows the position $y=s(t)$ of an object moving along a straight line.



a. Use the graph of the position function to determine the time intervals when the velocity is positive, negative, or zero.

b. Sketch the graph of the velocity function.

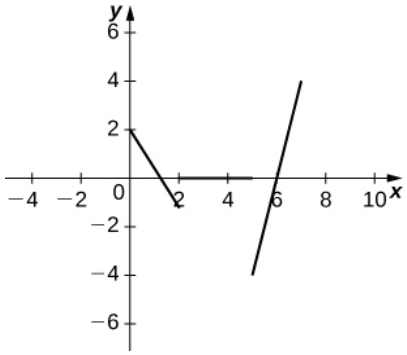
c. Use the graph of the velocity function to determine the time intervals when the acceleration is positive, negative, or zero.

d. Determine the time intervals when the object is speeding up or slowing down.

Answer

a. Velocity is positive on $(0,1.5)\cup(6,7)$, negative on $(1.5,2)\cup(5,6)$, and zero on $(2,5)$.

b.



c. Acceleration is positive on $(5,7)$, negative on $(0,2)$, and zero on $(2,5)$.

d. The object is speeding up on $(6,7)\cup(1.5,2)$ and slowing down on $(0,1.5)\cup(5,6)$.

11) The cost function, in dollars, of a company that manufactures food processors is given by $C(x) = 200 + \frac{7}{x} + \frac{x}{27}$, where x is the number of food processors manufactured.

a. Find the marginal cost function.

b. Find the marginal cost of manufacturing 12 food processors.

c. Find the actual cost of manufacturing the thirteenth food processor.

12) The price p (in dollars) and the demand x for a certain digital clock radio is given by the price–demand function $p = 10 - 0.001x$.

a. Find the revenue function $R(x)$

b. Find the marginal revenue function.

c. Find the marginal revenue at $x = 2000$ and 5000.

Answer

a. $R(x) = 10x - 0.001x^2$

b. $R'(x) = 10 - 0.002x$

c. \$6 per item, \$0 per item

13) [T] A profit is earned when revenue exceeds cost. Suppose the profit function for a skateboard manufacturer is given by $P(x) = 30x - 0.3x^2 - 250$, where x is the number of skateboards sold.

a. Find the exact profit from the sale of the thirtieth skateboard.

b. Find the marginal profit function and use it to estimate the profit from the sale of the thirtieth skateboard.

14) [T] In general, the profit function is the difference between the revenue and cost functions: $P(x) = R(x) - C(x)$.

Suppose the price-demand and cost functions for the production of cordless drills is given respectively by $p = 143 - 0.03x$ and $C(x) = 75,000 + 65x$, where x is the number of cordless drills that are sold at a price of p dollars per drill and $C(x)$ is the cost of producing x cordless drills.

a. Find the marginal cost function.

b. Find the revenue and marginal revenue functions.

c. Find $R'(1000)$ and $R'(4000)$. Interpret the results.

d. Find the profit and marginal profit functions.

e. Find $P'(1000)$ and $P'(4000)$. Interpret the results.

Answer

a. $C'(x) = 65$

b. $R(x) = 143x - 0.03x^2, R'(x) = 143 - 0.06x$

c. $R'(1000) = 83, \quad R'(4000) = -97$. At a production level of 1000 cordless drills, revenue is increasing at a rate of \$83 per drill; at a production level of 4000 cordless drills, revenue is decreasing at a rate of \$97 per drill.

d. $P(x) = -0.03x^2 + 78x - 75000, \quad P'(x) = -0.06x + 78$

e. $P'(1000) = 18, \quad P'(4000) = -162$. At a production level of 1000 cordless drills, profit is increasing at a rate of \$18 per drill; at a production level of 4000 cordless drills, profit is decreasing at a rate of \$162 per drill.

15) A small town in Ohio commissioned an actuarial firm to conduct a study that modeled the rate of change of the town's population. The study found that the town's population (measured in thousands of people) can be modeled by the function $P(t) = -\frac{1}{3}t^3 + 64t + 3000$, where t is measured in years.

a. Find the rate of change function $P'(t)$ of the population function.

b. Find $P'(1), P'(2), P'(3)$, and $P'(4)$. Interpret what the results mean for the town.

c. Find $P''(1), P''(2), P''(3)$, and $P''(4)$. Interpret what the results mean for the town's population.

16) [T] A culture of bacteria grows in number according to the function $N(t) = 3000(1 + \frac{4t}{t^2 + 100})$, where t is measured in hours.

a. Find the rate of change of the number of bacteria.

b. Find $N'(0), N'(10), N'(20)$, and $N'(30)$.

c. Interpret the results in (b).

d. Find $N''(0), N''(10), N''(20)$, and $N''(30)$. Interpret what the answers imply about the bacteria population growth.

Answer

a. $N'(t) = 3000\left(\frac{-4t^2 + 400}{(t^2 + 100)^2}\right)$

b. $120, 0, -14.4, -9.6$

c. The bacteria population increases from time 0 to 10 hours; afterwards, the bacteria population decreases.

d. $0, -6, 0.384, 0.432$. The rate at which the bacteria is increasing is decreasing during the first 10 hours. Afterwards, the bacteria population is decreasing at a decreasing rate.

17) The centripetal force of an object of mass m is given by $F(r) = \frac{mv^2}{r}$, where v is the speed of rotation and r is the distance from the center of rotation.

a. Find the rate of change of centripetal force with respect to the distance from the center of rotation.

b. Find the rate of change of centripetal force of an object with mass 1000 kilograms, velocity of 13.89 m/s, and a distance from the center of rotation of 200 meters.

The following questions concern the population (in millions) of London by decade in the 19th century, which is listed in the following table.

Year Since 1800	Population (millions)
1	0.8975
11	1.040
21	1.264
31	1.516
41	1.661
51	2.000
61	2.634
71	3.272
81	3.911
91	4.422

Population of LondonSource: http://en.Wikipedia.org/wiki/Demographics_of_London

18) [T]

a. Using a calculator or a computer program, find the best-fit linear function to measure the population.

b. Find the derivative of the equation in a. and explain its physical meaning.

c. Find the second derivative of the equation and explain its physical meaning.

Answer

a. $P(t) = 0.03983 + 0.4280$

b. $P'(t) = 0.03983$. The population is increasing.

c. $P''(t) = 0$. The rate at which the population is increasing is constant.

19) [T]

a. Using a calculator or a computer program, find the best-fit quadratic curve through the data.

b. Find the derivative of the equation and explain its physical meaning.

c. Find the second derivative of the equation and explain its physical meaning.

For the following exercises, consider an astronaut on a large planet in another galaxy. To learn more about the composition of this planet, the astronaut drops an electronic sensor into a deep trench. The sensor transmits its vertical position every second in relation to the astronaut's position. The summary of the falling sensor data is displayed in the following table.

Time after dropping (s)	Position (m)
0	0
1	−1
2	−2
3	−5
4	−7
5	−14

20) [T]

a. Using a calculator or computer program, find the best-fit quadratic curve to the data.

b. Find the derivative of the position function and explain its physical meaning.

c. Find the second derivative of the position function and explain its physical meaning.

Answer

a. $p(t) = -0.6071x^2 + 0.4357x - 0.3571$

b. $p'(t) = -1.214x + 0.4357$. This is the velocity of the sensor.

c. $p''(t) = -1.214$. This is the acceleration of the sensor; it is a constant acceleration downward.

21) [T]

a. Using a calculator or computer program, find the best-fit cubic curve to the data.

b. Find the derivative of the position function and explain its physical meaning.

c. Find the second derivative of the position function and explain its physical meaning.

d. Using the result from c. explain why a cubic function is not a good choice for this problem.

The following problems deal with the Holling type I, II, and III equations. These equations describe the ecological event of growth of a predator population given the amount of prey available for consumption.

22) [T] The **Holling type I equation** is described by $f(x) = ax$, where x is the amount of prey available and $a > 0$ is the rate at which the predator meets the prey for consumption.

a. Graph the Holling type I equation, given $a = 0.5$.

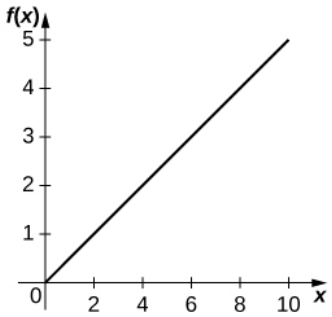
b. Determine the first derivative of the Holling type I equation and explain physically what the derivative implies.

c. Determine the second derivative of the Holling type I equation and explain physically what the derivative implies.

d. Using the interpretations from b. and c. explain why the Holling type I equation may not be realistic.

Answer

a.



b. $f'(x) = a$. The more increase in prey, the more growth for predators.

c. $f''(x) = 0$. As the amount of prey increases, the rate at which the predator population growth increases is constant.

d. This equation assumes that if there is more prey, the predator is able to increase consumption linearly. This assumption is unphysical because we would expect there to be some saturation point at which there is too much prey for the predator to consume adequately.

23) [T] The Holling type II equation is described by $f(x) = \frac{ax}{n+x}$, where x is the amount of prey available and $a > 0$ is the maximum consumption rate of the predator.

a. Graph the Holling type II equation given $a = 0.5$ and $n = 5$. What are the differences between the Holling type I and II equations?

b. Take the first derivative of the Holling type II equation and interpret the physical meaning of the derivative.

c. Show that $f(n) = \frac{1}{2}a$ and interpret the meaning of the parameter n.

d. Find and interpret the meaning of the second derivative. What makes the Holling type II function more realistic than the Holling type I function?

24) [T] The Holling type III equation is described by $f(x) = \frac{ax^2}{n^2+x^2}$, where x is the amount of prey available and $a > 0$ is the maximum consumption rate of the predator.

a. Graph the Holling type III equation given $a = 0.5$ and $n = 5$. What are the differences between the Holling type II and III equations?

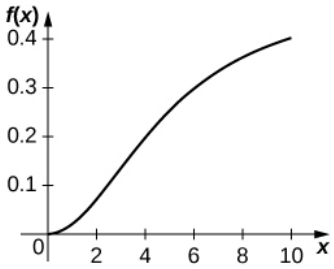
b. Take the first derivative of the Holling type III equation and interpret the physical meaning of the derivative.

c. Find and interpret the meaning of the second derivative (it may help to graph the second derivative).

d. What additional ecological phenomena does the Holling type III function describe compared with the Holling type II function?

Answer

a.



b. $f'(x) = \frac{2axn^2}{(n^2+x^2)^2}$. When the amount of prey increases, the predator growth increases.

c. $f''(x) = \frac{2an^2(n^2-3x^2)}{(n^2+x^2)^3}$. When the amount of prey is extremely small, the rate at which predator growth is increasing is increasing, but when the amount of prey reaches above a certain threshold, the rate at which predator growth is increasing begins to decrease.

d. At lower levels of prey, the prey is more easily able to avoid detection by the predator, so fewer prey individuals are consumed, resulting in less predator growth.

25) [T] The populations of the snowshoe hare (in thousands) and the lynx (in hundreds) collected over 7 years from 1937 to 1943 are shown in the following table. The snowshoe hare is the primary prey of the lynx.

Population of snowshoe hare (thousands)	Population of lynx (hundreds)
20	10
5	15
65	55
95	60

Snowshoe Hare and Lynx PopulationsSource: http://www.biotopics.co.uk/newgcse/predatorprey.html.

a. Graph the data points and determine which Holling-type function fits the data best.

b. Using the meanings of the parameters a and n, determine values for those parameters by examining a graph of the data. Recall that n measures what prey value results in the half-maximum of the predator value.

c. Plot the resulting Holling-type I, II, and III functions on top of the data. Was the result from part a. correct?

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3.5: Derivatives of Trigonometric Functions

Learning Objectives

- Find the derivatives of the sine and cosine function.
- Find the derivatives of the standard trigonometric functions.
- Calculate the higher-order derivatives of the sine and cosine.

One of the most important types of motion in physics is simple harmonic motion, which is associated with such systems as an object with mass oscillating on a spring. Simple harmonic motion can be described by using either sine or cosine functions. In this section we expand our knowledge of derivative formulas to include derivatives of these and other trigonometric functions. We begin with the derivatives of the sine and cosine functions and then use them to obtain formulas for the derivatives of the remaining four trigonometric functions. Being able to calculate the derivatives of the sine and cosine functions will enable us to find the velocity and acceleration of simple harmonic motion.

Derivatives of the Sine and Cosine Functions

We begin our exploration of the derivative for the sine function by using the formula to make a reasonable guess at its derivative. Recall that for a function $f(x)$,

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}.$$

Consequently, for values of h very close to 0,

$$f'(x) \approx \frac{f(x+h) - f(x)}{h}.$$

We see that by using $h = 0.01$,

$$\frac{d}{dx}(\sin x) \approx \frac{\sin(x + 0.01) - \sin x}{0.01}$$

By setting

$$D(x) = \frac{\sin(x + 0.01) - \sin x}{0.01}$$

and using a graphing utility, we can get a graph of an approximation to the derivative of $\sin x$ (Figure 3.5.1).

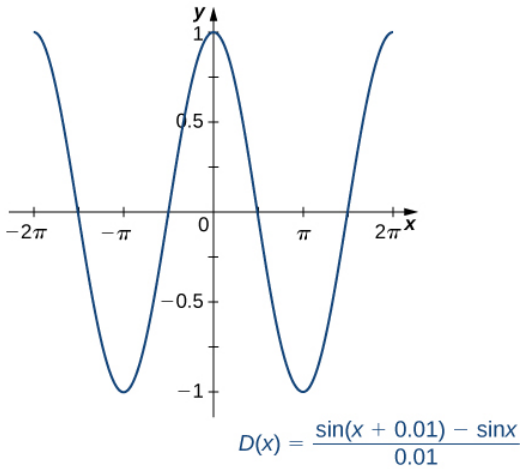


Figure 3.5.1: The graph of the function $D(x)$ looks a lot like a cosine curve.

Upon inspection, the graph of $D(x)$ appears to be very close to the graph of the cosine function. Indeed, we will show that

$$\frac{d}{dx}(\sin x) = \cos x.$$

If we were to follow the same steps to approximate the derivative of the cosine function, we would find that

$$\frac{d}{dx}(\cos x) = -\sin x.$$

The Derivatives of $\sin x$ and $\cos x$

The derivative of the sine function is the cosine and the derivative of the cosine function is the negative sine.

$$\frac{d}{dx}(\sin x) = \cos x$$

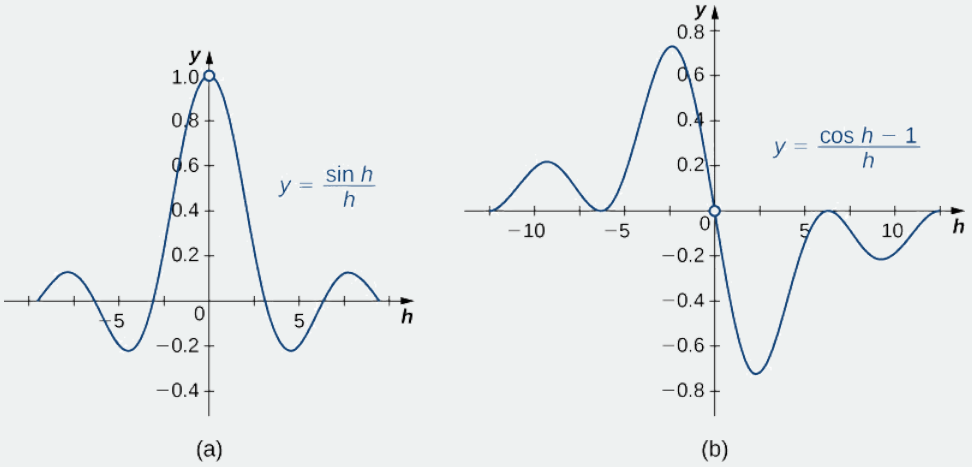
$$\frac{d}{dx}(\cos x) = -\sin x$$

Proof

Because the proofs for $\frac{d}{dx}(\sin x) = \cos x$ and $\frac{d}{dx}(\cos x) = -\sin x$ use similar techniques, we provide only the proof for $\frac{d}{dx}(\sin x) = \cos x$. Before beginning, recall two important trigonometric limits:

$$\lim_{h\to 0} \frac{\sin h}{h} = 1 \text{ and } \lim_{h\to 0} \frac{\cos h - 1}{h} = 0.$$

The graphs of $y = \frac{\sin h}{h}$ and $y = \frac{\cos h - 1}{h}$ are shown in Figure 3.5.2.



(a) (b)

Figure 3.5.2: These graphs show two important limits needed to establish the derivative formulas for the sine and cosine functions.

We also recall the following trigonometric identity for the sine of the sum of two angles:

$$\sin(x + h) = \sin x \cos h + \cos x \sin h.$$

Now that we have gathered all the necessary equations and identities, we proceed with the proof.

$$\begin{aligned}
\frac{d}{dx}(\sin x) &= \lim_{h\to 0}\frac{\sin(x+h)-\sin x}{h} && \text{Apply the definition of the derivative.}\\
&= \lim_{h\to 0}\frac{\sin x\cos h+\cos x\sin h-\sin x}{h} && \text{Use trig identity for the sine of the sum of two angles.}\\
&= \lim_{h\to 0}\left(\frac{\sin x\cos h-\sin x}{h}\right. && \text{Regroup.}\\
&\quad \left.+\frac{\cos x\sin h}{h}\right)\\
&= \lim_{h\to 0}\left(\sin x\left(\frac{\cos h-1}{h}\right)\right. && \text{Factor out } \sin x \text{ and } \cos x\\
&\quad \left.+(\cos x)\left(\frac{\sin h}{h}\right)\right)\\
&= (\sin x)\lim_{h\to 0}\left(\frac{\cos h-1}{h}\right) && \text{Factor } \sin x \text{ and } \cos x \text{ out of limits.}\\
&\quad +(\cos x)\lim_{h\to 0}\left(\frac{\sin h}{h}\right)\\
&= (\sin x)(0)+(\cos x)(1) && \text{Apply trig limit formulas.}\\
&= \cos x && \text{Simplify.}
\end{aligned}$$

□

Figure 3.5.3 shows the relationship between the graph of $f(x)=\sin x$ and its derivative $f'(x)=\cos x$. Notice that at the points where $f(x)=\sin x$ has a horizontal tangent, its derivative $f'(x)=\cos x$ takes on the value zero. We also see that where $f(x)=\sin x$ is increasing, $f'(x)=\cos x>0$ and where $f(x)=\sin x$ is decreasing, $f'(x)=\cos x<0$.

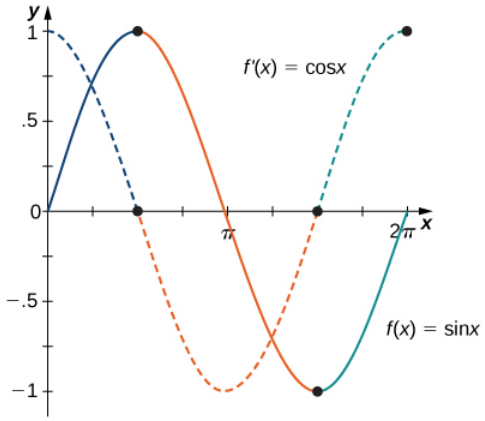


Figure 3.5.3: Where $f(x)$ has a maximum or a minimum, $f'(x)=0$ that is, $f'(x)=0$ where $f(x)$ has a horizontal tangent. These points are noted with dots on the graphs

Example 3.5.1: Differentiating a Function Containing $\sin x$

Find the derivative of $f(x)=5x^3\sin x$.

Solution

Using the product rule, we have

$$\begin{aligned}
f'(x) &= \frac{d}{dx}(5x^3)\cdot\sin x+\frac{d}{dx}(\sin x)\cdot 5x^3\\
&= 15x^2\cdot\sin x+\cos x\cdot 5x^3.
\end{aligned}$$

After simplifying, we obtain

$$f'(x)=15x^2\sin x+5x^3\cos x.$$

Exercise 3.5.1

Find the derivative of $f(x) = \sin x \cos x$.

Hint

Don't forget to use the product rule.

Answer

$$f'(x) = \cos^2 x - \sin^2 x$$

Example 3.5.2: Finding the Derivative of a Function Containing cos x

Find the derivative of $g(x) = \dfrac{\cos x}{4x^2}$.

Solution

By applying the quotient rule, we have

$$g'(x) = \frac{(-\sin x)4x^2 - 8x(\cos x)}{(4x^2)^2}.$$

Simplifying, we obtain

$$g'(x) = \frac{-4x^2 \sin x - 8x \cos x}{16x^4} = \frac{-x \sin x - 2 \cos x}{4x^3}.$$

Exercise 3.5.2

Find the derivative of $f(x) = \dfrac{x}{\cos x}$.

Hint

Use the quotient rule.

Answer

$$f'(x) = \frac{\cos x + x \sin x}{\cos^2 x}$$

Example 3.5.3: An Application to Velocity

A particle moves along a coordinate axis in such a way that its position at time t is given by $s(t) = 2 \sin t - t$ for $0 \leq t \leq 2\pi$. At what times is the particle at rest?

Solution

To determine when the particle is at rest, set $s'(t) = v(t) = 0$. Begin by finding $s'(t)$. We obtain

$$s'(t) = 2 \cos t - 1,$$

so we must solve

$$2 \cos t - 1 = 0 \text{ for } 0 \leq t \leq 2\pi.$$

The solutions to this equation are $t = \dfrac{\pi}{3}$ and $t = \dfrac{5\pi}{3}$. Thus the particle is at rest at times $t = \dfrac{\pi}{3}$ and $t = \dfrac{5\pi}{3}$.

Exercise 3.5.3

A particle moves along a coordinate axis. Its position at time t is given by $s(t) = \sqrt{3}t + 2\cos t$ for $0 \le t \le 2\pi$. At what times is the particle at rest?

Hint

Use the previous example as a guide.

Answer

$t = \dfrac{\pi}{3}, \quad t = \dfrac{2\pi}{3}$

Derivatives of Other Trigonometric Functions

Since the remaining four trigonometric functions may be expressed as quotients involving sine, cosine, or both, we can use the quotient rule to find formulas for their derivatives.

Example 3.5.4: The Derivative of the Tangent Function

Find the derivative of $f(x) = \tan x$.

Solution

Start by expressing $\tan x$ as the quotient of $\sin x$ and $\cos x$:

$$f(x) = \tan x = \frac{\sin x}{\cos x}.$$

Now apply the quotient rule to obtain

$$f'(x) = \frac{\cos x \cos x - (-\sin x)\sin x}{(\cos x)^2}.$$

Simplifying, we obtain

$$f'(x) = \frac{\cos^2 x + \sin^2 x}{\cos^2 x}.$$

Recognizing that $\cos^2 x + \sin^2 x = 1$, by the Pythagorean theorem, we now have

$$f'(x) = \frac{1}{\cos^2 x}$$

Finally, use the identity $\sec x = \dfrac{1}{\cos x}$ to obtain

$$f'(x) = \sec^2 x.$$

Exercise 3.5.4

Find the derivative of $f(x) = \cot x$.

Hint

Rewrite $\cot x$ as $\dfrac{\cos x}{\sin x}$ and use the quotient rule.

Answer

$f'(x) = -\csc^2 x$

The derivatives of the remaining trigonometric functions may be obtained by using similar techniques. We provide these formulas in the following theorem.

Derivatives of $\tan x$, $\cot x$, $\sec x$, and $\csc x$

The derivatives of the remaining trigonometric functions are as follows:

$$\frac{d}{dx}(\tan x) = \sec^2 x \tag{3.5.1}$$

$$\frac{d}{dx}(\cot x) = -\csc^2 x \tag{3.5.2}$$

$$\frac{d}{dx}(\sec x) = \sec x \tan x \tag{3.5.3}$$

$$\frac{d}{dx}(\csc x) = -\csc x \cot x. \tag{3.5.4}$$

Example 3.5.5: Finding the Equation of a Tangent Line

Find the equation of a line tangent to the graph of $f(x) = \cot x$ at $x = \frac{\pi}{4}$.

Solution

To find the equation of the tangent line, we need a point and a slope at that point. To find the point, compute

$$f\left(\tfrac{\pi}{4}\right) = \cot \tfrac{\pi}{4} = 1.$$

Thus the tangent line passes through the point $(\frac{\pi}{4}, 1)$. Next, find the slope by finding the derivative of $f(x) = \cot x$ and evaluating it at $\frac{\pi}{4}$:

$$f'(x) = -\csc^2 x \text{ and } f'\left(\tfrac{\pi}{4}\right) = -\csc^2\left(\tfrac{\pi}{4}\right) = -2.$$

Using the point-slope equation of the line, we obtain

$$y - 1 = -2\left(x - \tfrac{\pi}{4}\right)$$

or equivalently,

$$y = -2x + 1 + \tfrac{\pi}{2}.$$

Example 3.5.6: Finding the Derivative of Trigonometric Functions

Find the derivative of $f(x) = \csc x + x \tan x$.

Solution

To find this derivative, we must use both the sum rule and the product rule. Using the sum rule, we find

$$f'(x) = \frac{d}{dx}(\csc x) + \frac{d}{dx}(x \tan x).$$

In the first term, $\frac{d}{dx}(\csc x) = -\csc x \cot x$, and by applying the product rule to the second term we obtain

$$\frac{d}{dx}(x \tan x) = (1)(\tan x) + (\sec^2 x)(x).$$

Therefore, we have

$$f'(x) = -\csc x \cot x + \tan x + x \sec^2 x.$$

Exercise 3.5.5

Find the derivative of $f(x) = 2 \tan x - 3 \cot x$.

Hint

Use the rule for differentiating a constant multiple and the rule for differentiating a difference of two functions.

Answer

$f'(x) = 2\sec^2 x + 3\csc^2 x$

Exercise 3.5.6

Find the slope of the line tangent to the graph of $f(x) = \tan x$ at $x = \frac{\pi}{6}$.

Hint

Evaluate the derivative at $x = \frac{\pi}{6}$.

Answer

$\frac{4}{3}$

Higher-Order Derivatives

The higher-order derivatives of $\sin x$ and $\cos x$ follow a repeating pattern. By following the pattern, we can find any higher-order derivative of $\sin x$ and $\cos x$.

Example 3.5.7: Finding Higher-Order Derivatives of $y = \sin x$

Find the first four derivatives of $y = \sin x$.

Solution

Each step in the chain is straightforward:

$$\begin{aligned} y &= \sin x \\ \frac{dy}{dx} &= \cos x \\ \frac{d^2y}{dx^2} &= -\sin x \\ \frac{d^3y}{dx^3} &= -\cos x \\ \frac{d^4y}{dx^4} &= \sin x \end{aligned}$$

Analysis

Once we recognize the pattern of derivatives, we can find any higher-order derivative by determining the step in the pattern to which it corresponds. For example, every fourth derivative of $\sin x$ equals $\sin x$, so

$$\frac{d^4}{dx^4}(\sin x) = \frac{d^8}{dx^8}(\sin x) = \frac{d^{12}}{dx^{12}}(\sin x) = \ldots = \frac{d^{4n}}{dx^{4n}}(\sin x) = \sin x$$

$$\frac{d^5}{dx^5}(\sin x) = \frac{d^9}{dx^9}(\sin x) = \frac{d^{13}}{dx^{13}}(\sin x) = \ldots = \frac{d^{4n+1}}{dx^{4n+1}}(\sin x) = \cos x.$$

Exercise 3.5.7

For $y = \cos x$, find $\frac{d^4y}{dx^4}$.

Hint

See the previous example.

Answer

$\cos x$

Example 3.5.8: Using the Pattern for Higher-Order Derivatives of $y = \sin x$

Find $\dfrac{d^{74}}{dx^{74}}(\sin x)$.

Solution

We can see right away that for the 74th derivative of $\sin x$, $74 = 4(18) + 2$, so

$$\frac{d^{74}}{dx^{74}}(\sin x) = \frac{d^{72+2}}{dx^{72+2}}(\sin x) = \frac{d^2}{dx^2}(\sin x) = -\sin x.$$

Exercise 3.5.8

For $y = \sin x$, find $\dfrac{d^{59}}{dx^{59}}(\sin x)$.

Hint

$$\frac{d^{59}}{dx^{59}}(\sin x) = \frac{d^{4\cdot 14+3}}{dx^{4\cdot 14+3}}(\sin x)$$

Answer

$-\cos x$

Example 3.5.9: An Application to Acceleration

A particle moves along a coordinate axis in such a way that its position at time t is given by $s(t) = 2 - \sin t$. Find $v(\pi/4)$ and $a(\pi/4)$. Compare these values and decide whether the particle is speeding up or slowing down.

Solution

First find $v(t) = s'(t)$

$$v(t) = s'(t) = -\cos t.$$

Thus,

$$v\left(\tfrac{\pi}{4}\right) = -\frac{1}{\sqrt{2}} = -\frac{\sqrt{2}}{2}.$$

Next, find $a(t) = v'(t)$. Thus, $a(t) = v'(t) = \sin t$ and we have

$$a\left(\tfrac{\pi}{4}\right) = \frac{1}{\sqrt{2}} = \frac{\sqrt{2}}{2}.$$

Since $v\left(\tfrac{\pi}{4}\right) = -\dfrac{\sqrt{2}}{2} < 0$ and $a\left(\tfrac{\pi}{4}\right) = \dfrac{\sqrt{2}}{2} > 0$, we see that velocity and acceleration are acting in opposite directions; that is, the object is being accelerated in the direction opposite to the direction in which it is traveling. Consequently, the particle is slowing down.

Exercise 3.5.9

A block attached to a spring is moving vertically. Its position at time t is given by $s(t) = 2\sin t$. Find $v\left(\tfrac{5\pi}{6}\right)$ and $a\left(\tfrac{5\pi}{6}\right)$. Compare these values and decide whether the block is speeding up or slowing down.

Hint

Use Example 3.5.9 as a guide.

Answer

$v\left(\tfrac{5\pi}{6}\right) = -\sqrt{3} < 0$ and $a\left(\tfrac{5\pi}{6}\right) = -1 < 0$. The block is speeding up.

Key Concepts

- We can find the derivatives of $\sin x$ and $\cos x$ by using the definition of derivative and the limit formulas found earlier. The results are

  $$\frac{d}{dx}\big(\sin x\big) = \cos x \quad \text{and} \quad \frac{d}{dx}\big(\cos x\big) = -\sin x.$$

- With these two formulas, we can determine the derivatives of all six basic trigonometric functions.

Key Equations

- **Derivative of sine function**

  $$\frac{d}{dx}(\sin x) = \cos x$$

- **Derivative of cosine function**

  $$\frac{d}{dx}(\cos x) = -\sin x$$

- **Derivative of tangent function**

  $$\frac{d}{dx}(\tan x) = \sec^2 x$$

- **Derivative of cotangent function**

  $$\frac{d}{dx}(\cot x) = -\csc^2 x$$

- **Derivative of secant function**

  $$\frac{d}{dx}(\sec x) = \sec x \tan x$$

- **Derivative of cosecant function**

  $$\frac{d}{dx}(\csc x) = -\csc x \cot x$$

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3.5E: EXERCISES FOR SECTION 3.5

In exercises 1 - 10, find $\frac{dy}{dx}$ for the given functions.

1) $y = x^2 - \sec x + 1$

Answer

$$\frac{dy}{dx} = 2x - \sec x \tan x$$

2) $y = 3 \csc x + \frac{5}{x}$

3) $y = x^2 \cot x$

Answer

$$\frac{dy}{dx} = 2x \cot x - x^2 \csc^2 x$$

4) $y = x - x^3 \sin x$

5) $y = \frac{\sec x}{x}$

Answer

$$\frac{dy}{dx} = \frac{x \sec x \tan x - \sec x}{x^2}$$

6) $y = \sin x \tan x$

7) $y = (x + \cos x)(1 - \sin x)$

Answer

$$\frac{dy}{dx} = (1 - \sin x)(1 - \sin x) - \cos x(x + \cos x)$$

8) $y = \frac{\tan x}{1 - \sec x}$

9) $y = \frac{1 - \cot x}{1 + \cot x}$

Answer

$$\frac{dy}{dx} = \frac{2 \csc^2 x}{(1 + \cot x)^2}$$

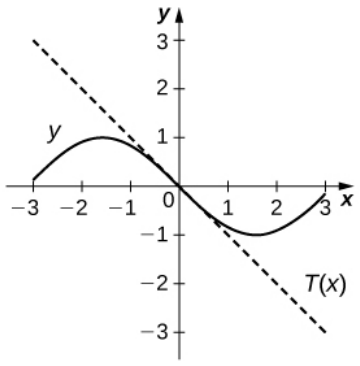
10) $y = (\cos x)(1 + \csc x)$

In exercises 11 - 16, find the equation of the tangent line to each of the given functions at the indicated values of x. Then use a calculator to graph both the function and the tangent line to ensure the equation for the tangent line is correct.

11) [T] $f(x) = -\sin x, \quad x = 0$

Answer

$y = -x$



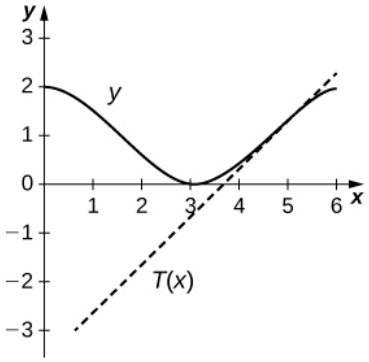
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12) [T] $f(x) = \csc x, \quad x = \frac{\pi}{2}$

13) [T] $f(x) = 1 + \cos x, \quad x = \frac{3\pi}{2}$

Answer

$y = x + \frac{2-3\pi}{2}$

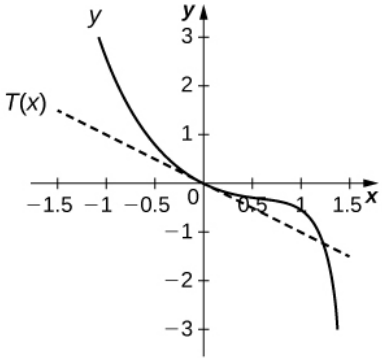


14) [T] $f(x) = \sec x, \quad x = \frac{\pi}{4}$

15) [T] $f(x) = x^2 - \tan x, \quad x = 0$

Answer

$y = -x$



16) [T] $f(x) = 5 \cot x, \quad x = \frac{\pi}{4}$

In exercises 17 - 22, find $\frac{d^2y}{dx^2}$ for the given functions.

17) $y = x \sin x - \cos x$

Answer

$\frac{d^2y}{dx^2} = 3 \cos x - x \sin x$

18) $y = \sin x \cos x$

19) $y = x - \frac{1}{2} \sin x$

Answer

$\frac{d^2y}{dx^2} = \frac{1}{2} \sin x$

20) $y = \frac{1}{x} + \tan x$

21) $y = 2 \csc x$

Answer

$$\frac{d^2y}{dx^2} = 2\csc(x)\left(\csc^2(x) + \cot^2(x)\right)$$

22) $y = \sec^2 x$

23) Find all x values on the graph of $f(x) = -3\sin x \cos x$ where the tangent line is horizontal.

Answer

$x = \frac{(2n+1)\pi}{4}$,where n is an integer

24) Find all x values on the graph of $f(x) = x - 2\cos x$ for $0 < x < 2\pi$ where the tangent line has slope 2.

25) Let $f(x) = \cot x$. Determine the points on the graph of f for $0 < x < 2\pi$ where the tangent line(s) is (are) parallel to the line $y = -2x$.

Answer

$\left(\frac{\pi}{4}, 1\right), \quad \left(\frac{3\pi}{4}, -1\right), \quad \left(\frac{5\pi}{4}, 1\right), \quad \left(\frac{7\pi}{4}, -1\right)$

26) [T] A mass on a spring bounces up and down in simple harmonic motion, modeled by the function $s(t) = -6\cos t$ where s is measured in inches and t is measured in seconds. Find the rate at which the spring is oscillating at $t = 5$ s.

27) Let the position of a swinging pendulum in simple harmonic motion be given by $s(t) = a\cos t + b\sin t$. Find the constants a and b such that when the velocity is 3 cm/s, $s = 0$ and $t = 0$.

Answer

$a = 0, \quad b = 3$

28) After a diver jumps off a diving board, the edge of the board oscillates with position given by $s(t) = -5\cos t$ cm at t seconds after the jump.

a. Sketch one period of the position function for $t \geq 0$.

b. Find the velocity function.

c. Sketch one period of the velocity function for $t \geq 0$.

d. Determine the times when the velocity is 0 over one period.

e. Find the acceleration function.

f. Sketch one period of the acceleration function for $t \geq 0$.

29) The number of hamburgers sold at a fast-food restaurant in Pasadena, California, is given by $y = 10 + 5\sin x$ where y is the number of hamburgers sold and x represents the number of hours after the restaurant opened at 11 a.m. until 11 p.m., when the store closes. Find y' and determine the intervals where the number of burgers being sold is increasing.

Answer

$y' = 5\cos(x)$, increasing on $\left(0, \frac{\pi}{2}\right)$, $\left(\frac{3\pi}{2}, \frac{5\pi}{2}\right)$, and $\left(\frac{7\pi}{2}, 12\right)$

30) [T] The amount of rainfall per month in Phoenix, Arizona, can be approximated by $y(t) = 0.5 + 0.3\cos t$, where t is months since January. Find $y'$and use a calculator to determine the intervals where the amount of rain falling is decreasing.

For exercises 31 - 33, use the quotient rule to derive the given equations.

31) $\frac{d}{dx}(\cot x) = -\csc^2 x$

32) $\frac{d}{dx}(\sec x) = \sec x \tan x$

33) $\frac{d}{dx}(\csc x) = -\csc x \cot x$

34) Use the definition of derivative and the identity $\cos(x + h) = \cos x \cos h - \sin x \sin h$ to prove that $\frac{d}{dx}(\cos x) = -\sin x$.

For exercises 35 - 39, find the requested higher-order derivative for the given functions.

35) $\frac{d^3y}{dx^3}$ of $y = 3\cos x$

Answer

$\frac{d^3y}{dx^3} = 3\sin x$

36) $\frac{d^2y}{dx^2}$ of $y = 3\sin x + x^2 \cos x$

37) $\frac{d^4y}{dx^4}$ of $y = 5\cos x$

Answer

$$\frac{d^4y}{dx^4} = 5\cos x$$

38) $\frac{d^2y}{dx^2}$ of $y = \sec x + \cot x$

39) $\frac{d^3y}{dx^3}$ of $y = x^{10} - \sec x$

Answer

$$\frac{d^3y}{dx^3} = 720x^7 - 5\tan(x)\sec^3(x) - \tan^3(x)\sec(x)$$

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3.6: The Chain Rule

Learning Objectives

- State the chain rule for the composition of two functions.
- Apply the chain rule together with the power rule.
- Apply the chain rule and the product/quotient rules correctly in combination when both are necessary.
- Recognize the chain rule for a composition of three or more functions.
- Describe the proof of the chain rule.

We have seen the techniques for differentiating basic functions ($x^n, \sin x, \cos x,$ etc.) as well as sums, differences, products, quotients, and constant multiples of these functions. However, these techniques do not allow us to differentiate compositions of functions, such as $h(x) = \sin(x^3)$ or $k(x) = \sqrt{3x^2 + 1}$. In this section, we study the rule for finding the derivative of the composition of two or more functions.

Deriving the Chain Rule

When we have a function that is a composition of two or more functions, we could use all of the techniques we have already learned to differentiate it. However, using all of those techniques to break down a function into simpler parts that we are able to differentiate can get cumbersome. Instead, we use the *chain rule*, which states that the derivative of a composite function is the derivative of the outer function evaluated at the inner function times the derivative of the inner function.

To put this rule into context, let's take a look at an example: $h(x) = \sin(x^3)$. We can think of the derivative of this function with respect to x as the rate of change of $\sin(x^3)$ relative to the change in x. Consequently, we want to know how $\sin(x^3)$ changes as x changes. We can think of this event as a chain reaction: As x changes, x^3 changes, which leads to a change in $\sin(x^3)$. This chain reaction gives us hints as to what is involved in computing the derivative of $\sin(x^3)$. First of all, a change in x forcing a change in x^3 suggests that somehow the derivative of x^3 is involved. In addition, the change in x^3 forcing a change in $\sin(x^3)$ suggests that the derivative of $\sin(u)$ with respect to u, where $u = x^3$, is also part of the final derivative.

We can take a more formal look at the derivative of $h(x) = \sin(x^3)$ by setting up the limit that would give us the derivative at a specific value a in the domain of $h(x) = \sin(x^3)$.

$$h'(a) = \lim_{x \to a} \frac{\sin(x^3) - \sin(a^3)}{x - a}$$

This expression does not seem particularly helpful; however, we can modify it by multiplying and dividing by the expression $x^3 - a^3$ to obtain

$$h'(a) = \lim_{x \to a} \frac{\sin(x^3) - \sin(a^3)}{x^3 - a^3} \cdot \frac{x^3 - a^3}{x - a}.$$

From the definition of the derivative, we can see that the second factor is the derivative of x^3 at $x = a$. That is,

$$\lim_{x \to a} \frac{x^3 - a^3}{x - a} = \frac{d}{dx}(x^3)\Big|_{x=a} = 3a^2.$$

However, it might be a little more challenging to recognize that the first term is also a derivative. We can see this by letting $u = x^3$ and observing that as $x \to a, u \to a^3$:

$$\begin{aligned} \lim_{x \to a} \frac{\sin(x^3) - \sin(a^3)}{x^3 - a^3} &= \lim_{u \to a^3} \frac{\sin u - \sin(a^3)}{u - a^3} \\ &= \frac{d}{du}(\sin u)\Big|_{u=a^3} \\ &= \cos(a^3) \end{aligned}.$$

Thus, $h'(a) = \cos(a^3) \cdot 3a^2$.

In other words, if $h(x) = \sin(x^3)$, then $h'(x) = \cos(x^3) \cdot 3x^2$. Thus, if we think of $h(x) = \sin(x^3)$ as the composition $(f \circ g)(x) = f\big(g(x)\big)$ where $f(x) = \sin x$ and $g(x) = x^3$, then the derivative of $h(x) = \sin(x^3)$ is the product of the derivative of $g(x) = x^3$ and the derivative of the function $f(x) = \sin x$ evaluated at the function $g(x) = x^3$. At this point, we anticipate that for $h(x) = \sin\big(g(x)\big)$, it is quite likely that $h'(x) = \cos\big(g(x)\big)g'(x)$. As we determined above, this is the case for $h(x) = \sin(x^3)$.

Now that we have derived a special case of the chain rule, we state the general case and then apply it in a general form to other composite functions. An informal proof is provided at the end of the section.

Rule: The Chain Rule

Let f and g be functions. For all x in the domain of g for which g is differentiable at x and f is differentiable at $g(x)$, the derivative of the composite function

$$h(x) = (f \circ g)(x) = f\big(g(x)\big)$$

is given by

$$h'(x) = f'\big(g(x)\big) \cdot g'(x).$$

Alternatively, if y is a function of u, and u is a function of x, then

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}.$$

Problem-Solving Strategy: Applying the Chain Rule

1. To differentiate $h(x) = f\big(g(x)\big)$, begin by identifying $f(x)$ and $g(x)$.
2. Find $f'(x)$ and evaluate it at $g(x)$ to obtain $f'\big(g(x)\big)$.
3. Find $g'(x)$.
4. Write $h'(x) = f'\big(g(x)\big) \cdot g'(x)$.

Note: When applying the chain rule to the composition of two or more functions, keep in mind that we work our way from the outside function in. It is also useful to remember that the derivative of the composition of two functions can be thought of as having two parts; the derivative of the composition of three functions has three parts; and so on. Also, remember that we never evaluate a derivative at a derivative.

The Chain and Power Rules Combined

We can now apply the chain rule to composite functions, but note that we often need to use it with other rules. For example, to find derivatives of functions of the form $h(x) = \big(g(x)\big)^n$, we need to use the chain rule combined with the power rule. To do so, we can think of $h(x) = \big(g(x)\big)^n$ as $f\big(g(x)\big)$ where $f(x) = x^n$. Then $f'(x) = nx^{n-1}$. Thus, $f'\big(g(x)\big) = n\big(g(x)\big)^{n-1}$. This leads us to the derivative of a power function using the chain rule,

$$h'(x) = n\big(g(x)\big)^{n-1} \cdot g'(x)$$

Rule: Power Rule for Composition of Functions (General Power Rule)

For all values of x for which the derivative is defined, if

$$h(x) = \big(g(x)\big)^n,$$

Then

$$h'(x) = n\big(g(x)\big)^{n-1} \cdot g'(x). \tag{3.6.1}$$

Example 3.6.1: Using the Chain and Power Rules

Find the derivative of $h(x) = \dfrac{1}{(3x^2+1)^2}$.

Solution

First, rewrite $h(x) = \dfrac{1}{(3x^2+1)^2} = (3x^2+1)^{-2}$.

Applying the power rule with $g(x) = 3x^2+1$, we have

$$h'(x) = -2(3x^2+1)^{-3} \cdot 6x.$$

Rewriting back to the original form gives us

$$h'(x) = \frac{-12x}{(3x^2+1)^3}$$

Exercise 3.6.1

Find the derivative of $h(x) = (2x^3+2x-1)^4$.

Hint

Use the General Power Rule (Equation 3.6.1) with $g(x) = 2x^3+2x-1$.

Answer

$h'(x) = 4(2x^3+2x-1)^3(6x^2+2) = 8(3x^2+1)(2x^3+2x-1)^3$

Example 3.6.2: Using the Chain and Power Rules with a Trigonometric Function

Find the derivative of $h(x) = \sin^3 x$.

Solution

First recall that $\sin^3 x = (\sin x)^3$, so we can rewrite $h(x) = \sin^3 x$ as $h(x) = (\sin x)^3$.

Applying the power rule with $g(x) = \sin x$, we obtain

$$h'(x) = 3(\sin x)^2 \cos x = 3\sin^2 x \cos x.$$

Example 3.6.3: Finding the Equation of a Tangent Line

Find the equation of a line tangent to the graph of $h(x) = \dfrac{1}{(3x-5)^2}$ at $x = 2$.

Solution

Because we are finding an equation of a line, we need a point. The x-coordinate of the point is 2. To find the y-coordinate, substitute 2 into $h(x)$. Since $h(2) = \dfrac{1}{(3(2)-5)^2} = 1$, the point is $(2, 1)$.

For the slope, we need $h'(2)$. To find $h'(x)$, first we rewrite $h(x) = (3x-5)^{-2}$ and apply the power rule to obtain

$$h'(x) = -2(3x-5)^{-3}(3) = -6(3x-5)^{-3}.$$

By substituting, we have $h'(2) = -6(3(2)-5)^{-3} = -6$.

Therefore, the line has equation $y - 1 = -6(x-2)$. Rewriting, the equation of the line is $y = -6x + 13$.

Exercise 3.6.2

Find the equation of the line tangent to the graph of $f(x) = (x^2 - 2)^3$ at $x = -2$.

Hint

Use the preceding example as a guide.

Answer

$y = -48x - 88$

Combining the Chain Rule with Other Rules

Now that we can combine the chain rule and the power rule, we examine how to combine the chain rule with the other rules we have learned. In particular, we can use it with the formulas for the derivatives of trigonometric functions or with the product rule.

Example 3.6.4: Using the Chain Rule on a General Cosine Function

Find the derivative of $h(x) = \cos\big(g(x)\big)$.

Solution

Think of $h(x) = \cos\big(g(x)\big)$ as $f\big(g(x)\big)$ where $f(x) = \cos x$. Since $f'(x) = -\sin x$, we have $f'\big(g(x)\big) = -\sin\big(g(x)\big)$. Then we do the following calculation.

$$\begin{aligned} h'(x) &= f'\big(g(x)\big) \cdot g'(x) && \text{Apply the chain rule.} \\ &= -\sin\big(g(x)\big) \cdot g'(x) && \text{Substitute } f'\big(g(x)\big) = -\sin\big(g(x)\big). \end{aligned}$$

Thus, the derivative of $h(x) = \cos\big(g(x)\big)$ is given by $h'(x) = -\sin\big(g(x)\big) \cdot g'(x)$.

In the following example we apply the rule that we have just derived.

Example 3.6.5: Using the Chain Rule on a Cosine Function

Find the derivative of $h(x) = \cos(5x^2)$.

Solution

Let $g(x) = 5x^2$. Then $g'(x) = 10x$. Using the result from the previous example,

$$h'(x) = -\sin(5x^2) \cdot 10x = -10x\sin(5x^2)$$

Example 3.6.6: Using the Chain Rule on Another Trigonometric Function

Find the derivative of $h(x) = \sec(4x^5 + 2x)$.

Solution

Apply the chain rule to $h(x) = \sec\big(g(x)\big)$ to obtain

$$h'(x) = \sec(g(x))\tan\big(g(x)\big) \cdot g'(x).$$

In this problem, $g(x) = 4x^5 + 2x$, so we have $g'(x) = 20x^4 + 2$. Therefore, we obtain

$$h'(x) = \sec(4x^5 + 2x)\tan(4x^5 + 2x)(20x^4 + 2) = (20x^4 + 2)\sec(4x^5 + 2x)\tan(4x^5 + 2x).$$

Exercise 3.6.3

Find the derivative of $h(x) = \sin(7x + 2)$.

Hint

Apply the chain rule to $h(x) = \sin\big(g(x)\big)$ first and then use $g(x) = 7x + 2$.

Answer

$h'(x) = 7\cos(7x + 2)$

At this point we provide a list of derivative formulas that may be obtained by applying the chain rule in conjunction with the formulas for derivatives of trigonometric functions. Their derivations are similar to those used in the examples above. For convenience, formulas are also given in Leibniz's notation, which some students find easier to remember. (We discuss the chain rule using Leibniz's notation at the end of this section.) It is not absolutely necessary to memorize these as separate formulas as they are all applications of the chain rule to previously learned formulas.

Using the Chain Rule with Trigonometric Functions

For all values of x for which the derivative is defined,

$\frac{d}{dx}\Big(\sin(g(x))\Big) = \cos(g(x)) \cdot g'(x)$	$\frac{d}{dx}\Big(\sin u\Big) = \cos u \cdot \frac{du}{dx}$
$\frac{d}{dx}\Big(\cos(g(x))\Big) = -\sin(g(x)) \cdot g'(x)$	$\frac{d}{dx}\Big(\cos u\Big) = -\sin u \cdot \frac{du}{dx}$
$\frac{d}{dx}\Big(\tan(g(x))\Big) = \sec^2(g(x)) \cdot g'(x)$	$\frac{d}{dx}\Big(\tan u\Big) = \sec^2 u \cdot \frac{du}{dx}$
$\frac{d}{dx}\Big(\cot(g(x))\Big) = -\csc^2(g(x)) \cdot g'(x)$	$\frac{d}{dx}\Big(\cot u\Big) = -\csc^2 u \cdot \frac{du}{dx}$
$\frac{d}{dx}\Big(\sec(g(x))\Big) = \sec(g(x))\tan(g(x)) \cdot g'(x)$	$\frac{d}{dx}\Big(\sec u\Big) = \sec u \tan u \cdot \frac{du}{dx}$
$\frac{d}{dx}\Big(\csc(g(x))\Big) = -\csc(g(x))\cot(g(x)) \cdot g'(x)$	$\frac{d}{dx}\Big(\csc u\Big) = -\csc u \cot u \cdot \frac{du}{dx}.$

Example 3.6.7: Combining the Chain Rule with the Product Rule

Find the derivative of $h(x) = (2x + 1)^5(3x - 2)^7$.

Solution

First apply the product rule, then apply the chain rule to each term of the product.

$$
\begin{aligned}
h'(x) &= \frac{d}{dx}\big((2x+1)^5\big) \cdot (3x-2)^7 + \frac{d}{dx}\big((3x-2)^7\big) \cdot (2x+1)^5 && \text{Apply the product rule.} \\
&= 5(2x+1)^4 \cdot 2 \cdot (3x-2)^7 + 7(3x-2)^6 \cdot 3 \cdot (2x+1)^5 && \text{Apply the chain rule.} \\
&= 10(2x+1)^4(3x-2)^7 + 21(3x-2)^6(2x+1)^5 && \text{Simplify.} \\
&= (2x+1)^4(3x-2)^6(10(3x-2) + 21(2x+1)) && \text{Factor out } (2x+1)^4(3x-2)^6 \\
&= (2x+1)^4(3x-2)^6(72x+1) && \text{Simplify.}
\end{aligned}
$$

Exercise 3.6.4

Find the derivative of $h(x) = \dfrac{x}{(2x+3)^3}$.

Hint

Start out by applying the quotient rule. Remember to use the chain rule to differentiate the denominator.

Answer

$h'(x) = \dfrac{3 - 4x}{(2x+3)^4}$

Composites of Three or More Functions

We can now combine the chain rule with other rules for differentiating functions, but when we are differentiating the composition of three or more functions, we need to apply the chain rule more than once. If we look at this situation in general terms, we can generate a formula, but we do not need to remember it, as we can simply apply the chain rule multiple times.

In general terms, first we let

$$k(x) = h\Big(f\big(g(x)\big)\Big).$$

Then, applying the chain rule once we obtain

$$k'(x) = \frac{d}{dx}\Big(h\big(f\big(g(x)\big)\big)\Big) = h'\Big(f\big(g(x)\big)\Big) \cdot \frac{d}{dx}\Big(f\big(g(x)\big)\Big).$$

Applying the chain rule again, we obtain

$$k'(x) = h'\Big(f\big(g(x)\big)\Big) \cdot f'\big(g(x)\big) \cdot g'(x).$$

Rule: Chain Rule for a Composition of Three Functions

For all values of x for which the function is differentiable, if

$$k(x) = h\Big(f\big(g(x)\big)\Big),$$

then

$$k'(x) = h'\Big(f\big(g(x)\big)\Big) \cdot f'\big(g(x)\big) \cdot g'(x).$$

In other words, we are applying the chain rule twice.

Notice that the derivative of the composition of three functions has three parts. (Similarly, the derivative of the composition of four functions has four parts, and so on.) Also, *remember, we can always work from the outside in, taking one derivative at a time.*

Example 3.6.8: Differentiating a Composite of Three Functions

Find the derivative of $k(x) = \cos^4(7x^2 + 1)$.

Solution

First, rewrite $k(x)$ as

$$k(x) = \big(\cos(7x^2 + 1)\big)^4.$$

Then apply the chain rule several times.

$$\begin{aligned} k'(x) &= 4(\cos(7x^2+1))^3 \cdot \frac{d}{dx}\big(\cos(7x^2+1)\big) && \text{Apply the chain rule.} \\ &= 4(\cos(7x^2+1))^3(-\sin(7x^2+1)) \cdot \frac{d}{dx}\big(7x^2+1\big) && \text{Apply the chain rule.} \\ &= 4(\cos(7x^2+1))^3(-\sin(7x^2+1))(14x) && \text{Apply the chain rule.} \\ &= -56x\sin(7x^2+1)\cos^3(7x^2+1) && \text{Simplify} \end{aligned}$$

Exercise 3.6.5

Find the derivative of $h(x) = \sin^6(x^3)$.

Hint

Rewrite $h(x) = \sin^6(x^3) = \big(\sin(x^3)\big)^6$ and use Example 3.6.8 as a guide.

Answer

$h'(x) = 18x^2 \sin^5(x^3)\cos(x^3)$

Example 3.6.9: Using the Chain Rule in a Velocity Problem

A particle moves along a coordinate axis. Its position at time t is given by $s(t) = \sin(2t) + \cos(3t)$. What is the velocity of the particle at time $t = \frac{\pi}{6}$?

Solution

To find $v(t)$, the velocity of the particle at time t, we must differentiate $s(t)$. Thus,

$$v(t) = s'(t) = 2\cos(2t) - 3\sin(3t).$$

Proof of Chain Rule

At this point, we present a very informal proof of the chain rule. For simplicity's sake we ignore certain issues: For example, we assume that $g(x) \neq g(a)$ for $x \neq a$ in some open interval containing a. We begin by applying the limit definition of the derivative to the function $h(x)$ to obtain $h'(a)$:

$$h'(a) = \lim_{x \to a} \frac{f\big(g(x)\big) - f\big(g(a)\big)}{x - a}.$$

Rewriting, we obtain

$$h'(a) = \lim_{x \to a} \frac{f\big(g(x)\big) - f\big(g(a)\big)}{g(x) - g(a)} \cdot \frac{g(x) - g(a)}{x - a}.$$

Although it is clear that

$$\lim_{x \to a} \frac{g(x) - g(a)}{x - a} = g'(a),$$

it is not obvious that

$$\lim_{x \to a} \frac{f\big(g(x)\big) - f\big(g(a)\big)}{g(x) - g(a)} = f'\big(g(a)\big).$$

To see that this is true, first recall that since g is differentiable at a, g is also continuous at a. Thus,

$$\lim_{x \to a} g(x) = g(a).$$

Next, make the substitution $y = g(x)$and $b = g(a)$ and use change of variables in the limit to obtain

$$\lim_{x \to a} \frac{f\big(g(x)\big) - f\big(g(a)\big)}{g(x) - g(a)} = \lim_{y \to b} \frac{f(y) - f(b)}{y - b} = f'(b) = f'\big(g(a)\big).$$

Finally,

$$h'(a) = \lim_{x \to a} \frac{f\big(g(x)\big) - f\big(g(a)\big)}{g(x) - g(a)} \cdot \frac{g(x) - g(a)}{x - a} = f'\big(g(a)\big) \cdot g'(a).$$

□

Example 3.6.10: Using the Chain Rule with Functional Values

Let $h(x) = f\big(g(x)\big)$. If $g(1) = 4, g'(1) = 3$, and $f'(4) = 7$, find $h'(1)$.

Solution

Use the chain rule, then substitute.

$$\begin{aligned} h'(1) &= f'\big(g(1)\big)\cdot g'(1) && \text{Apply the chain rule.} \\ &= f'(4)\cdot 3 && \text{Substitute } g(1)=4 \text{ and } g'(1)=3. \\ &= 7\cdot 3 && \text{Substitute } f'(4)=7. \\ &= 21 && \text{Simplify.} \end{aligned}$$

? Exercise 3.6.6

Given $h(x)=f(g(x))$. If $g(2)=-3, g'(2)=4$, and $f'(-3)=7$, find $h'(2)$.

Hint

Follow Example 3.6.10.

Answer

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The Chain Rule Using Leibniz's Notation

As with other derivatives that we have seen, we can express the chain rule using Leibniz's notation. This notation for the chain rule is used heavily in physics applications.

For $h(x)=f(g(x))$, let $u=g(x)$ and $y=h(x)=f(u)$. Thus,

$$h'(x)=\frac{dy}{dx}$$

$$f'(g(x))=f'(u)=\frac{dy}{du}$$

and

$$g'(x)=\frac{du}{dx}.$$

Consequently,

$$\frac{dy}{dx}=h'(x)=f'\big(g(x)\big)\cdot g'(x)=\frac{dy}{du}\cdot\frac{du}{dx}.$$

Rule: Chain Rule Using Leibniz's Notation

If y is a function of u, and u is a function of x, then

$$\frac{dy}{dx}=\frac{dy}{du}\cdot\frac{du}{dx}.$$

Example 3.6.11: Taking a Derivative Using Leibniz's Notation I

Find the derivative of $y=\left(\frac{x}{3x+2}\right)^5$.

Solution

First, let $u=\frac{x}{3x+2}$. Thus, $y=u^5$. Next, find $\frac{du}{dx}$ and $\frac{dy}{du}$. Using the quotient rule,

$$\frac{du}{dx}=\frac{2}{(3x+2)^2}$$

and

$$\frac{dy}{du}=5u^4.$$

Finally, we put it all together.

$$\begin{aligned}
\frac{dy}{dx} &= \frac{dy}{du}\cdot\frac{du}{dx} && \text{Apply the chain rule.}\\
&= 5u^4\cdot\frac{2}{(3x+2)^2} && \text{Substitute } \frac{dy}{du}=5u^4 \text{ and } \frac{du}{dx}=\frac{2}{(3x+2)^2}.\\
&= 5\left(\frac{x}{3x+2}\right)^4\cdot\frac{2}{(3x+2)^2} && \text{Substitute } u=\frac{x}{3x+2}.\\
&= \frac{10x^4}{(3x+2)^6} && \text{Simplify.}
\end{aligned}$$

It is important to remember that, when using the Leibniz form of the chain rule, the final answer must be expressed entirely in terms of the original variable given in the problem.

Example 3.6.12: Taking a Derivative Using Leibniz's Notation II

Find the derivative of $y=\tan(4x^2-3x+1)$.

Solution

First, let $u=4x^2-3x+1$. Then $y=\tan u$. Next, find $\frac{du}{dx}$ and $\frac{dy}{du}$:

$$\frac{du}{dx}=8x-3 \text{ and } \frac{dy}{du}=\sec^2 u.$$

Finally, we put it all together.

$$\begin{aligned}
\frac{dy}{dx} &= \frac{dy}{du}\cdot\frac{du}{dx} && \text{Apply the chain rule.}\\
&= \sec^2 u\cdot(8x-3) && \text{Use } \frac{du}{dx}=8x-3 \text{ and } \frac{dy}{du}=\sec^2 u.\\
&= \sec^2(4x^2-3x+1)\cdot(8x-3) && \text{Substitute } u=4x^2-3x+1.
\end{aligned}$$

Exercise 3.6.7

Use Leibniz's notation to find the derivative of $y=\cos(x^3)$. Make sure that the final answer is expressed entirely in terms of the variable x.

Hint

Let $u=x^3$.

Answer

$\frac{dy}{dx}=-3x^2\sin(x^3)$.

Key Concepts

- The chain rule allows us to differentiate compositions of two or more functions. It states that for $h(x)=f\big(g(x)\big)$,

$$h'(x)=f'\big(g(x)\big)\cdot g'(x).$$

  In Leibniz's notation this rule takes the form

$$\frac{dy}{dx}=\frac{dy}{du}\cdot\frac{du}{dx}.$$

- We can use the chain rule with other rules that we have learned, and we can derive formulas for some of them.
- The chain rule combines with the power rule to form a new rule:

$$\text{If } h(x)=\big(g(x)\big)^n, \text{ then } h'(x)=n\big(g(x)\big)^{n-1}\cdot g'(x).$$

- When applied to the composition of three functions, the chain rule can be expressed as follows: If $h(x)=f\Big(g\big(k(x)\big)\Big)$, then $h'(x)=f'\Big(g\big(k(x)\big)\Big)\cdot g'\big(k(x)\big)\cdot k'(x)$.

Key Equations

- **The chain rule**

$$h'(x) = f'\big(g(x)\big) \cdot g'(x)$$

- **The power rule for functions**

$$h'(x) = n\big(g(x)\big)^{n-1} \cdot g'(x)$$

Glossary

chain rule

the chain rule defines the derivative of a composite function as the derivative of the outer function evaluated at the inner function times the derivative of the inner function

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3.6E: EXERCISES FOR SECTION 3.6

In exercises 1 - 6, given $y = f(u)$ and $u = g(x)$, find $\frac{dy}{dx}$ by using Leibniz's notation for the chain rule: $\frac{dy}{dx} = \frac{dy}{du}\frac{du}{dx}$.

1) $y = 3u - 6, \quad u = 2x^2$

2) $y = 6u^3, \quad u = 7x - 4$

Answer

$\frac{dy}{dx} = 18u^2 \cdot 7 = 18(7x-4)^2 \cdot 7 = 126(7x-4)^2$

3) $y = \sin u, \quad u = 5x - 1$

4) $y = \cos u, \quad u = -\frac{x}{8}$

Answer

$\frac{dy}{dx} = -\sin u \cdot \left(-\frac{1}{8}\right) = \frac{1}{8}\sin\left(-\frac{x}{8}\right)$

5) $y = \tan u, \quad u = 9x + 2$

6) $y = \sqrt{4u+3}, \quad u = x^2 - 6x$

Answer

$\frac{dy}{dx} = \frac{8x-24}{2\sqrt{4u+3}} = \frac{4x-12}{\sqrt{4x^2-24x+3}}$

For each of the following exercises,

a. decompose each function in the form $y = f(u)$ and $u = g(x)$, and

b. find $\frac{dy}{dx}$ as a function of x.

7) $y = (3x-2)^6$

8) $y = (3x^2+1)^3$

Answer

a. $f(u) = u^3, \quad u = 3x^2 + 1;$

b. $\frac{dy}{dx} = 18x(3x^2+1)^2$

9) $y = \sin^5(x)$

10) $y = \left(\frac{x}{7} + \frac{7}{x}\right)^7$

Answer

a. $f(u) = u^7, \quad u = \frac{x}{7} + \frac{7}{x};$

b. $\frac{dy}{dx} = 7\left(\frac{x}{7} + \frac{7}{x}\right)^6 \cdot \left(\frac{1}{7} - \frac{7}{x^2}\right)$

11) $y = \tan(\sec x)$

12) $y = \csc(\pi x + 1)$

Answer

a. $f(u) = \csc u, \quad u = \pi x + 1;$

b. $\frac{dy}{dx} = -\pi \csc(\pi x+1) \cdot \cot(\pi x+1)$

13) $y = \cot^2 x$

14) $y = -6\sin^{-3} x$

Answer

a. $f(u) = -6u^{-3}, \quad u = \sin x;$

b. $\dfrac{dy}{dx} = 18\sin^{-4} x \cdot \cos x$

In exercises 15 - 24, find $\dfrac{dy}{dx}$ for each function.

15) $y = (3x^2 + 3x - 1)^4$

16) $y = (5 - 2x)^{-2}$

Answer

$$\frac{dy}{dx} = \frac{4}{(5-2x)^3}$$

17) $y = \cos^3(\pi x)$

18) $y = (2x^3 - x^2 + 6x + 1)^3$

Answer

$$\frac{dy}{dx} = 6(2x^3 - x^2 + 6x + 1)^2 \cdot (3x^2 - x + 3)$$

19) $y = \dfrac{1}{\sin^2(x)}$

20) $y = \Big(\tan x + \sin x\Big)^{-3}$

Answer

$$\frac{dy}{dx} = -3\Big(\tan x + \sin x\Big)^{-4} \cdot (\sec^2 x + \cos x)$$

21) $y = x^2 \cos^4 x$

22) $y = \sin(\cos 7x)$

Answer

$$\frac{dy}{dx} = -7\cos(\cos 7x) \cdot \sin 7x$$

23) $y = \sqrt{6 + \sec \pi x^2}$

24) $y = \cot^3(4x + 1)$

Answer

$$\frac{dy}{dx} = -12\cot^2(4x+1) \cdot \csc^2(4x+1)$$

25) Let $y = \big[f(x)\big]^3$ and suppose that $f'(1) = 4$ and $\frac{dy}{dx} = 10$ for $x = 1$. Find $f(1)$.

26) Let $y = \Big(f(x) + 5x^2\Big)^4$ and suppose that $f(-1) = -4$ and $\frac{dy}{dx} = 3$ when $x = -1$. Find $f'(-1)$

Answer

$f'(-1) = 10\frac{3}{4}$

27) Let $y = (f(u) + 3x)^2$ and $u = x^3 - 2x$. If $f(4) = 6$ and $\frac{dy}{dx} = 18$ when $x = 2$, find $f'(4)$.

28) [T] Find the equation of the tangent line to $y = -\sin(\frac{x}{2})$ at the origin. Use a calculator to graph the function and the tangent line together.

Answer

$y = -\frac{1}{2}x$

29) [T] Find the equation of the tangent line to $y = \Big(3x + \frac{1}{x}\Big)^2$ at the point $(1, 16)$. Use a calculator to graph the function and the tangent line together.

30) Find the x -coordinates at which the tangent line to $y = \Big(x - \frac{6}{x}\Big)^8$ is horizontal.

Answer

$x = \pm\sqrt{6}$

31) [T] Find an equation of the line that is normal to $g(\theta) = \sin^2(\pi\theta)$ at the point $\left(\frac{1}{4}, \frac{1}{2}\right)$. Use a calculator to graph the function and the normal line together.

For exercises 32 - 39, use the information in the following table to find $h'(a)$ at the given value for a.

x	$f(x)$	$f'(x)$	$g(x)$	$g'(x)$
0	2	5	0	2
1	1	−2	3	0
2	4	4	1	−1
3	3	−3	2	3

32) $h(x) = f\big(g(x)\big); \quad a = 0$

Answer

$h'(0) = 10$

33) $h(x) = g\big(f(x)\big); \quad a = 0$

34) $h(x) = \big(x^4 + g(x)\big)^{-2}; \quad a = 1$

Answer

$h'(1) = -\frac{1}{8}$

35) $h(x) = \left(\dfrac{f(x)}{g(x)}\right)^2; \quad a = 3$

36) $h(x) = f\big(x + f(x)\big); \quad a = 1$

Answer

$h'(1) = -4$

37) $h(x) = \big(1 + g(x)\big)^3; \quad a = 2$

38) $h(x) = g\big(2 + f(x^2)\big); \quad a = 1$

Answer

$h'(1) = -12$

39) $h(x) = f\big(g(\sin x)\big); \quad a = 0$

40) [T] The position function of a freight train is given by $s(t) = 100(t+1)^{-2}$, with s in meters and t in seconds. At time $t = 6$ s, find the train's

a. velocity and

b. acceleration.

c. Considering your results in parts a. and b., is the train speeding up or slowing down?

Answer

a. $v(6) = -\frac{200}{343}$ m/s,

b. $a(6) = \frac{600}{2401}\ \text{m/s}^2$,

c. The train is slowing down since velocity and acceleration have opposite signs.

41) [T] A mass hanging from a vertical spring is in simple harmonic motion as given by the following position function, where t is measured in seconds and s is in inches:

$$s(t) = -3\cos\left(\pi t + \frac{\pi}{4}\right).$$

a. Determine the position of the spring at $t = 1.5$ s.

b. Find the velocity of the spring at $t = 1.5$ s.

42) [T] The total cost to produce x boxes of Thin Mint Girl Scout cookies is C dollars, where $C = 0.0001x^3 - 0.02x^2 + 3x + 300$. In t weeks production is estimated to be $x = 1600 + 100t$ boxes.

a. Find the marginal cost $C'(x)$.

b. Use Leibniz's notation for the chain rule, $\frac{dC}{dt} = \frac{dC}{dx} \cdot \frac{dx}{dt}$, to find the rate with respect to time t that the cost is changing.

c. Use your result in part b. to determine how fast costs are increasing when $t = 2$ weeks. Include units with the answer.

Answer

a. $C'(x) = 0.0003x^2 - 0.04x + 3$

b. $\frac{dC}{dt} = 100 \cdot (0.0003x^2 - 0.04x + 3) = 100 \cdot (0.0003(1600 + 100t)^2 - 0.04(1600 + 100t) + 3) = 300t^2 + 9200t + 70700$

c. Approximately $90,300 per week

43) [T] The formula for the area of a circle is $A = \pi r^2$, where r is the radius of the circle. Suppose a circle is expanding, meaning that both the area A and the radius r (in inches) are expanding.

a. Suppose $r = 2 - \frac{100}{(t+7)^2}$ where t is time in seconds. Use the chain rule $\frac{dA}{dt} = \frac{dA}{dr} \cdot \frac{dr}{dt}$ to find the rate at which the area is expanding.

b. Use your result in part a. to find the rate at which the area is expanding at $t = 4$ s.

44) [T] The formula for the volume of a sphere is $S = \frac{4}{3}\pi r^3$, where r (in feet) is the radius of the sphere. Suppose a spherical snowball is melting in the sun.

a. Suppose $r = \frac{1}{(t+1)^2} - \frac{1}{12}$ where t is time in minutes. Use the chain rule $\frac{dS}{dt} = \frac{dS}{dr} \cdot \frac{dr}{dt}$ to find the rate at which the snowball is melting.

b. Use your result in part a. to find the rate at which the volume is changing at $t = 1$ min.

Answer

a. $\frac{dS}{dt} = -\frac{8\pi r^2}{(t+1)^3} = -\frac{8\pi\left(\frac{1}{(t+1)^2} - \frac{1}{12}\right)^2}{(t+1)^3}$

b. The volume is decreasing at a rate of $-\frac{\pi}{36}$ ft^3/min

45) [T] The daily temperature in degrees Fahrenheit of Phoenix in the summer can be modeled by the function $T(x) = 94 - 10\cos\left[\frac{\pi}{12}(x-2)\right]$, where x is hours after midnight. Find the rate at which the temperature is changing at 4 p.m.

46) [T] The depth (in feet) of water at a dock changes with the rise and fall of tides. The depth is modeled by the function $D(t) = 5\sin\left(\frac{\pi}{6}t - \frac{7\pi}{6}\right) + 8$, where t is the number of hours after midnight. Find the rate at which the depth is changing at 6 a.m.

Answer

2.3 ft/hr

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3.7: Derivatives of Inverse Functions

Learning Objectives

- Calculate the derivative of an inverse function.
- Recognize the derivatives of the standard inverse trigonometric functions.

In this section we explore the relationship between the derivative of a function and the derivative of its inverse. For functions whose derivatives we already know, we can use this relationship to find derivatives of inverses without having to use the limit definition of the derivative. In particular, we will apply the formula for derivatives of inverse functions to trigonometric functions. This formula may also be used to extend the power rule to rational exponents.

The Derivative of an Inverse Function

We begin by considering a function and its inverse. If $f(x)$ is both invertible and differentiable, it seems reasonable that the inverse of $f(x)$ is also differentiable. Figure 3.7.1 shows the relationship between a function $f(x)$ and its inverse $f^{-1}(x)$. Look at the point $(a,\ f^{-1}(a))$ on the graph of $f^{-1}(x)$ having a tangent line with a slope of

$$\left(f^{-1}\right)'(a) = \frac{p}{q}.$$

This point corresponds to a point $(f^{-1}(a),\ a)$ on the graph of $f(x)$ having a tangent line with a slope of

$$f'\left(f^{-1}(a)\right) = \frac{q}{p}.$$

Thus, if $f^{-1}(x)$ is differentiable at a, then it must be the case that

$$\left(f^{-1}\right)'(a) = \frac{1}{f'\left(f^{-1}(a)\right)}.$$

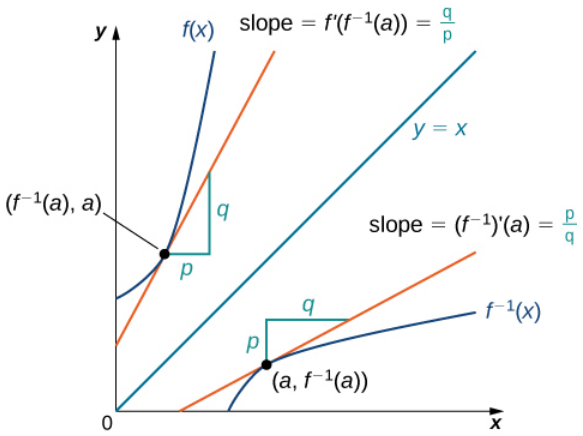


Figure 3.7.1:The tangent lines of a function and its inverse are related; so, too, are the derivatives of these functions.

We may also derive the formula for the derivative of the inverse by first recalling that $x = f\left(f^{-1}(x)\right)$. Then by differentiating both sides of this equation (using the chain rule on the right), we obtain

$$1 = f'\left(f^{-1}(x)\right)\left(f^{-1}\right)'(x)).$$

Solving for $\left(f^{-1}\right)'(x)$, we obtain

$$\left(f^{-1}\right)'(x) = \frac{1}{f'\left(f^{-1}(x)\right)}.$$

We summarize this result in the following theorem.

Inverse Function Theorem

Let $f(x)$ be a function that is both invertible and differentiable. Let $y = f^{-1}(x)$ be the inverse of $f(x)$. For all x satisfying $f'\left(f^{-1}(x)\right) \neq 0$,

$$\frac{dy}{dx} = \frac{d}{dx}\left(f^{-1}(x)\right) = \left(f^{-1}\right)'(x) = \frac{1}{f'\left(f^{-1}(x)\right)}. \tag{3.7.1}$$

Alternatively, if $y = g(x)$ is the inverse of $f(x)$, then

$$g'(x) = \frac{1}{f'\left(g(x)\right)}. \tag{3.7.2}$$

Example 3.7.1: Applying the Inverse Function Theorem

Use the inverse function theorem to find the derivative of $g(x) = \dfrac{x+2}{x}$. Compare the resulting derivative to that obtained by differentiating the function directly.

Solution

The inverse of $g(x) = \dfrac{x+2}{x}$ is $f(x) = \dfrac{2}{x-1}$.

We will use Equation 3.7.2 and begin by finding $f'(x)$. Thus,

$$f'(x) = \frac{-2}{(x-1)^2}$$

and

$$f'\left(g(x)\right) = \frac{-2}{(g(x)-1)^2} = \frac{-2}{\left(\dfrac{x+2}{x} - 1\right)^2} = -\frac{x^2}{2}.$$

Finally,

$$g'(x) = \frac{1}{f'\left(g(x)\right)} = -\frac{2}{x^2}.$$

We can verify that this is the correct derivative by applying the quotient rule to $g(x)$ to obtain

$$g'(x) = -\frac{2}{x^2}.$$

Exercise 3.7.1

Use the inverse function theorem to find the derivative of $g(x) = \dfrac{1}{x+2}$. Compare the result obtained by differentiating $g(x)$ directly.

Hint

Use the preceding example as a guide.

Answer

$$g'(x) = -\frac{1}{(x+2)^2}$$

Example 3.7.2: Applying the Inverse Function Theorem

Use the inverse function theorem to find the derivative of $g(x) = \sqrt[3]{x}$.

Solution

The function $g(x) = \sqrt[3]{x}$ is the inverse of the function $f(x) = x^3$. Since $g'(x) = \dfrac{1}{f'\big(g(x)\big)}$, begin by finding $f'(x)$. Thus,

$$f'(x) = 3x^2$$

and

$$f'\big(g(x)\big) = 3\big(\sqrt[3]{x}\big)^2 = 3x^{2/3}$$

Finally,

$$g'(x) = \frac{1}{3x^{2/3}}.$$

If we were to differentiate $g(x)$ directly, using the power rule, we would first rewrite $g(x) = \sqrt[3]{x}$ as a power of x to get,

$$g(x) = x^{1/3}$$

Then we would differentiate using the power rule to obtain

$$g'(x) = \tfrac{1}{3}x^{-2/3} = \frac{1}{3x^{2/3}}.$$

Exercise 3.7.2

Find the derivative of $g(x) = \sqrt[5]{x}$ by applying the inverse function theorem.

Hint

$g(x)$ is the inverse of $f(x) = x^5$.

Answer

$g(x) = \tfrac{1}{5}x^{-4/5}$

From the previous example, we see that we can use the inverse function theorem to extend the power rule to exponents of the form $\dfrac{1}{n}$, where n is a positive integer. This extension will ultimately allow us to differentiate x^q, where q is any rational number.

Extending the Power Rule to Rational Exponents

The power rule may be extended to rational exponents. That is, if n is a positive integer, then

$$\frac{d}{dx}\left(x^{1/n}\right) = \frac{1}{n}x^{(1/n)-1}.$$

Also, if n is a positive integer and m is an arbitrary integer, then

$$\frac{d}{dx}\left(x^{m/n}\right) = \frac{m}{n}x^{(m/n)-1}.$$

Proof

The function $g(x) = x^{1/n}$ is the inverse of the function $f(x) = x^n$. Since $g'(x) = \dfrac{1}{f'\big(g(x)\big)}$, begin by finding $f'(x)$. Thus,

$$f'(x) = nx^{n-1} \text{ and } f'\big(g(x)\big) = n\left(x^{1/n}\right)^{n-1} = nx^{(n-1)/n}.$$

Finally,

$$g'(x) = \frac{1}{nx^{(n-1)/n}} = \frac{1}{n}x^{(1-n)/n} = \frac{1}{n}x^{(1/n)-1}.$$

To differentiate $x^{m/n}$ we must rewrite it as $(x^{1/n})^m$ and apply the chain rule. Thus,

$$\frac{d}{dx}\left(x^{m/n}\right) = \frac{d}{dx}\left(\left(x^{1/n}\right)^m\right) = m\left(x^{1/n}\right)^{m-1} \cdot \frac{1}{n}x^{(1/n)-1} = \frac{m}{n}x^{(m/n)-1}.$$

□

Example 3.7.3: Applying the Power Rule to a Rational Power

Find the equation of the line tangent to the graph of $y = x^{2/3}$ at $x = 8$.

Solution

First find $\frac{dy}{dx}$ and evaluate it at $x = 8$. Since

$$\frac{dy}{dx} = \frac{2}{3}x^{-1/3}$$

and

$$\left.\frac{dy}{dx}\right|_{x=8} = \frac{1}{3}$$

the slope of the tangent line to the graph at $x = 8$ is $\frac{1}{3}$.

Substituting $x = 8$ into the original function, we obtain $y = 4$. Thus, the tangent line passes through the point $(8, 4)$. Substituting into the point-slope formula for a line, we obtain the tangent line

$$y = \tfrac{1}{3}x + \tfrac{4}{3}.$$

Exercise 3.7.3

Find the derivative of $s(t) = \sqrt{2t+1}$.

Hint

Use the chain rule.

Answer

$s'(t) = (2t+1)^{-1/2}$

Derivatives of Inverse Trigonometric Functions

We now turn our attention to finding derivatives of inverse trigonometric functions. These derivatives will prove invaluable in the study of integration later in this text. The derivatives of inverse trigonometric functions are quite surprising in that their derivatives are actually algebraic functions. Previously, derivatives of algebraic functions have proven to be algebraic functions and derivatives of trigonometric functions have been shown to be trigonometric functions. Here, for the first time, we see that the derivative of a function need not be of the same type as the original function.

Example 3.7.4A: Derivative of the Inverse Sine Function

Use the inverse function theorem to find the derivative of $g(x) = \sin^{-1} x$.

Solution

Since for x in the interval $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$, $f(x) = \sin x$ is the inverse of $g(x) = \sin^{-1} x$, begin by finding $f'(x)$. Since

$$f'(x) = \cos x$$

and

$$f'\big(g(x)\big) = \cos\left(\sin^{-1} x\right) = \sqrt{1-x^2}$$

we see that

$$g'(x) = \frac{d}{dx}\left(\sin^{-1} x\right) = \frac{1}{f'\big(g(x)\big)} = \frac{1}{\sqrt{1-x^2}}$$

Analysis

To see that $\cos(\sin^{-1} x) = \sqrt{1-x^2}$, consider the following argument. Set $\sin^{-1} x = \theta$. In this case, $\sin\theta = x$ where $-\frac{\pi}{2} \le \theta \le \frac{\pi}{2}$. We begin by considering the case where $0 < \theta < \frac{\pi}{2}$. Since θ is an acute angle, we may construct a right triangle having acute angle θ, a hypotenuse of length 1 and the side opposite angle θ having length x. From the Pythagorean theorem, the side adjacent to angle θ has length $\sqrt{1-x^2}$. This triangle is shown in Figure 3.7.2 Using the triangle, we see that $\cos(\sin^{-1} x) = \cos\theta = \sqrt{1-x^2}$.

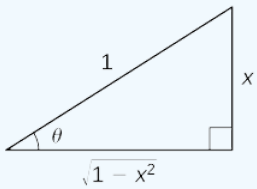


Figure 3.7.2: Using a right triangle having acute angle θ, a hypotenuse of length 1, and the side opposite angle θ having length x, we can see that $\cos(\sin^{-1} x) = \cos\theta = \sqrt{1-x^2}$.

In the case where $-\frac{\pi}{2} < \theta < 0$, we make the observation that $0 < -\theta < \frac{\pi}{2}$ and hence

$$\cos\left(\sin^{-1} x\right) = \cos\theta = \cos(-\theta) = \sqrt{1-x^2}.$$

Now if $\theta = \frac{\pi}{2}$ or $\theta = -\frac{\pi}{2}, x = 1$ or $x = -1$, and since in either case $\cos\theta = 0$ and $\sqrt{1-x^2} = 0$, we have

$$\cos\left(\sin^{-1} x\right) = \cos\theta = \sqrt{1-x^2}.$$

Consequently, in all cases,

$$\cos\left(\sin^{-1} x\right) = \sqrt{1-x^2}.$$

✔ Example 3.7.4B: Applying the Chain Rule to the Inverse Sine Function

Apply the chain rule to the formula derived in Example 3.7.4A to find the derivative of $h(x) = \sin^{-1}\big(g(x)\big)$ and use this result to find the derivative of $h(x) = \sin^{-1}(2x^3)$.

Solution

Applying the chain rule to $h(x) = \sin^{-1}\big(g(x)\big)$, we have

$$h'(x) = \frac{1}{\sqrt{1-\big(g(x)\big)^2}} g'(x).$$

Now let $g(x) = 2x^3$, so $g'(x) = 6x^2$. Substituting into the previous result, we obtain

$$\begin{aligned} h'(x) &= \frac{1}{\sqrt{1-4x^6}} \cdot 6x^2 \\ &= \frac{6x^2}{\sqrt{1-4x^6}} \end{aligned}$$

? Exercise 3.7.4

Use the inverse function theorem to find the derivative of $g(x) = \tan^{-1} x$.

Hint

The inverse of $g(x)$ is $f(x) = \tan x$. Use Example 3.7.4A as a guide.

Answer

$$g'(x) = \frac{1}{1+x^2}$$

The derivatives of the remaining inverse trigonometric functions may also be found by using the inverse function theorem. These formulas are provided in the following theorem.

Derivatives of Inverse Trigonometric Functions

$$\frac{d}{dx}\left(\sin^{-1} x\right) = \frac{1}{\sqrt{1-x^2}} \tag{3.7.3}$$

$$\frac{d}{dx}\left(\cos^{-1} x\right) = \frac{-1}{\sqrt{1-x^2}} \tag{3.7.4}$$

$$\frac{d}{dx}\left(\tan^{-1} x\right) = \frac{1}{1+x^2} \tag{3.7.5}$$

$$\frac{d}{dx}\left(\cot^{-1} x\right) = \frac{-1}{1+x^2} \tag{3.7.6}$$

$$\frac{d}{dx}\left(\sec^{-1} x\right) = \frac{1}{|x|\sqrt{x^2-1}} \tag{3.7.7}$$

$$\frac{d}{dx}\left(\csc^{-1} x\right) = \frac{-1}{|x|\sqrt{x^2-1}} \tag{3.7.8}$$

Example 3.7.5A: Applying Differentiation Formulas to an Inverse Tangent Function

Find the derivative of $f(x) = \tan^{-1}(x^2)$.

Solution

Let $g(x) = x^2$, so $g'(x) = 2x$. Substituting into Equation 3.7.5, we obtain

$$f'(x) = \frac{1}{1+(x^2)^2} \cdot (2x).$$

Simplifying, we have

$$f'(x) = \frac{2x}{1+x^4}.$$

Example 3.7.5B: Applying Differentiation Formulas to an Inverse Sine Function

Find the derivative of $h(x) = x^2 \sin^{-1} x$.

Solution

By applying the product rule, we have

$$h'(x) = 2x \sin^{-1} x + \frac{1}{\sqrt{1-x^2}} \cdot x^2$$

Exercise 3.7.5

Find the derivative of $h(x) = \cos^{-1}(3x-1)$.

Hint

Use Equation 3.7.4. with $g(x) = 3x - 1$

Answer

$$h'(x) = \frac{-3}{\sqrt{6x - 9x^2}}$$

Example 3.7.6: Applying the Inverse Tangent Function

The position of a particle at time t is given by $s(t) = \tan^{-1}\left(\frac{1}{t}\right)$ for $t \geq \frac{1}{2}$. Find the velocity of the particle at time $t = 1$.

Solution

Begin by differentiating $s(t)$ in order to find $v(t)$.Thus,

$$v(t) = s'(t) = \frac{1}{1 + \left(\frac{1}{t}\right)^2} \cdot \frac{-1}{t^2}.$$

Simplifying, we have

$$v(t) = -\frac{1}{t^2 + 1}.$$

Thus, $v(1) = -\frac{1}{2}$.

Exercise 3.7.6

Find the equation of the line tangent to the graph of $f(x) = \sin^{-1} x$ at $x = 0$.

Hint

$f'(0)$ is the slope of the tangent line.

Answer

$y = x$

Key Concepts

- The inverse function theorem allows us to compute derivatives of inverse functions without using the limit definition of the derivative.
- We can use the inverse function theorem to develop differentiation formulas for the inverse trigonometric functions.

Key Equations

- **Inverse function theorem**

  $(f^{-1})'(x) = \frac{1}{f'\left(f^{-1}(x)\right)}$ whenever $f'\left(f^{-1}(x)\right) \neq 0$ and $f(x)$ is differentiable.

- **Power rule with rational exponents**

  $\frac{d}{dx}\left(x^{m/n}\right) = \frac{m}{n}x^{(m/n)-1}$.

- **Derivative of inverse sine function**

  $\frac{d}{dx}\left(\sin^{-1} x\right) = \frac{1}{\sqrt{1 - x^2}}$

- **Derivative of inverse cosine function**

  $\frac{d}{dx}\left(\cos^{-1} x\right) = \frac{-1}{\sqrt{1 - x^2}}$

Derivative of inverse tangent function

$\frac{d}{dx}\left(\tan^{-1} x\right) = \frac{1}{1 + x^2}$

Derivative of inverse cotangent function

$$\frac{d}{dx}\left(\cot^{-1}x\right)=\frac{-1}{1+x^2}$$

Derivative of inverse secant function

$$\frac{d}{dx}\left(\sec^{-1}x\right)=\frac{1}{|x|\sqrt{x^2-1}}$$

Derivative of inverse cosecant function

$$\frac{d}{dx}\left(\csc^{-1}x\right)=\frac{-1}{|x|\sqrt{x^2-1}}$$

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- Paul Seeburger (Monroe Community College) added the second half of Example 3.7.2.

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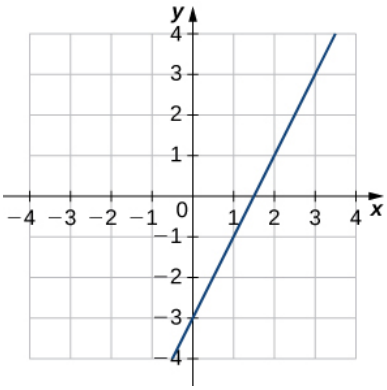
3.7E: EXERCISES FOR SECTION 3.7

In exercises 1 - 4, use the graph of $y = f(x)$ to

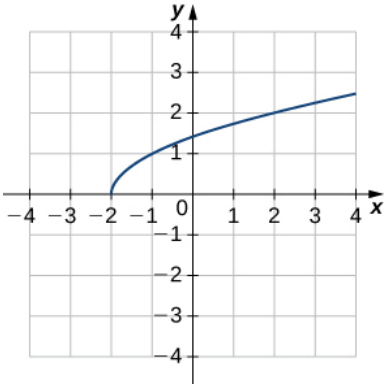
a. sketch the graph of $y = f^{-1}(x)$, and

b. use part a. to estimate $\left(f^{-1}\right)'(1)$.

1)

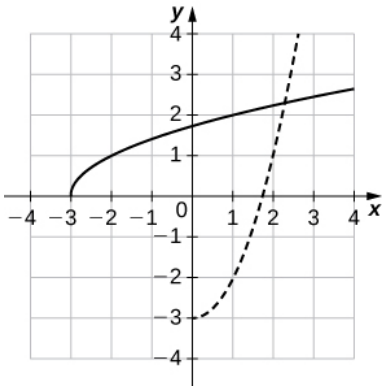


2)



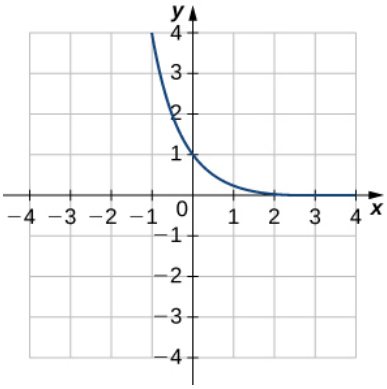
Answer

a.

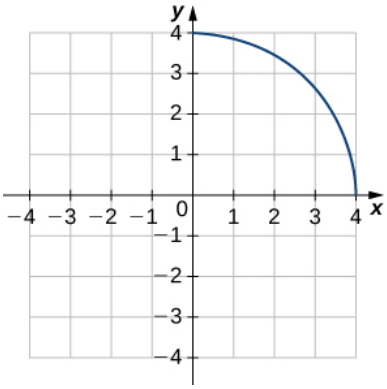


b. $(f^{-1})'(1) \approx 2$

3)

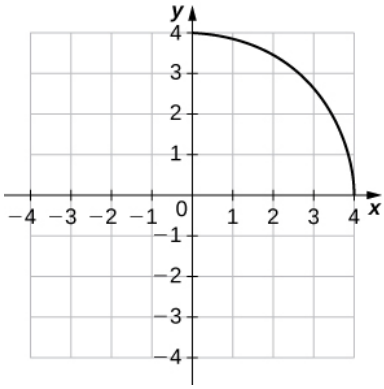


4)



Answer

a.



b. $(f^{-1})'(1) \approx -1/\sqrt{3}$

For exercises 5 - 8, use the given function $y = f(x)$ to find

a. $\frac{df}{dx}$ at $x = a$ and

b. $x = f^{-1}(y)$.

c. Then use part b. to find $\frac{df^{-1}}{dy}$ at $y = f(a)$.

5) $f(x) = 6x - 1,\ x = -2$

6) $f(x) = 2x^3 - 3,\ x = 1$

Answer

a. $\frac{df}{dx} = 6$
b. $x = f^{-1}(y) = \left(\frac{y+3}{2}\right)^{1/3}$
c. $\frac{df^{-1}}{dy} = \frac{1}{6}$

7) $f(x) = 9 - x^2,\ 0 \le x \le 3, x = 2$

8) $f(x) = \sin x,\ x = 0$

Answer

a. $\frac{df}{dx} = 1$
b. $x = f^{-1}(y) = \sin^{-1} y$
c. $\frac{df^{-1}}{dy} = 1$

For each function in exercises 9 - 14, find $\left(f^{-1}\right)'(a)$.

9) $f(x) = x^2 + 3x + 2,\ x \ge -1,\ a = 2$

10 $f(x) = x^3 + 2x + 3,\ a = 0$

Answer

$\left(f^{-1}\right)'(1) = \frac{1}{5}$

11) $f(x) = x + \sqrt{x},\ a = 2$

12) $f(x) = x - \frac{2}{x},\ x < 0,\ a = 1$

Answer

$\left(f^{-1}\right)'(1) = \frac{1}{3}$

13) $f(x) = x + \sin x,\ a = 0$

14) $f(x) = \tan x + 3x^2,\ a = 0$

Answer

$\left(f^{-1}\right)'(0) = 1$

For each function $y = f(x)$, given in exercises 15 - 19,

a. find the slope of the tangent line to its inverse function f^{-1} at the indicated point P, and

b. find the equation of the tangent line to the graph of f^{-1} at the indicated point.

15) $f(x) = \frac{4}{1+x^2},\quad P(2,1)$

16) $f(x) = \sqrt{x-4},\quad P(2,8)$

Answer

a. 4
b. $y = 4x$

17) $f(x) = (x^3+1)^4,\quad P(16,1)$

18) $f(x) = -x^3 - x + 2,\quad P(-8,2)$

Answer

a. $-\frac{1}{96}$
b. $y = -\frac{1}{13}x + \frac{18}{13}$

19) $f(x) = x^5 + 3x^3 - 4x - 8,\quad P(-8,1)$

In exercises 20 - 29, find $\frac{dy}{dx}$ for the given function.

20) $y = \sin^{-1}(x^2)$

Answer

$$\frac{dy}{dx} = \frac{2x}{\sqrt{1-x^4}}$$

21) $y = \cos^{-1}(\sqrt{x})$

22) $y = \sec^{-1}\left(\frac{1}{x}\right)$

Answer

$$\frac{dy}{dx} = \frac{-1}{\sqrt{1-x^2}}$$

23) $y = \sqrt{\csc^{-1} x}$

24) $y = (1 + \tan^{-1} x)^3$

Answer

$$\frac{dy}{dx} = \frac{3(1+\tan^{-1} x)^2}{1+x^2}$$

25) $y = \cos^{-1}(2x) \cdot \sin^{-1}(2x)$

26) $y = \dfrac{1}{\tan^{-1}(x)}$

Answer

$$\frac{dy}{dx} = \frac{-1}{(1+x^2)(\tan^{-1} x)^2}$$

27) $y = \sec^{-1}(-x)$

28) $y = \cot^{-1}\sqrt{4-x^2}$

Answer

$$\frac{dy}{dx} = \frac{x}{(5-x^2)\sqrt{4-x^2}}$$

29) $y = x \cdot \csc^{-1} x$

In exercises 30 - 35, use the given values to find $\left(f^{-1}\right)'(a)$.

30) $f(\pi) = 0, f'(\pi) = -1, a = 0$

Answer

$$\left(f^{-1}\right)'(0) = -1$$

31) $f(6) = 2,\ f'(6) = \frac{1}{3},\ a = 2$

32) $f\left(\frac{1}{3}\right) = -8,\ f'\left(\frac{1}{3}\right) = 2,\ a = -8$

Answer

$$\left(f^{-1}\right)'(-8) = \tfrac{1}{2}$$

33) $f(\sqrt{3}) = \frac{1}{2}, f'(\sqrt{3}) = \frac{2}{3}, a = \frac{1}{2}$

34) $f(1) = -3,\ f'(1) = 10,\ a = -3$

Answer

$$\left(f^{-1}\right)'(-3) = \tfrac{1}{10}$$

35) $f(1) = 0,\ f'(1) = -2,\ a = 0$

36) [T] The position of a moving hockey puck after t seconds is $s(t) = tan^{-1}t$ where s is in meters.

a. Find the velocity of the hockey puck at any time t.

b. Find the acceleration of the puck at any time t.

c. Evaluate parts a. and b. for $t = 2, 4$,and 6 seconds.

d. What conclusion can be drawn from the results in c.?

Answer

a. $v(t) = \dfrac{1}{1+t^2}$

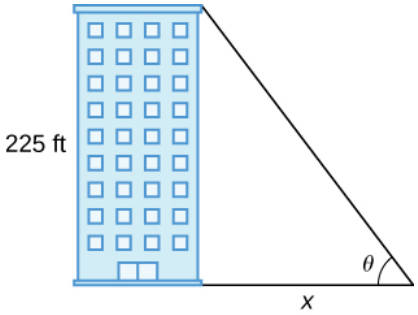
b. $a(t) = \dfrac{-2t}{(1+t^2)^2}$

c. (a) $0.2,\ 0.06,\ 0.03$; (b) $-0.16,\ -0.028,\ -0.0088$

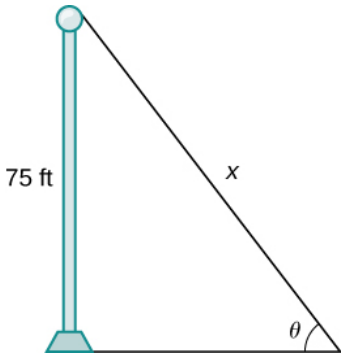
d. The hockey puck is decelerating/slowing down at 2, 4, and 6 seconds.

Solution:

37) [T] A building that is 225 feet tall casts a shadow of various lengths x as the day goes by. An angle of elevation θ is formed by lines from the top and bottom of the building to the tip of the shadow, as seen in the following figure. Find the rate of change of the angle of elevation $\frac{d\theta}{dx}$ when $x = 272$ feet.



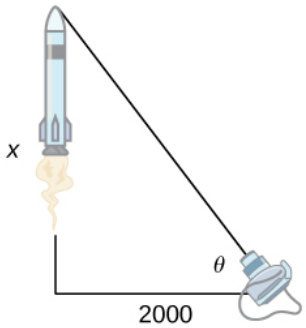
38) [T] A pole stands 75 feet tall. An angle θ is formed when wires of various lengths of x feet are attached from the ground to the top of the pole, as shown in the following figure. Find the rate of change of the angle $\frac{d\theta}{dx}$ when a wire of length 90 feet is attached.



Answer

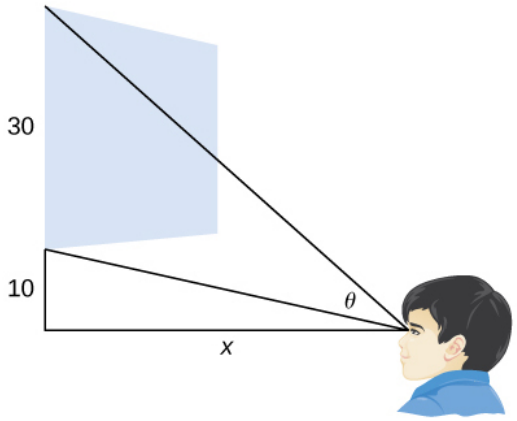
-0.0168 radians per foot

39) [T] A television camera at ground level is 2000 feet away from the launching pad of a space rocket that is set to take off vertically, as seen in the following figure. The angle of elevation of the camera can be found by $\theta = \tan^{-1}\left(\frac{x}{2000}\right)$, where x is the height of the rocket. Find the rate of change of the angle of elevation after launch when the camera and the rocket are 5000 feet apart.



40) [T] A local movie theater with a 30-foot-high screen that is 10 feet above a person's eye level when seated has a viewing angle θ (in radians) given by $\theta = \cot^{-1}\frac{x}{40} - \cot^{-1}\frac{x}{10}$,

where x is the distance in feet away from the movie screen that the person is sitting, as shown in the following figure.



a. Find $\frac{d\theta}{dx}$.

b. Evaluate $\frac{d\theta}{dx}$ for $x = 5, 10, 15,$ and 20.

c. Interpret the results in part b.

d. Evaluate $\frac{d\theta}{dx}$ for $x = 25, 30, 35$, and 40.

e. Interpret the results in part d. At what distance x should the person stand to maximize his or her viewing angle?

Answer

a. $\frac{d\theta}{dx} = \frac{10}{100 + x^2} - \frac{40}{1600 + x^2}$

b. $\frac{18}{325}, \frac{9}{340}, \frac{42}{4745}, 0$

c. As a person moves farther away from the screen, the viewing angle is increasing, which implies that as he or she moves farther away, his or her screen vision is widening. d. $-\frac{54}{12905}, -\frac{3}{500}, -\frac{198}{29945}, -\frac{9}{1360}$

e. As the person moves beyond 20 feet from the screen, the viewing angle is decreasing. The optimal distance the person should stand for maximizing the viewing angle is 20 feet.

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3.8: Implicit Differentiation

Learning Objectives

- Find the derivative of a complicated function by using implicit differentiation.
- Use implicit differentiation to determine the equation of a tangent line.

We have already studied how to find equations of tangent lines to functions and the rate of change of a function at a specific point. In all these cases we had the explicit equation for the function and differentiated these functions explicitly. Suppose instead that we want to determine the equation of a tangent line to an arbitrary curve or the rate of change of an arbitrary curve at a point. In this section, we solve these problems by finding the derivatives of functions that define y implicitly in terms of x.

Implicit Differentiation

In most discussions of math, if the dependent variable y is a function of the independent variable x, we express y in terms of x. If this is the case, we say that y is an explicit function of x. For example, when we write the equation $y = x^2 + 1$, we are defining y explicitly in terms of x. On the other hand, if the relationship between the function y and the variable x is expressed by an equation where y is not expressed entirely in terms of x, we say that the equation defines y implicitly in terms of x. For example, the equation $y - x^2 = 1$ defines the function $y = x^2 + 1$ implicitly.

Implicit differentiation allows us to find slopes of tangents to curves that are clearly not functions (they fail the vertical line test). We are using the idea that portions of y are functions that satisfy the given equation, but that y is not actually a function of x.

In general, an equation defines a function implicitly if the function satisfies that equation. An equation may define many different functions implicitly. For example, the functions

$$y = \sqrt{25 - x^2}$$

and

$$y = \begin{cases} \sqrt{25 - x^2}, & \text{if } -5 \leq x < 0 \\ -\sqrt{25 - x^2}, & \text{if } 0 \leq x \leq 5 \end{cases}$$

which are illustrated in Figure 3.8.1, are just two of the many functions defined implicitly by the equation $x^2 + y^2 = 25$.

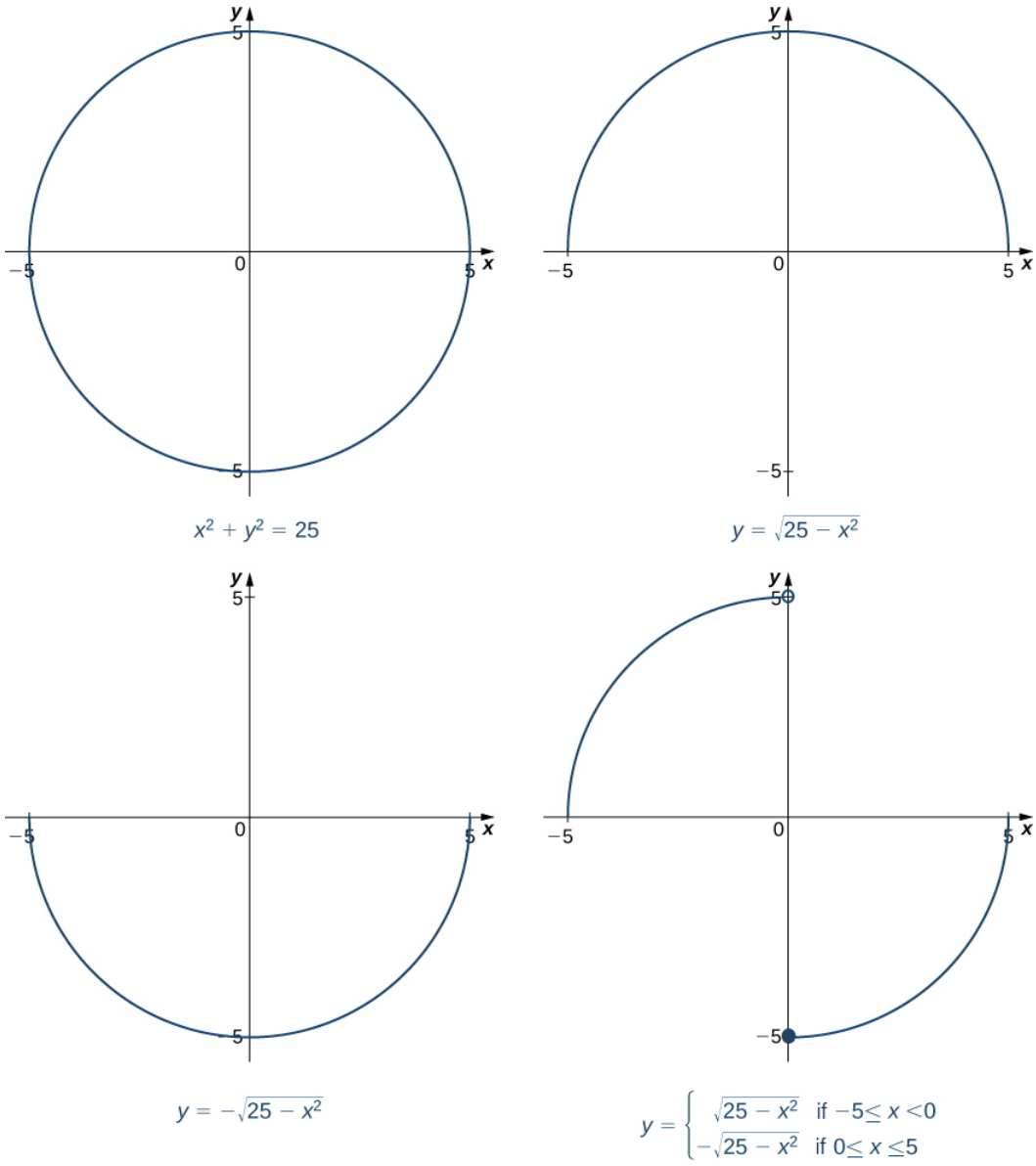


Figure 3.8.1:The equation $x^2 + y^2 = 25$ defines many functions implicitly.

If we want to find the slope of the line tangent to the graph of $x^2 + y^2 = 25$ at the point $(3, 4)$, we could evaluate the derivative of the function $y = \sqrt{25 - x^2}$ at $x = 3$. On the other hand, if we want the slope of the tangent line at the point $(3, -4)$, we could use the derivative of $y = -\sqrt{25 - x^2}$. However, it is not always easy to solve for a function defined implicitly by an equation. Fortunately, the technique of **implicit differentiation** allows us to find the derivative of an implicitly defined function without ever solving for the function explicitly. The process of finding $\frac{dy}{dx}$ using implicit differentiation is described in the following problem-solving strategy.

Problem-Solving Strategy: Implicit Differentiation

To perform *implicit differentiation* on an equation that defines a function y implicitly in terms of a variable x, use the following steps:

1. Take the derivative of both sides of the equation. Keep in mind that y is a function of x. Consequently, whereas

$$\frac{d}{dx}(\sin x) = \cos x$$

and

$$\frac{d}{dx}(\sin y) = \cos y \cdot \frac{dy}{dx}$$

because we must use the chain rule to differentiate $\sin y$ with respect to x.

2. Rewrite the equation so that all terms containing dy/dx are on the left and all terms that do not contain dy/dx are on the right.
3. Factor out dy/dx on the left.
4. Solve for dy/dx by dividing both sides of the equation by an appropriate algebraic expression.

Example 3.8.1: Using Implicit Differentiation

Assuming that y is defined implicitly by the equation $x^2 + y^2 = 25$, find $\frac{dy}{dx}$.

Solution

Follow the steps in the problem-solving strategy.

$\frac{d}{dx}(x^2 + y^2) = \frac{d}{dx}(25)$	Step 1. Differentiate both sides of the equation.
$\frac{d}{dx}(x^2) + \frac{d}{dx}(y^2) = 0$	Step 1.1. Use the sum rule on the left. On the right $\frac{d}{dx}(25) = 0$.
$2x + 2y\frac{dy}{dx} = 0$	Step 1.2. Take the derivatives, so $\frac{d}{dx}(x^2) = 2x$ and $\frac{d}{dx}(y^2) = 2y\frac{dy}{dx}$.
$2y\frac{dy}{dx} = -2x$	Step 2. Keep the terms with $\frac{dy}{dx}$ on the left. Move the remaining terms to the right.
$\frac{dy}{dx} = -\frac{x}{y}$	Step 4. Divide both sides of the equation by $2y$. (Step 3 does not apply in this case.)

Analysis

Note that the resulting expression for $\frac{dy}{dx}$ is in terms of both the independent variable x and the dependent variable y. Although in some cases it may be possible to express $\frac{dy}{dx}$ in terms of x only, it is generally not possible to do so.

Example 3.8.2: Using Implicit Differentiation and the Product Rule

Assuming that y is defined implicitly by the equation $x^3 \sin y + y = 4x + 3$, find $\frac{dy}{dx}$.

Solution

$\frac{d}{dx}(x^3 \sin y + y) = \frac{d}{dx}(4x + 3)$	Step 1: Differentiate both sides of the equation.
$\frac{d}{dx}(x^3 \sin y) + \frac{d}{dx}(y) = 4$	Step 1.1: Apply the sum rule on the left. On the right, $\frac{d}{dx}(4x + 3) = 4$.
$\left(\frac{d}{dx}(x^3) \cdot \sin y + \frac{d}{dx}(\sin y) \cdot x^3\right) + \frac{dy}{dx} = 4$	Step 1.2: Use the product rule to find $\frac{d}{dx}(x^3 \sin y)$. Observe that $\frac{d}{dx}(y) = \frac{dy}{dx}$.
$3x^2 \sin y + (\cos y \frac{dy}{dx}) \cdot x^3 + \frac{dy}{dx} = 4$	Step 1.3: We know $\frac{d}{dx}(x^3) = 3x^2$. Use the chain rule to obtain $\frac{d}{dx}(\sin y) = \cos y \frac{dy}{dx}$.
$x^3 \cos y \frac{dy}{dx} + \frac{dy}{dx} = 4 - 3x^2 \sin y$	Step 2: Keep all terms containing $\frac{dy}{dx}$ on the left. Move all other terms to the right.
$\frac{dy}{dx}(x^3 \cos y + 1) = 4 - 3x^2 \sin y$	Step 3: Factor out $\frac{dy}{dx}$ on the left.
$\frac{dy}{dx} = \frac{4 - 3x^2 \sin y}{x^3 \cos y + 1}$	Step 4: Solve for $\frac{dy}{dx}$ by dividing both sides of the equation by $x^3 \cos y + 1$.

Example 3.8.3: Using Implicit Differentiation to Find a Second Derivative

Find $\frac{d^2y}{dx^2}$ if $x^2 + y^2 = 25$.

Solution

In Example 3.8.1, we showed that $\frac{dy}{dx} = -\frac{x}{y}$. We can take the derivative of both sides of this equation to find $\frac{d^2y}{dx^2}$.

$$\frac{d^2y}{dx^2} = \frac{d}{dy}\left(-\frac{x}{y}\right) \quad \text{Differentiate both sides of } \frac{dy}{dx} = -\frac{x}{y}.$$

$$= -\frac{\left(1 \cdot y - x\frac{dy}{dx}\right)}{y^2} \quad \text{Use the quotient rule to find } \frac{d}{dy}\left(-\frac{x}{y}\right).$$

$$= \frac{-y + x\frac{dy}{dx}}{y^2} \quad \text{Simplify.}$$

$$= \frac{-y + x\left(-\frac{x}{y}\right)}{y^2} \quad \text{Substitute } \frac{dy}{dx} = -\frac{x}{y}.$$

$$= \frac{-y^2 - x^2}{y^3} \quad \text{Simplify.}$$

At this point we have found an expression for $\frac{d^2y}{dx^2}$. If we choose, we can simplify the expression further by recalling that $x^2 + y^2 = 25$ and making this substitution in the numerator to obtain $\frac{d^2y}{dx^2} = -\frac{25}{y^3}$.

Exercise 3.8.1

Find $\frac{dy}{dx}$ for y defined implicitly by the equation $4x^5 + \tan y = y^2 + 5x$.

Hint

Follow the problem solving strategy, remembering to apply the chain rule to differentiate $\tan y$ and y^2.

Answer

$$\frac{dy}{dx} = \frac{5 - 20x^4}{\sec^2 y - 2y}$$

Finding Tangent Lines Implicitly

Now that we have seen the technique of implicit differentiation, we can apply it to the problem of finding equations of tangent lines to curves described by equations.

Example 3.8.4: Finding a Tangent Line to a Circle

Find the equation of the line tangent to the curve $x^2 + y^2 = 25$ at the point $(3, -4)$.

Solution

Although we could find this equation without using implicit differentiation, using that method makes it much easier. In Example 3.8.1, we found $\frac{dy}{dx} = -\frac{x}{y}$.

The slope of the tangent line is found by substituting $(3, -4)$ into this expression. Consequently, the slope of the tangent line is $\left.\frac{dy}{dx}\right|_{(3,-4)} = -\frac{3}{-4} = \frac{3}{4}$.

Using the point $(3, -4)$ and the slope $\frac{3}{4}$ in the point-slope equation of the line, we obtain the equation $y = \frac{3}{4}x - \frac{25}{4}$ (Figure 3.8.2).

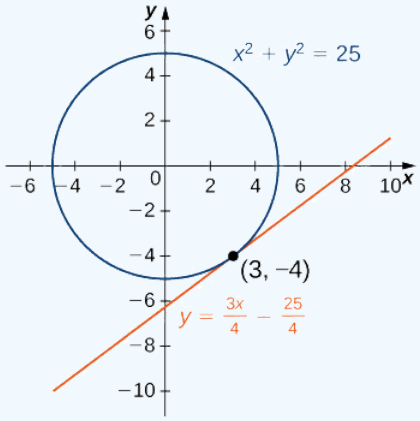


Figure 3.8.2: The line $y = \frac{3}{4}x - \frac{25}{4}$ is tangent to $x^2 + y^2 = 25$ at the point (3, −4).

Example 3.8.5: Finding the Equation of the Tangent Line to a Curve

Find the equation of the line tangent to the graph of $y^3 + x^3 - 3xy = 0$ at the point $\left(\frac{3}{2}, \frac{3}{2}\right)$ (Figure 3.8.3). This curve is known as the folium (or leaf) of Descartes.

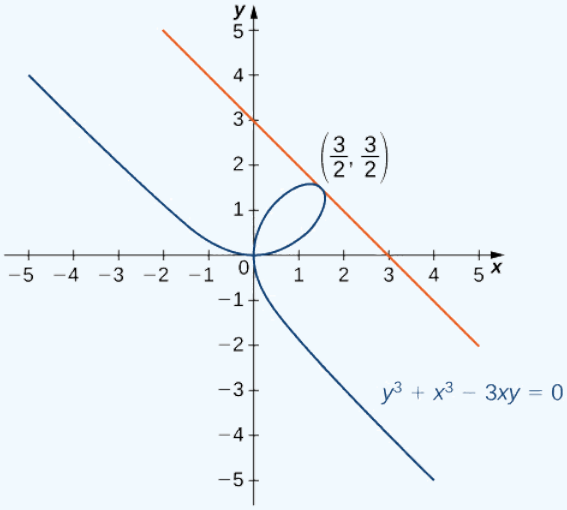


Figure 3.8.3: Finding the tangent line to the folium of Descartes at $\left(\frac{3}{2}, \frac{3}{2}\right)$.

Solution

Begin by finding $\frac{dy}{dx}$.

$$\frac{d}{dx}\left(y^3 + x^3 - 3xy\right) = \frac{d}{dx}\left(0\right)$$
$$3y^2\frac{dy}{dx} + 3x^2 - \left(3y + 3x\frac{dy}{dx}\right) = 0$$
$$3y^2\frac{dy}{dx} + 3x^2 - 3y - 3x\frac{dy}{dx} = 0$$
$$\left(3y^2 - 3x\right)\frac{dy}{dx} = 3y - 3x^2$$
$$\frac{dy}{dx} = \frac{3y - 3x^2}{3y^2 - 3x}.$$

Next, substitute $\left(\frac{3}{2}, \frac{3}{2}\right)$ into $\frac{dy}{dx} = \frac{3y - 3x^2}{3y^2 - 3x}$ to find the slope of the tangent line:

$$\left.\frac{dy}{dx}\right|_{\left(\frac{3}{2}, \frac{3}{2}\right)} = -1.$$

Finally, substitute into the point-slope equation of the line to obtain

$$y = -x + 3.$$

Example 3.8.6: Applying Implicit Differentiation

In a simple video game, a rocket travels in an elliptical orbit whose path is described by the equation $4x^2 + 25y^2 = 100$. The rocket can fire missiles along lines tangent to its path. The object of the game is to destroy an incoming asteroid traveling along the positive x-axis toward $(0, 0)$. If the rocket fires a missile when it is located at $\left(3, \frac{8}{5}\right)$, where will it intersect the x-axis?

Solution

To solve this problem, we must determine where the line tangent to the graph of

$4x^2 + 25y^2 = 100$ at $\left(3, \frac{8}{5}\right)$ intersects the x-axis. Begin by finding $\frac{dy}{dx}$ implicitly.

Differentiating, we have

$$8x + 50y\frac{dy}{dx} = 0.$$

Solving for $\frac{dy}{dx}$,

we have

$$\frac{dy}{dx} = -\frac{4x}{25y}.$$

The slope of the tangent line is $\left.\frac{dy}{dx}\right|_{(3,\frac{8}{5})} = -\frac{3}{10}$. The equation of the tangent line is $y = -\frac{3}{10}x + \frac{5}{2}$. To determine where the line intersects the x-axis, solve $0 = -\frac{3}{10}x + \frac{5}{2}$. The solution is $x = \frac{25}{3}$. The missile intersects the x-axis at the point $\left(\frac{25}{3}, 0\right)$.

? Exercise 3.8.2

Find the equation of the line tangent to the hyperbola $x^2 - y^2 = 16$ at the point $(5, 3)$.

Hint

$\frac{dy}{dx} = \frac{x}{y}$

Answer

$y = \frac{5}{3}x - \frac{16}{3}$

Key Concepts

- We use implicit differentiation to find derivatives of implicitly defined functions (functions defined by equations).
- By using implicit differentiation, we can find the equation of a tangent line to the graph of a curve.

Glossary

implicit differentiation

is a technique for computing $\frac{dy}{dx}$ for a function defined by an equation, accomplished by differentiating both sides of the equation (remembering to treat the variable y as a function) and solving for $\frac{dy}{dx}$

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3.8E: EXERCISES FOR SECTION 3.8

In exercises 1 - 10, use implicit differentiation to find $\frac{dy}{dx}$.

1) $x^2 - y^2 = 4$

2) $6x^2 + 3y^2 = 12$

Answer

$$\frac{dy}{dx} = \frac{-2x}{y}$$

3) $x^2 y = y - 7$

4) $3x^3 + 9xy^2 = 5x^3$

Answer

$$\frac{dy}{dx} = \frac{x}{3y} - \frac{y}{2x}$$

5) $xy - \cos(xy) = 1$

6) $y\sqrt{x+4} = xy + 8$

Answer

$$\frac{dy}{dx} = \frac{y - \frac{y}{2\sqrt{x+4}}}{\sqrt{x+4} - x}$$

7) $-xy - 2 = \frac{x}{7}$

8) $y\sin(xy) = y^2 + 2$

Answer

$$\frac{dy}{dx} = \frac{y^2\cos(xy)}{2y - \sin(xy) - xy\cos(xy)}$$

9) $(xy)^2 + 3x = y^2$

10) $x^3 y + xy^3 = -8$

Answer

$$\frac{dy}{dx} = \frac{-3x^2 y - y^3}{x^3 + 3xy^2}$$

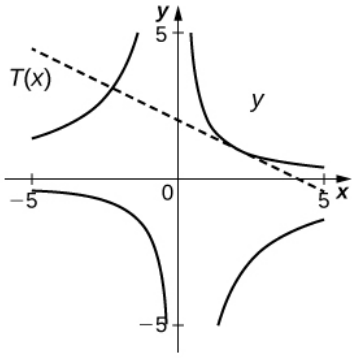
For exercises 11 - 16, find the equation of the tangent line to the graph of the given equation at the indicated point. Use a calculator or computer software to graph the function and the tangent line.

11) [T] $x^4 y - xy^3 = -2, \quad (-1, -1)$

12) [T] $x^2 y^2 + 5xy = 14, \quad (2, 1)$

Answer

$y = -\frac{1}{2}x + 2$

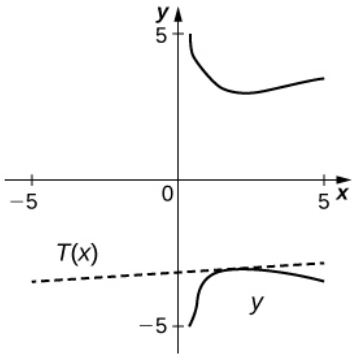


13) [T] $\tan(xy) = y, \quad (\frac{\pi}{4}, 1)$

14) [T] $xy^2 + \sin(\pi y) - 2x^2 = 10, \quad (2, -3)$

Answer

$y = \frac{1}{\pi+12}x - \frac{3\pi+38}{\pi+12}$

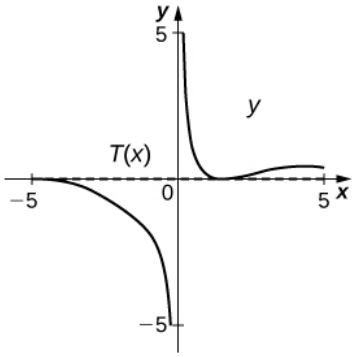


15) [T] $\frac{x}{y} + 5x - 7 = -\frac{3}{4}y, \quad (1, 2)$

16) [T] $xy + \sin(x) = 1, \quad (\frac{\pi}{2}, 0)$

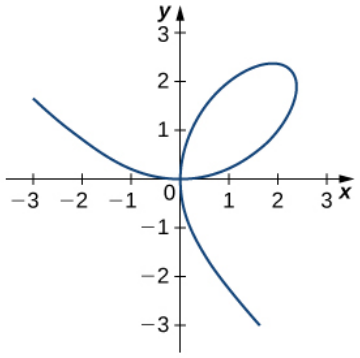
Answer

$y = 0$



17) [T] The graph of a folium of Descartes with equation $2x^3 + 2y^3 - 9xy = 0$ is given in the following graph.

https://math.libretexts.org/@go/page/51417



a. Find the equation of the tangent line at the point $(2,1)$. Graph the tangent line along with the folium.

b. Find the equation of the normal line to the tangent line in a. at the point $(2,1)$.

18) For the equation $x^2+2xy-3y^2=0$,

a. Find the equation of the normal to the tangent line at the point $(1,1)$.

b. At what other point does the normal line in a. intersect the graph of the equation?

Answer

a. $y=-x+2$

b. $(3,-1)$

19) Find all points on the graph of $y^3-27y=x^2-90$ at which the tangent line is vertical.

20) For the equation $x^2+xy+y^2=7$,

a. Find the x-intercept(s).

b.Find the slope of the tangent line(s) at the x-intercept(s).

c. What does the value(s) in part b. indicate about the tangent line(s)?

Answer

a. $\left(\pm\sqrt{7},0\right)$

b. -2

c. They are parallel since the slope is the same at both intercepts.

21) Find the equation of the tangent line to the graph of the equation $\sin^{-1}x+\sin^{-1}y=\frac{\pi}{6}$ at the point $\left(0,\frac{1}{2}\right)$.

22) Find the equation of the tangent line to the graph of the equation $\tan^{-1}(x+y)=x^2+\frac{\pi}{4}$ at the point $(0,1)$.

Answer

$y=-x+1$

23) Find y' and y'' for $x^2+6xy-2y^2=3$.

24) [T] The number of cell phones produced when x dollars is spent on labor and y dollars is spent on capital invested by a manufacturer can be modeled by the equation $60x^{3/4}y^{1/4}=3240$.

a. Find $\frac{dy}{dx}$ and evaluate at the point $(81,16)$.

b. Interpret the result of a.

Answer

a. $\frac{dy}{dx}=-0.5926$

b. When \$81 is spent on labor and \$16 is spent on capital, the amount spent on capital is decreasing by \$0.5926 per \$1 spent on labor.

25) [T] The number of cars produced when x dollars is spent on labor and y dollars is spent on capital invested by a manufacturer can be modeled by the equation $30x^{1/3}y^{2/3}=360$.

(Both $x$and y are measured in thousands of dollars.)

a. Find $\frac{dy}{dx}$ and evaluate at the point $(27, 8)$.

b. Interpret the result of part a.

26) The volume of a right circular cone of radius x and height y is given by $V = \frac{1}{3}\pi x^2 y$. Suppose that the volume of the cone is $85\pi \text{cm}^3$. Find $\frac{dy}{dx}$ when $x = 4$ and $y = 16$.

Answer

$$\frac{dy}{dx} = -8$$

For exercises 27 - 28, consider a closed rectangular box with a square base with side x and height y.

27) Find an equation for the surface area of the rectangular box, $S(x, y)$.

28) If the surface area of the rectangular box is 78 square feet, find $\frac{dy}{dx}$ when $x = 3$ feet and $y = 5$ feet.

Answer

$$\frac{dy}{dx} = -2.67$$

In exercises 29 - 31, use implicit differentiation to determine y'. Does the answer agree with the formulas we have previously determined?

29) $x = \sin y$

30) $x = \cos y$

Answer

$$y' = -\frac{1}{\sqrt{1 - x^2}}$$

31) $x = \tan y$

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3.9: Derivatives of Exponential and Logarithmic Functions

Learning Objectives

- Find the derivative of exponential functions.
- Find the derivative of logarithmic functions.
- Use logarithmic differentiation to determine the derivative of a function.

So far, we have learned how to differentiate a variety of functions, including trigonometric, inverse, and implicit functions. In this section, we explore derivatives of exponential and logarithmic functions. As we discussed in Introduction to Functions and Graphs, exponential functions play an important role in modeling population growth and the decay of radioactive materials. Logarithmic functions can help rescale large quantities and are particularly helpful for rewriting complicated expressions.

Derivative of the Exponential Function

Just as when we found the derivatives of other functions, we can find the derivatives of exponential and logarithmic functions using formulas. As we develop these formulas, we need to make certain basic assumptions. The proofs that these assumptions hold are beyond the scope of this course.

First of all, we begin with the assumption that the function $B(x) = b^x$, $b > 0$, is defined for every real number and is continuous. In previous courses, the values of exponential functions for all rational numbers were defined—beginning with the definition of b^n, where n is a positive integer—as the product of b multiplied by itself n times. Later, we defined $b^0 = 1, b^{-n} = \frac{1}{b^n}$, for a positive integer n, and $b^{s/t} = (\sqrt[t]{b})^s$ for positive integers s and t. These definitions leave open the question of the value of b^r where r is an arbitrary real number. By assuming the continuity of $B(x) = b^x, b > 0$, we may interpret b^r as $\lim_{x \to r} b^x$ where the values of x as we take the limit are rational. For example, we may view 4^π as the number satisfying

$$4^3 < 4^\pi < 4^4, \quad 4^{3.1} < 4^\pi < 4^{3.2}, \quad 4^{3.14} < 4^\pi < 4^{3.15},$$

$$4^{3.141} < 4^\pi < 4^{3.142}, \quad 4^{3.1415} < 4^\pi < 4^{3.1416}, \quad \ldots.$$

As we see in the following table, $4^\pi \approx 77.88$.

x	4^x	x	4^x
4^3	64	$4^{3.141593}$	77.8802710486
$4^{3.1}$	73.5166947198	$4^{3.1416}$	77.8810268071
$4^{3.14}$	77.7084726013	$4^{3.142}$	77.9242251944
$4^{3.141}$	77.8162741237	$4^{3.15}$	78.7932424541
$4^{3.1415}$	77.8702309526	$4^{3.2}$	84.4485062895
$4^{3.14159}$	77.8799471543	4^4	256

Approximating a Value of 4^π

We also assume that for $B(x) = b^x$, $b > 0$, the value $B'(0)$ of the derivative exists. In this section, we show that by making this one additional assumption, it is possible to prove that the function $B(x)$ is differentiable everywhere.

We make one final assumption: that there is a unique value of $b > 0$ for which $B'(0) = 1$. We define e to be this unique value, as we did in Introduction to Functions and Graphs. Figure 3.9.1 provides graphs of the functions $y = 2^x$, $y = 3^x$, $y = 2.7^x$, and $y = 2.8^x$. A visual estimate of the slopes of the tangent lines to these functions at 0 provides evidence that the value of e lies somewhere between 2.7 and 2.8. The function $E(x) = e^x$ is called the **natural exponential function**. Its inverse, $L(x) = \log_e x = \ln x$ is called the **natural logarithmic function.**

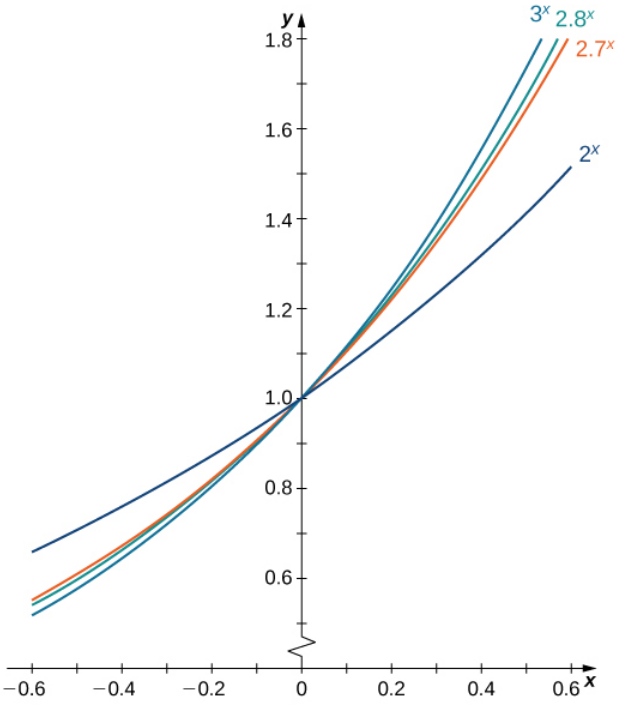


Figure 3.9.1: The graph of $E(x) = e^x$ is between $y = 2^x$ and $y = 3^x$.

For a better estimate of e, we may construct a table of estimates of $B'(0)$ for functions of the form $B(x) = b^x$. Before doing this, recall that

$$B'(0) = \lim_{x \to 0} \frac{b^x - b^0}{x - 0} = \lim_{x \to 0} \frac{b^x - 1}{x} \approx \frac{b^x - 1}{x}$$

for values of x very close to zero. For our estimates, we choose $x = 0.00001$ and $x = -0.00001$

to obtain the estimate

$$\frac{b^{-0.00001} - 1}{-0.00001} < B'(0) < \frac{b^{0.00001} - 1}{0.00001}.$$

See the following table.

Table : Estimating the value of e

b	$\frac{b^{-0.00001}-1}{-0.00001} < B'(0) < \frac{b^{0.00001}-1}{0.00001}$.	b	$\frac{b^{-0.00001}-1}{-0.00001} < B'(0) < \frac{b^{0.00001}-1}{0.00001}$.
2	$0.693145 < B'(0) < 0.69315$	2.7183	$1.000002 < B'(0) < 1.000012$
2.7	$0.993247 < B'(0) < 0.993257$	2.719	$1.000259 < B'(0) < 1.000269$
2.71	$0.996944 < B'(0) < 0.996954$	2.72	$1.000627 < B'(0) < 1.000637$
2.718	$0.999891 < B'(0) < 0.999901$	2.8	$1.029614 < B'(0) < 1.029625$
2.7182	$0.999965 < B'(0) < 0.999975$	3	$1.098606 < B'(0) < 1.098618$

The evidence from the table suggests that $2.7182 < e < 2.7183$.

The graph of $E(x) = e^x$ together with the line $y = x + 1$ are shown in Figure 3.9.2. This line is tangent to the graph of $E(x) = e^x$ at $x = 0$.

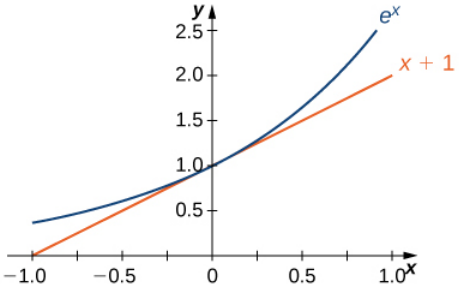


Figure 3.9.2: The tangent line to $E(x) = e^x$ at $x = 0$ has slope 1.

Now that we have laid out our basic assumptions, we begin our investigation by exploring the derivative of $B(x) = b^x$, $b > 0$. Recall that we have assumed that $B'(0)$ exists. By applying the limit definition to the derivative we conclude that

$$B'(0) = \lim_{h \to 0} \frac{b^{0+h} - b^0}{h} = \lim_{h \to 0} \frac{b^h - 1}{h}$$

Turning to $B'(x)$, we obtain the following.

$$\begin{aligned}
B'(x) &= \lim_{h \to 0} \frac{b^{x+h} - b^x}{h} && \text{Apply the limit definition of the derivative.} \\
&= \lim_{h \to 0} \frac{b^x b^h - b^x}{h} && \text{Note that } b^{x+h} = b^x b^h. \\
&= \lim_{h \to 0} \frac{b^x (b^h - 1)}{h} && \text{Factor out } b^x. \\
&= b^x \lim_{h \to 0} \frac{b^h - 1}{h} && \text{Apply a property of limits.} \\
&= b^x B'(0) && \text{Use } B'(0) = \lim_{h \to 0} \frac{b^{0+h} - b^0}{h} = \lim_{h \to 0} \frac{b^h - 1}{h}.
\end{aligned}$$

We see that on the basis of the assumption that $B(x) = b^x$ is differentiable at $0, B(x)$ is not only differentiable everywhere, but its derivative is

$$B'(x) = b^x B'(0).$$

For $E(x) = e^x$, $E'(0) = 1$. Thus, we have $E'(x) = e^x$. (The value of $B'(0)$ for an arbitrary function of the form $B(x) = b^x$, $b > 0$, will be derived later.)

Derivative of the Natural Exponential Function

Let $E(x) = e^x$ be the natural exponential function. Then

$$E'(x) = e^x.$$

In general,

$$\frac{d}{dx}\left(e^{g(x)}\right) = e^{g(x)} g'(x)$$

Example 3.9.1: Derivative of an Exponential Function

Find the derivative of $f(x) = e^{\tan(2x)}$.

Solution:

Using the derivative formula and the chain rule,

$$f'(x) = e^{\tan(2x)} \frac{d}{dx}\left(\tan(2x)\right) = e^{\tan(2x)} \sec^2(2x) \cdot 2$$

Example 3.9.2: Combining Differentiation Rules

Find the derivative of $y = \dfrac{e^{x^2}}{x}$.

Solution

Use the derivative of the natural exponential function, the quotient rule, and the chain rule.

$$y' = \frac{(e^{x^2} \cdot 2)x \cdot x - 1 \cdot e^{x^2}}{x^2} \qquad \text{Apply the quotient rule.}$$

$$= \frac{e^{x^2}(2x^2 - 1)}{x^2} \qquad \text{Simplify.}$$

Exercise 3.9.1

Find the derivative of $h(x) = xe^{2x}$.

Hint

Don't forget to use the product rule.

Answer

$h'(x) = e^{2x} + 2xe^{2x}$

Example 3.9.3: Applying the Natural Exponential Function

A colony of mosquitoes has an initial population of 1000. After t days, the population is given by $A(t) = 1000e^{0.3t}$. Show that the ratio of the rate of change of the population, $A'(t)$, to the population, $A(t)$ is constant.

Solution

First find $A'(t)$. By using the chain rule, we have $A'(t) = 300e^{0.3t}$. Thus, the ratio of the rate of change of the population to the population is given by

$$\frac{A'(t)}{A(t)} = \frac{300e^{0.3t}}{1000e^{0.3t}} = 0.3.$$

The ratio of the rate of change of the population to the population is the constant 0.3.

Exercise 3.9.2

If $A(t) = 1000e^{0.3t}$ describes the mosquito population after t days, as in the preceding example, what is the rate of change of $A(t)$ after 4 days?

Hint

Find $A'(4)$.

Answer

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Derivative of the Logarithmic Function

Now that we have the derivative of the natural exponential function, we can use implicit differentiation to find the derivative of its inverse, the natural logarithmic function.

Definition: The Derivative of the Natural Logarithmic Function

If $x > 0$ and $y = \ln x$, then

$$\frac{dy}{dx} = \frac{1}{x}.$$

More generally, let $g(x)$ be a differentiable function. For all values of x for which $g'(x) > 0$, the derivative of $h(x) = \ln(g(x))$ is given by

$$h'(x) = \frac{1}{g(x)}g'(x).$$

Proof

If $x > 0$ and $y = \ln x$, then $e^y = x$. Differentiating both sides of this equation results in the equation

$$e^y \frac{dy}{dx} = 1.$$

Solving for $\frac{dy}{dx}$ yields

$$\frac{dy}{dx} = \frac{1}{e^y}.$$

Finally, we substitute $x = e^y$ to obtain

$$\frac{dy}{dx} = \frac{1}{x}.$$

We may also derive this result by applying the inverse function theorem, as follows. Since $y = g(x) = \ln x$ is the inverse of $f(x) = e^x$, by applying the inverse function theorem we have

$$\frac{dy}{dx} = \frac{1}{f'(g(x))} = \frac{1}{e^{\ln x}} = \frac{1}{x}.$$

Using this result and applying the chain rule to $h(x) = \ln(g(x))$ yields

$$h'(x) = \frac{1}{g(x)}g'(x). \tag{3.9.1}$$

□

The graph of $y = \ln x$ and its derivative $\frac{dy}{dx} = \frac{1}{x}$ are shown in Figure 3.9.3.

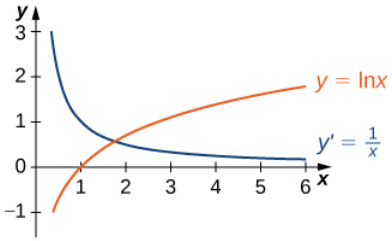


Figure 3.9.3: The function $y = \ln x$ is increasing on $(0, +\infty)$. Its derivative $y' = \frac{1}{x}$ is greater than zero on $(0, +\infty)$

Example 3.9.4: Taking a Derivative of a Natural Logarithm

Find the derivative of $f(x) = \ln(x^3 + 3x - 4)$.

Solution

Use Equation 3.9.1 directly.

$f'(x) = \dfrac{1}{x^3+3x-4} \cdot (3x^2+3)$ Use $g(x) = x^3+3x-4$ in $h'(x) = \dfrac{1}{g(x)}g'(x)$.

$= \dfrac{3x^2+3}{x^3+3x-4}$ Rewrite.

Example 3.9.5: Using Properties of Logarithms in a Derivative

Find the derivative of $f(x) = \ln\left(\dfrac{x^2 \sin x}{2x+1}\right)$.

Solution

At first glance, taking this derivative appears rather complicated. However, by using the properties of logarithms prior to finding the derivative, we can make the problem much simpler.

$f(x) = \ln\left(\dfrac{x^2 \sin x}{2x+1}\right) = 2\ln x + \ln(\sin x) - \ln(2x+1)$ Apply properties of logarithms.

$f'(x) = \dfrac{2}{x} + \cot x - \dfrac{2}{2x+1}$ Apply sum rule and $h'(x) = \dfrac{1}{g(x)}g'(x)$.

Exercise 3.9.3

Differentiate: $f(x) = \ln(3x+2)^5$.

Hint

Use a property of logarithms to simplify before taking the derivative.

Answer

$f'(x) = \dfrac{15}{3x+2}$

Now that we can differentiate the natural logarithmic function, we can use this result to find the derivatives of $y = \log_b x$ and $y = b^x$ for $b > 0,\ b \neq 1$.

Derivatives of General Exponential and Logarithmic Functions

Let $b > 0, b \neq 1$, and let $g(x)$ be a differentiable function.

i. If $y = \log_b x$, then

$$\frac{dy}{dx} = \frac{1}{x \ln b}.$$

More generally, if $h(x) = \log_b(g(x))$, then for all values of x for which $g(x) > 0$,

$$h'(x) = \frac{g'(x)}{g(x)\ln b}. \tag{3.9.2}$$

ii. If $y = b^x$, then

$$\frac{dy}{dx} = b^x \ln b.$$

More generally, if $h(x) = b^{g(x)}$, then

$$h'(x) = b^{g(x)} g'(x) \ln b \tag{3.9.3}$$

Proof

If $y = \log_b x$, then $b^y = x$. It follows that $\ln(b^y) = \ln x$. Thus $y \ln b = \ln x$. Solving for y, we have $y = \dfrac{\ln x}{\ln b}$. Differentiating and keeping in mind that $\ln b$ is a constant, we see that

$$\frac{dy}{dx} = \frac{1}{x \ln b}.$$

The derivative in Equation 3.9.2 now follows from the chain rule.

If $y = b^x$. then $\ln y = x \ln b$. Using implicit differentiation, again keeping in mind that $\ln b$ is constant, it follows that $\dfrac{1}{y}\dfrac{dy}{dx} = \ln b$. Solving for $\dfrac{dy}{dx}$ and substituting $y = b^x$, we see that

$$\frac{dy}{dx} = y \ln b = b^x \ln b.$$

The more general derivative (Equation 3.9.3) follows from the chain rule.

□

Example 3.9.6: Applying Derivative Formulas

Find the derivative of $h(x) = \dfrac{3^x}{3^x + 2}$.

Solution

Use the quotient rule and Note.

$$h'(x) = \frac{3^x \ln 3(3^x + 2) - 3^x \ln 3(3^x)}{(3^x + 2)^2} \qquad \text{Apply the quotient rule.}$$

$$= \frac{2 \cdot 3^x \ln 3}{(3x + 2)^2} \qquad \text{Simplify.}$$

Example 3.9.7: Finding the Slope of a Tangent Line

Find the slope of the line tangent to the graph of $y = \log_2(3x + 1)$ at $x = 1$.

Solution

To find the slope, we must evaluate $\dfrac{dy}{dx}$ at $x = 1$. Using Equation 3.9.2, we see that

$$\frac{dy}{dx} = \frac{3}{(3x + 1) \ln 2}.$$

By evaluating the derivative at $x = 1$, we see that the tangent line has slope

$$\left.\frac{dy}{dx}\right|_{x=1} = \frac{3}{4 \ln 2} = \frac{3}{\ln 16}.$$

Exercise 3.9.4

Find the slope for the line tangent to $y = 3^x$ at $x = 2$.

Hint

Evaluate the derivative at $x = 2$.

Answer

$9 \ln(3)$

Logarithmic Differentiation

At this point, we can take derivatives of functions of the form $y = (g(x))^n$ for certain values of n, as well as functions of the form $y = b^{g(x)}$, where $b > 0$ and $b \neq 1$. Unfortunately, we still do not know the derivatives of functions such as $y = x^x$ or $y = x^\pi$. These functions require a technique called **logarithmic differentiation**, which allows us to differentiate any function of the form $h(x) = g(x)^{f(x)}$. It can also be used to convert a very complex differentiation problem into a simpler one, such as finding the derivative of $y = \dfrac{x\sqrt{2x+1}}{e^x \sin^3 x}$. We outline this technique in the following problem-solving strategy.

Problem-Solving Strategy: Using Logarithmic Differentiation

1. To differentiate $y = h(x)$ using logarithmic differentiation, take the natural logarithm of both sides of the equation to obtain $\ln y = \ln(h(x))$.
2. Use properties of logarithms to expand $\ln(h(x))$ as much as possible.
3. Differentiate both sides of the equation. On the left we will have $\dfrac{1}{y}\dfrac{dy}{dx}$.
4. Multiply both sides of the equation by y to solve for $\dfrac{dy}{dx}$.
5. Replace y by $h(x)$.

Example 3.9.8: Using Logarithmic Differentiation

Find the derivative of $y = (2x^4 + 1)^{\tan x}$.

Solution

Use logarithmic differentiation to find this derivative.

$$\ln y = \ln (2x^4 + 1)^{\tan x}$$ Step 1. Take the natural logarithm of both sides.

$$\ln y = \tan x \ln(2x^4 + 1)$$ Step 2. Expand using properties of logarithms.

$$\frac{1}{y}\frac{dy}{dx} = \sec^2 x \ln(2x^4 + 1) + \frac{8x^3}{2x^4 + 1} \cdot \tan x$$ Step 3. Differentiate both sides. Use the product rule on the right.

$$\frac{dy}{dx} = y \cdot (\sec^2 x \ln(2x^4 + 1) + \frac{8x^3}{2x^4 + 1} \cdot \tan x)$$ Step 4. Multiply by y on both sides.

$$\frac{dy}{dx} = (2x^4 + 1)^{\tan x}(\sec^2 x \ln(2x^4 + 1) + \frac{8x^3}{2x^4 + 1} \tan x)$$ Step 5. Substitute $y = (2x^4 + 1)^{\tan x}$.

Example 3.9.9: Extending the Power Rule

Find the derivative of $y = \dfrac{x\sqrt{2x+1}}{e^x \sin^3 x}$.

Solution

This problem really makes use of the properties of logarithms and the differentiation rules given in this chapter.

$\ln y = \ln \dfrac{x\sqrt{2x+1}}{e^x \sin^3 x}$	Step 1. Take the natural logarithm of both sides.
$\ln y = \ln x + \frac{1}{2}\ln(2x+1) - x \ln e - 3 \ln \sin x$	Step 2. Expand using properties of logarithms.
$\dfrac{1}{y}\dfrac{dy}{dx} = \dfrac{1}{x} + \dfrac{1}{2x+1} - 1 - 3\dfrac{\cos x}{\sin x}$	Step 3. Differentiate both sides.
$\dfrac{dy}{dx} = y\left(\dfrac{1}{x} + \dfrac{1}{2x+1} - 1 - 3\cot x\right)$	Step 4. Multiply by y on both sides.
$\dfrac{dy}{dx} = \dfrac{x\sqrt{2x+1}}{e^x \sin^3 x}\left(\dfrac{1}{x} + \dfrac{1}{2x+1} - 1 - 3\cot x\right)$	Step 5. Substitute $y = \dfrac{x\sqrt{2x+1}}{e^x \sin^3 x}$.

Exercise 3.9.5

Use logarithmic differentiation to find the derivative of $y = x^x$.

Hint

Follow the problem solving strategy.

Answer

Solution: $\frac{dy}{dx} = x^x(1 + \ln x)$

Exercise 3.9.6

Find the derivative of $y = (\tan x)^\pi$.

Hint

Use the power rule (since the exponent π is a constant) and the chain rule.

Answer

$y' = \pi(\tan x)^{\pi-1} \sec^2 x$

Key Concepts

- On the basis of the assumption that the exponential function $y = b^x$, $b > 0$ is continuous everywhere and differentiable at 0, this function is differentiable everywhere and there is a formula for its derivative.
- We can use a formula to find the derivative of $y = \ln x$, and the relationship $\log_b x = \frac{\ln x}{\ln b}$ allows us to extend our differentiation formulas to include logarithms with arbitrary bases.
- Logarithmic differentiation allows us to differentiate functions of the form $y = g(x)^{f(x)}$ or very complex functions by taking the natural logarithm of both sides and exploiting the properties of logarithms before differentiating.

Key Equations

- **Derivative of the natural exponential function**

  $$\frac{d}{dx}\left(e^{g(x)}\right) = e^{g(x)}g'(x)$$

- **Derivative of the natural logarithmic function**

  $$\frac{d}{dx}\left(\ln g(x)\right) = \frac{1}{g(x)}g'(x)$$

- **Derivative of the general exponential function**

  $$\frac{d}{dx}\left(b^{g(x)}\right) = b^{g(x)}g'(x)\ln b$$

- **Derivative of the general logarithmic function**

  $$\frac{d}{dx}\left(\log_b g(x)\right) = \frac{g'(x)}{g(x)\ln b}$$

Glossary

logarithmic differentiation

is a technique that allows us to differentiate a function by first taking the natural logarithm of both sides of an equation, applying properties of logarithms to simplify the equation, and differentiating implicitly

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3.9E: EXERCISES FOR SECTION 3.9

In exercises 1 - 15, find $f'(x)$ for each function.

1) $f(x) = x^2 e^x$

Answer

$f'(x) = 2xe^x + x^2 e^x$

2) $f(x) = \dfrac{e^{-x}}{x}$

3) $f(x) = e^{x^3 \ln x}$

Answer

$f'(x) = e^{x^3 \ln x}\left(3x^2 \ln x + x^2\right)$

4) $f(x) = \sqrt{e^{2x} + 2x}$

5) $f(x) = \dfrac{e^x - e^{-x}}{e^x + e^{-x}}$

Answer

$f'(x) = \dfrac{4}{(e^x + e^{-x})^2}$

6) $f(x) = \dfrac{10^x}{\ln 10}$

7) $f(x) = 2^{4x} + 4x^2$

Answer

$f'(x) = 2^{4x+2} \cdot \ln 2 + 8x$

8) $f(x) = 3^{\sin 3x}$

9) $f(x) = x^\pi \cdot \pi^x$

Answer

$f'(x) = \pi x^{\pi-1} \cdot \pi^x + x^\pi \cdot \pi^x \ln \pi$

10) $f(x) = \ln(4x^3 + x)$

11) $f(x) = \ln \sqrt{5x - 7}$

Answer

$f'(x) = \dfrac{5}{2(5x - 7)}$

12) $f(x) = x^2 \ln 9x$

13) $f(x) = \log(\sec x)$

Answer

$f'(x) = \dfrac{\tan x}{\ln 10}$

14) $f(x) = \log_7 (6x^4 + 3)^5$

15) $f(x) = 2^x \cdot \log_3 7^{x^2-4}$

Answer

$f'(x) = 2^x \cdot \ln 2 \cdot \log_3 7^{x^2-4} + 2^x \cdot \dfrac{2x \ln 7}{\ln 3}$

For exercises 16 - 23, use logarithmic differentiation to find $\dfrac{dy}{dx}$.

16) $y = x^{\sqrt{x}}$

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17) $y = (\sin 2x)^{4x}$

Answer

$$\frac{dy}{dx} = (\sin 2x)^{4x}\left[4 \cdot \ln(\sin 2x) + 8x \cdot \cot 2x\right]$$

18) $y = (\ln x)^{\ln x}$

19) $y = x^{\log_2 x}$

Answer

$$\frac{dy}{dx} = x^{\log_2 x} \cdot \frac{2 \ln x}{x \ln 2}$$

20) $y = (x^2 - 1)^{\ln x}$

21) $y = x^{\cot x}$

Answer

$$\frac{dy}{dx} = x^{\cot x} \cdot \left[-\csc^2 x \cdot \ln x + \frac{\cot x}{x}\right]$$

22) $y = \dfrac{x + 11}{\sqrt[3]{x^2 - 4}}$

23) $y = x^{-1/2}(x^2 + 3)^{2/3}(3x - 4)^4$

Answer

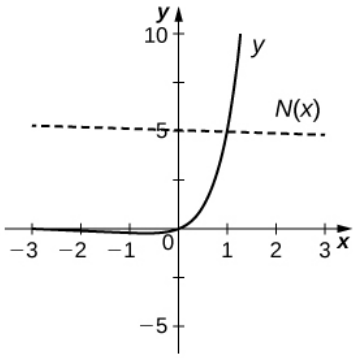
$$\frac{dy}{dx} = x^{-1/2}(x^2 + 3)^{2/3}(3x - 4)^4 \cdot \left[\frac{-1}{2x} + \frac{4x}{3(x^2 + 3)} + \frac{12}{3x - 4}\right]$$

24) [T] Find an equation of the tangent line to the graph of $f(x) = 4xe^{(x^2-1)}$ at the point where $x = -1$. Graph both the function and the tangent line.

25) [T] Find the equation of the line that is normal to the graph of $f(x) = x \cdot 5^x$ at the point where $x = 1$. Graph both the function and the normal line.

Answer

$y = \frac{-1}{5+5\ln 5}x + \left(5 + \frac{1}{5+5\ln 5}\right)$



26) [T] Find the equation of the tangent line to the graph of $x^3 - x \ln y + y^3 = 2x + 5$ at the point where $x = 2$. (Hint: Use implicit differentiation to find $\frac{dy}{dx}$.) Graph both the curve and the tangent line.

27) Consider the function $y = x^{1/x}$ for $x > 0$.

a. Determine the points on the graph where the tangent line is horizontal.

b. Determine the points on the graph where $y' > 0$ and those where $y' < 0$.

Answer

a. $x = e \approx 2.718$

b. $y' > 0$ for $(0, e)$ and $y' < 0$ for (e, ∞).

28) The formula $I(t) = \frac{\sin t}{e^t}$ is the formula for a decaying alternating current.

a. Complete the following table with the appropriate values.

t	$\frac{\sin t}{e^t}$
0	(i)
$\pi/2$	(ii)
π	(iii)
$3\pi/2$	(vi)
2π	(v)
2π	(vi)
3π	(vii)

b. Using only the values in the table, determine where the tangent line to the graph of $I(t)$ is horizontal.

29) [T] The population of Toledo, Ohio, in 2000 was approximately 500,000. Assume the population is increasing at a rate of 5% per year.

a. Write the exponential function that relates the total population as a function of t.

b. Use part a. to determine the rate at which the population is increasing in t years.

c. Use part b. to determine the rate at which the population is increasing in 10 years

Answer

a. $P = 500,000(1.05)^t$ individuals

b. $P'(t) = 24395 \cdot (1.05)^t$ individuals per year

c. $39,737$ individuals per year

30)[T] An isotope of the element erbium has a half-life of approximately 12 hours. Initially there are 9 grams of the isotope present.

a. Write the exponential function that relates the amount of substance remaining as a function of t, measured in hours.

b. Use a. to determine the rate at which the substance is decaying in t hours.

c. Use b. to determine the rate of decay at $t = 4$ hours.

31) [T] The number of cases of influenza in New York City from the beginning of 1960 to the beginning of 1964 is modeled by the function $N(t) = 5.3e^{0.093t^2 - 0.87t}, (0 \le t \le 4)$, where $N(t)$ gives the number of cases (in thousands) and t is measured in years, with $t = 0$ corresponding to the beginning of 1960.

a. Show work that evaluates $N(0)$ and $N(4)$. Briefly describe what these values indicate about the disease in New York City.

b. Show work that evaluates $N'(0)$ and $N'(3)$. Briefly describe what these values indicate about the disease in the United States.

Answer

a. At the beginning of 1960 there were 5.3 thousand cases of the disease in New York City. At the beginning of 1963 there were approximately 723 cases of the disease in the United States.

b. At the beginning of 1960 the number of cases of the disease was decreasing at rate of -4.611 thousand per year; at the beginning of 1963, the number of cases of the disease was decreasing at a rate of -0.2808 thousand per year.

32) [T] The relative rate of change of a differentiable function $y = f(x)$ is given by $\frac{100 \cdot f'(x)}{f(x)}$ One model for population growth is a Gompertz growth function, given by $P(x) = ae^{-b \cdot e^{-cx}}$ where a, b, and c are constants.

a. Find the relative rate of change formula for the generic Gompertz function.

b. Use part a. to find the relative rate of change of a population in $x = 20$ months when $a = 204,\ b = 0.0198$, and $c = 0.15$.

c. Briefly interpret what the result of part b. means.

For exercises 33 - 36, use the population of New York City from 1790 to 1860, given in the following table.

Year since 1790	Population
0	33,131
10	60,515
20	96,373
30	123,706
40	202,300
50	312,710
60	515,547
70	813,669

New York City Population Over TimeSource: http://en.Wikipedia.org/wiki/Largest..._United_States_by_population_by_decade

33) [T] Using a computer program or a calculator, fit a growth curve to the data of the form $p = ab^t$.

Answer

$p = 35741(1.045)^t$

34) [T] Using the exponential best fit for the data, write a table containing the derivatives evaluated at each year.

35) [T] Using the exponential best fit for the data, write a table containing the second derivatives evaluated at each year.

Answer

Year since 1790	P''
0	69.25
10	107.5
20	167.0
30	259.4
40	402.8
50	625.5
60	971.4
70	1508.5

36) [T] Using the tables of first and second derivatives and the best fit, answer the following questions:

a. Will the model be accurate in predicting the future population of New York City? Why or why not?

b. Estimate the population in 2010. Was the prediction correct from part a.?

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3.R: Chapter 3 Review Exercises

Chapter Review Exercises

True or False? Justify the answer with a proof or a counterexample.

1) Every function has a derivative.

Answer

False

2) A continuous function has a continuous derivative.

3) A continuous function has a derivative.

Answer

False

4) If a function is differentiable, it is continuous.

In exercises 5 and 6, use the limit definition of the derivative to exactly evaluate the derivative.

5) $f(x) = \sqrt{x+4}$

Answer

$$f'(x) = \frac{1}{2\sqrt{x+4}}$$

6) $f(x) = \dfrac{3}{x}$

In exercises 7 - 15, find the derivatives of the given functions.

7) $f(x) = 3x^3 - \dfrac{4}{x^2}$

Answer

$f'(x) = 9x^2 + \frac{8}{x^3}$

9) $f(x) = (4 - x^2)^3$

10) $f(x) = e^{\sin x}$

Answer

$f'(x) = e^{\sin x} \cos x$

11) $f(x) = \ln(x+2)$

12) $f(x) = x^2 \cos x + x \tan x$

Answer

$f'(x) = x \sec^2 x + 2x \cos x + \tan x - x^2 \sin x$

13) $f(x) = \sqrt{3x^2+2}$

14) $f(x) = \dfrac{x}{4} \sin^{-1}(x)$

Answer

$f'(x) = \frac{1}{4}\left(\frac{x}{\sqrt{1-x^2}} + \sin^{-1} x\right)$

15) $x^2 y = (y+2) + xy \sin x$

In exercises 16 - 18, find the indicated derivatives of various orders.

16) First derivative of $y = x(\ln x)\cos x$

Answer

$$\frac{dy}{dx} = \cos x \cdot (\ln x + 1) - x(\ln x)\sin x$$

17) Third derivative of $y = (3x + 2)^2$

18) Second derivative of $y = 4^x + x^2 \sin x$

Answer

$$\frac{d^2y}{dx^2} = 4^x(\ln 4)^2 + 2\sin x + 4x\cos x - x^2 \sin x$$

In exercises 19 and 20, find the equation of the tangent line to the following equations at the specified point.

19) $y = \cos^{-1}(x) + x$ at $x = 0$

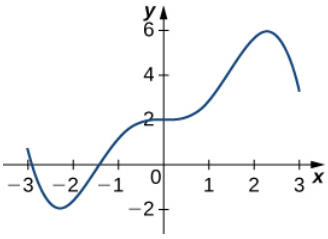
20) $y = x + e^x - \frac{1}{x}$ at $x = 1$

Answer

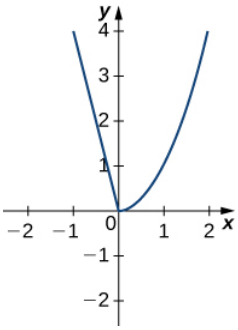
$y = (2 + e)x - 2$

In exercises 21 and 22, draw the derivative of the functions with the given graphs.

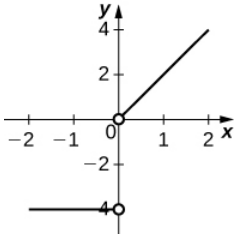
21)



22)



Answer



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Questions 22 and 23 concern the water level in Ocean City, New Jersey, in January, which can be approximated by $w(t) = 1.9 + 2.9\cos(\frac{\pi}{6}t)$, where t is measured in hours after midnight, and the height is measured in feet.

22) Find and graph the derivative. What is the physical meaning?

23) Find $w'(3)$. What is the physical meaning of this value?

Answer

$w'(3) = -\frac{2.9\pi}{6}$. At 3 a.m. the tide is decreasing at a rate of 1.514 ft/hr.

Questions 24 and 25 consider the wind speeds of Hurricane Katrina, which affected New Orleans, Louisiana, in August 2005. The data are displayed in a table.

Hours after Midnight, August 26	Wind Speed (mph)
1	45
5	75
11	100
29	115
49	145
58	175
73	155
81	125
85	95
107	35

Wind Speeds of Hurricane KatrinaSource: news.nationalgeographic.com/n..._timeline.html.

24) Using the table, estimate the derivative of the wind speed at hour 39. What is the physical meaning?

25) Estimate the derivative of the wind speed at hour 83. What is the physical meaning?

Answer

-7.5. The wind speed is decreasing at a rate of 7.5 mph/hr

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CHAPTER OVERVIEW

4: Applications of Derivatives

A rocket launch involves two related quantities that change over time. Being able to solve this type of problem is just one application of derivatives introduced in this chapter. We also look at how derivatives are used to find maximum and minimum values of functions. As a result, we will be able to solve applied optimization problems, such as maximizing revenue and minimizing surface area. In addition, we examine how derivatives are used to evaluate complicated limits, to approximate roots of functions, and to provide accurate graphs of functions.

4.0: Prelude to Applications of Derivatives
4.1: Related Rates
4.1E: Exercises for Section 4.1
4.2: Linear Approximations and Differentials
4.2E: Exercises for Section 4.2
4.3: Maxima and Minima
4.3E: Exercises for Section 4.3
4.4: The Mean Value Theorem
4.4E: Exercises for Section 4.4
4.5: Derivatives and the Shape of a Graph
4.5E: Exercises for Section 4.5
4.6: Limits at Infinity and Asymptotes
4.6E: Exercises for Section 4.6
4.7: Applied Optimization Problems
4.7E: Exercises for Section 4.7
4.8: L'Hôpital's Rule
4.8E: Exercises for Section 4.8
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4.10: Antiderivatives
4.10E: Exercises for Section 4.10
4.R: Chapter 4 Review Exercises

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4.0: Prelude to Applications of Derivatives

A rocket is being launched from the ground and cameras are recording the event. A video camera is located on the ground a certain distance from the launch pad. At what rate should the angle of inclination (the angle the camera makes with the ground) change to allow the camera to record the flight of the rocket as it heads upward?



Figure 4.0.1: As a rocket is being launched, at what rate should the angle of a video camera change to continue viewing the rocket? (credit: modification of work by Steve Jurvetson, Wikimedia Commons)

A rocket launch involves two related quantities that change over time. Being able to solve this type of problem is just one application of derivatives introduced in this chapter. We also look at how derivatives are used to find maximum and minimum values of functions. As a result, we will be able to solve applied optimization problems, such as maximizing revenue and minimizing surface area. In addition, we examine how derivatives are used to evaluate complicated limits, to approximate roots of functions, and to provide accurate graphs of functions.

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4.1: Related Rates

Learning Objectives

- Express changing quantities in terms of derivatives.
- Find relationships among the derivatives in a given problem.
- Use the chain rule to find the rate of change of one quantity that depends on the rate of change of other quantities.

We have seen that for quantities that are changing over time, the rates at which these quantities change are given by derivatives. If two related quantities are changing over time, the rates at which the quantities change are related. For example, if a balloon is being filled with air, both the radius of the balloon and the volume of the balloon are increasing. In this section, we consider several problems in which two or more related quantities are changing and we study how to determine the relationship between the rates of change of these quantities.

Setting up Related-Rates Problems

In many real-world applications, related quantities are changing with respect to time. For example, if we consider the balloon example again, we can say that the rate of change in the volume, V, is related to the rate of change in the radius, r. In this case, we say that $\frac{dV}{dt}$ and $\frac{dr}{dt}$ are **related rates** because V is related to r. Here we study several examples of related quantities that are changing with respect to time and we look at how to calculate one rate of change given another rate of change.

Example 4.1.1: Inflating a Balloon

A spherical balloon is being filled with air at the constant rate of $2\ \text{cm}^3/\text{sec}$ (Figure 4.1.1). How fast is the radius increasing when the radius is 3 cm?

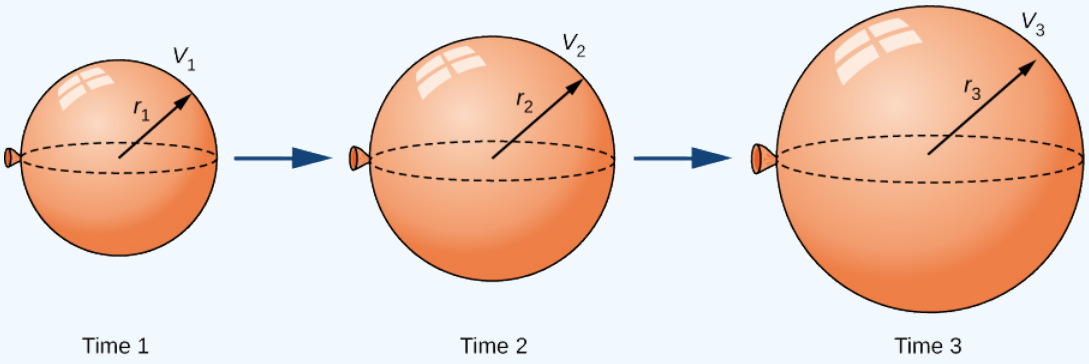


Figure 4.1.1: As the balloon is being filled with air, both the radius and the volume are increasing with respect to time.

Solution

The volume of a sphere of radius r centimeters is

$$V = \tfrac{4}{3}\pi r^3\ \text{cm}^3.$$

Since the balloon is being filled with air, both the volume and the radius are functions of time. Therefore, t seconds after beginning to fill the balloon with air, the volume of air in the balloon is

$$V(t) = \tfrac{4}{3}\pi\big[r(t)\big]^3\text{cm}^3.$$

Differentiating both sides of this equation with respect to time and applying the chain rule, we see that the rate of change in the volume is related to the rate of change in the radius by the equation

$$V'(t) = 4\pi\big[r(t)\big]^2 r'(t).$$

The balloon is being filled with air at the constant rate of $2\,\text{cm}^3/\text{sec}$, so $V'(t) = 2\,\text{cm}^3/\text{sec}$. Therefore,

$$2\,\text{cm}^3/\text{sec} = \Big(4\pi\big[r(t)\big]^2\ \text{cm}^2\Big)\cdot\Big(r'(t)\ \text{cm/s}\Big),$$

which implies

$$r'(t) = \frac{1}{2\pi\left[r(t)\right]^2} \text{ cm/sec.}$$

When the radius $r = 3$ cm,

$$r'(t) = \frac{1}{18\pi} \text{ cm/sec.}$$

Exercise 4.1.1

What is the instantaneous rate of change of the radius when $r = 6$ cm?

Hint

$\frac{dr}{dt} = \frac{1}{2\pi r^2}$

Answer

$\frac{1}{72\pi}$ cm/sec, or approximately 0.0044 cm/sec

Before looking at other examples, let's outline the problem-solving strategy we will be using to solve related-rates problems.

Problem-Solving Strategy: Solving a Related-Rates Problem

1. Assign symbols to all variables involved in the problem. Draw a figure if applicable.
2. State, in terms of the variables, the information that is given and the rate to be determined.
3. Find an equation relating the variables introduced in step 1.
4. Using the chain rule, differentiate both sides of the equation found in step 3 with respect to the independent variable. This new equation will relate the derivatives.
5. Substitute all known values into the equation from step 4, then solve for the unknown rate of change

Note that when solving a related-rates problem, it is crucial not to substitute known values too soon. For example, if the value for a changing quantity is substituted into an equation before both sides of the equation are differentiated, then that quantity will behave as a constant and its derivative will not appear in the new equation found in step 4. We examine this potential error in the following example.

Examples of the Process

Let's now implement the strategy just described to solve several related-rates problems. The first example involves a plane flying overhead. The relationship we are studying is between the speed of the plane and the rate at which the distance between the plane and a person on the ground is changing.

Example 4.1.2: An Airplane Flying at a Constant Elevation

An airplane is flying overhead at a constant elevation of 4000 ft. A man is viewing the plane from a position 3000 ft from the base of a radio tower. The airplane is flying horizontally away from the man. If the plane is flying at the rate of 600 ft/sec, at what rate is the distance between the man and the plane increasing when the plane passes over the radio tower?

Solution

Step 1. Draw a picture, introducing variables to represent the different quantities involved.

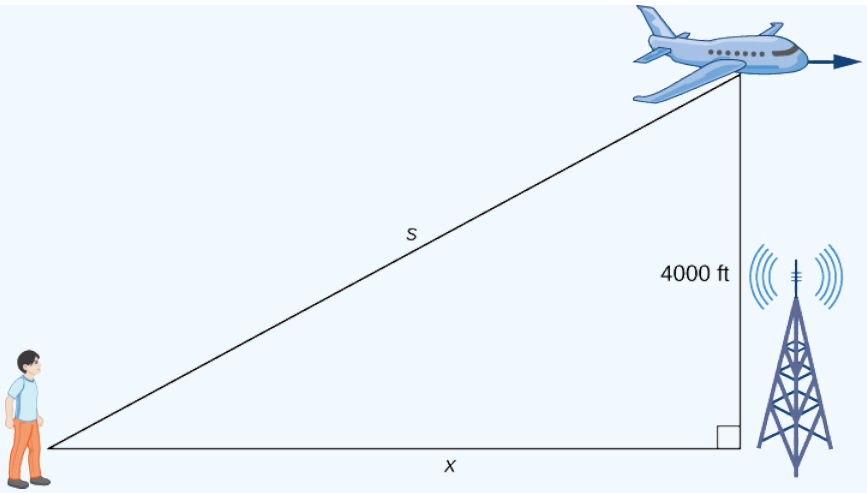


Figure 4.1.2: An airplane is flying at a constant height of 4000 ft. The distance between the person and the airplane and the person and the place on the ground directly below the airplane are changing. We denote those quantities with the variables s and x, respectively.

As shown, x denotes the distance between the man and the position on the ground directly below the airplane. The variable s denotes the distance between the man and the plane. Note that both x and s are functions of time. We do not introduce a variable for the height of the plane because it remains at a constant elevation of 4000 ft. Since an object's height above the ground is measured as the shortest distance between the object and the ground, the line segment of length 4000 ft is perpendicular to the line segment of length x feet, creating a right triangle.

Step 2. Since x denotes the horizontal distance between the man and the point on the ground below the plane, dx/dt represents the speed of the plane. We are told the speed of the plane is 600 ft/sec. Therefore, $\frac{dx}{dt} = 600$ ft/sec. Since we are asked to find the rate of change in the distance between the man and the plane when the plane is directly above the radio tower, we need to find ds/dt when $x = 3000$ ft.

Step 3. From the figure, we can use the **Pythagorean theorem** to write an equation relating x and s:

$$[x(t)]^2 + 4000^2 = [s(t)]^2.$$

Step 4. Differentiating this equation with respect to time and using the fact that the derivative of a constant is zero, we arrive at the equation

$$x\frac{dx}{dt} = s\frac{ds}{dt}.$$

Step 5. Find the rate at which the distance between the man and the plane is increasing when the plane is directly over the radio tower. That is, find $\frac{ds}{dt}$ when $x = 3000$ ft. Since the speed of the plane is 600 ft/sec, we know that $\frac{dx}{dt} = 600$ ft/sec. We are not given an explicit value for s; however, since we are trying to find $\frac{ds}{dt}$ when $x = 3000$ ft, we can use the Pythagorean theorem to determine the distance s when $x = 3000$ ft and the height is 4000 ft. Solving the equation

$$3000^2 + 4000^2 = s^2$$

for s, we have $s = 5000$ ft at the time of interest. Using these values, we conclude that ds/dt

is a solution of the equation

$$(3000)(600) = (5000) \cdot \frac{ds}{dt}.$$

Therefore,

$$\frac{ds}{dt} = \frac{3000 \cdot 600}{5000} = 360 \text{ ft/sec}.$$

Note: When solving related-rates problems, it is important not to substitute values for the variables too soon. For example, in step 3, we related the variable quantities $x(t)$ and $s(t)$ by the equation

$$[x(t)]^2 + 4000^2 = [s(t)]^2.$$

Since the plane remains at a constant height, it is not necessary to introduce a variable for the height, and we are allowed to use the constant 4000 to denote that quantity. However, the other two quantities are changing. If we mistakenly substituted $x(t) = 3000$ into the equation before differentiating, our equation would have been

$$3000^2 + 4000^2 = [s(t)]^2.$$

After differentiating, our equation would become

$$0 = s(t)\frac{ds}{dt}.$$

As a result, we would incorrectly conclude that $\frac{ds}{dt} = 0$.

Exercise 4.1.2

What is the speed of the plane if the distance between the person and the plane is increasing at the rate of 300 ft/sec?

Hint

$\frac{ds}{dt} = 300$ ft/sec

Answer

500 ft/sec

We now return to the problem involving the rocket launch from the beginning of the chapter.

Example 4.1.3: Chapter Opener - A Rocket Launch

A rocket is launched so that it rises vertically. A camera is positioned 5000 ft from the launch pad. When the rocket is 1000 ft above the launch pad, its velocity is 600 ft/sec.



Figure 4.1.3: (credit: modification of work by Steve Jurvetson, Wikimedia Commons)

Find the necessary rate of change of the camera's angle as a function of time so that it stays focused on the rocket.

Solution

Step 1. Draw a picture introducing the variables.

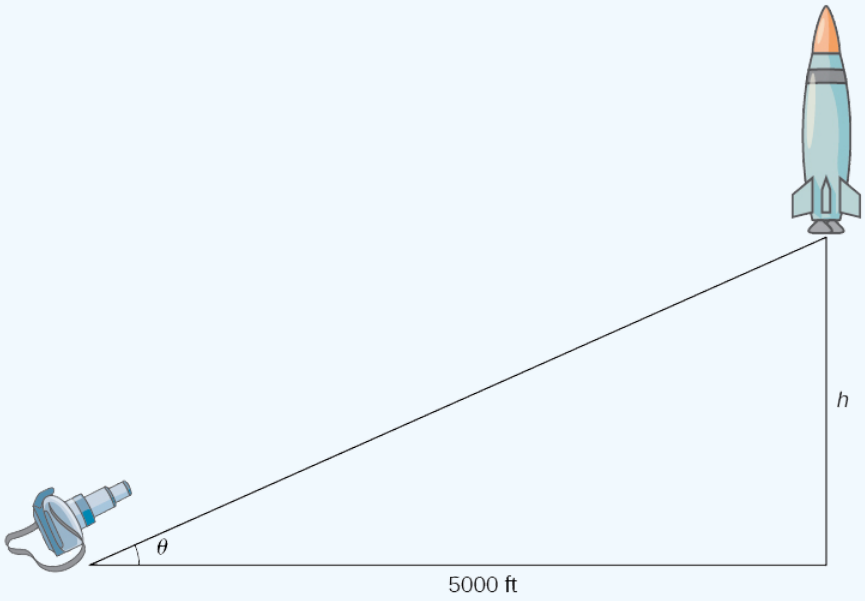


Figure 4.1.4: A camera is positioned 5000 ft from the launch pad of the rocket. The height of the rocket and the angle of the camera are changing with respect to time. We denote those quantities with the variables h and θ,respectively.

Let h denote the height of the rocket above the launch pad and θ be the angle between the camera lens and the ground.

Step 2. We are trying to find the rate of change in the angle of the camera with respect to time when the rocket is 1000 ft off the ground. That is, we need to find $\frac{d\theta}{dt}$ when $h = 1000$ ft. At that time, we know the velocity of the rocket is $\frac{dh}{dt} = 600$ ft/sec.

Step 3. Now we need to find an equation relating the two quantities that are changing with respect to time: h and θ. How can we create such an equation? Using the fact that we have drawn a right triangle, it is natural to think about trigonometric functions. Recall that $\tan\theta$ is the ratio of the length of the opposite side of the triangle to the length of the adjacent side. Thus, we have

$$\tan\theta = \frac{h}{5000}.$$

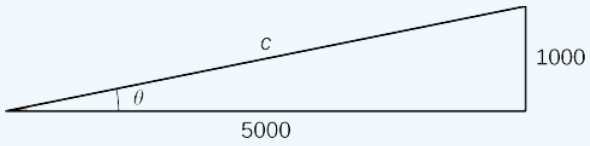
This gives us the equation

$$h = 5000\tan\theta.$$

Step 4. Differentiating this equation with respect to time t, we obtain

$$\frac{dh}{dt} = 5000\sec^2\theta\frac{d\theta}{dt}.$$

Step 5. We want to find $\frac{d\theta}{dt}$ when $h = 1000$ ft. At this time, we know that $\frac{dh}{dt} = 600$ ft/sec. We need to determine $\sec^2\theta$. Recall that $\sec\theta$ is the ratio of the length of the hypotenuse to the length of the adjacent side. We know the length of the adjacent side is 5000 ft. To determine the length of the hypotenuse, we use the Pythagorean theorem, where the length of one leg is 5000 ft, the length of the other leg is $h = 1000$ ft, and the length of the hypotenuse is c feet as shown in the following figure.



We see that

$$1000^2 + 5000^2 = c^2$$

and we conclude that the hypotenuse is

$$c = 1000\sqrt{26}\,\text{ft}.$$

Therefore, when $h = 1000$, we have

$$\sec^2\theta = \left(\frac{1000\sqrt{26}}{5000}\right)^2 = \frac{26}{25}.$$

Recall from step 4 that the equation relating $\frac{d\theta}{dt}$ to our known values is

$$\frac{dh}{dt} = 5000\sec^2\theta\frac{d\theta}{dt}.$$

When $h = 1000$ ft, we know that $\frac{dh}{dt} = 600$ ft/sec and $\sec^2\theta = \frac{26}{25}$. Substituting these values into the previous equation, we arrive at the equation

$$600 = 5000\left(\tfrac{26}{25}\right)\frac{d\theta}{dt}.$$

Therefore, $\frac{d\theta}{dt} = \frac{3}{26}$ rad/sec.

Exercise 4.1.3

What rate of change is necessary for the elevation angle of the camera if the camera is placed on the ground at a distance of 4000 ft from the launch pad and the velocity of the rocket is 500 ft/sec when the rocket is 2000 ft off the ground?

Hint

Find $\frac{d\theta}{dt}$ when $h = 2000$ ft. At that time, $\frac{dh}{dt} = 500$ ft/sec.

Answer

$\frac{1}{10}$ rad/sec

In the next example, we consider water draining from a cone-shaped funnel. We compare the rate at which the level of water in the cone is decreasing with the rate at which the volume of water is decreasing.

Example 4.1.4: Water Draining from a Funnel

Water is draining from the bottom of a cone-shaped funnel at the rate of 0.03 ft^3/sec. The height of the funnel is 2 ft and the radius at the top of the funnel is 1 ft. At what rate is the height of the water in the funnel changing when the height of the water is $\frac{1}{2}$ ft?

Solution

Step 1: Draw a picture introducing the variables.

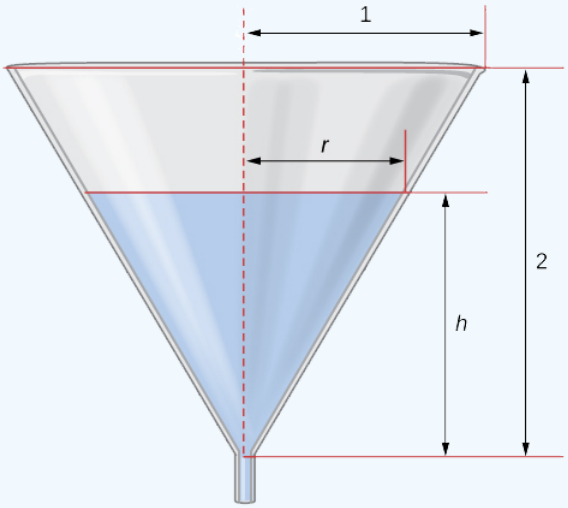


Figure 4.1.5: Water is draining from a funnel of height 2 ft and radius 1 ft. The height of the water and the radius of water are changing over time. We denote these quantities with the variables h and r, respectively.

Let h denote the height of the water in the funnel, r denote the radius of the water at its surface, and V denote the volume of the water.

Step 2: We need to determine $\frac{dh}{dt}$ when $h = \frac{1}{2}$ ft. We know that $\frac{dV}{dt} = -0.03$ ft/sec.

Step 3: The volume of water in the cone is

$$V = \tfrac{1}{3}\pi r^2 h.$$

From the figure, we see that we have similar triangles. Therefore, the ratio of the sides in the two triangles is the same. Therefore, $\frac{r}{h} = \frac{1}{2}$ or $r = \frac{h}{2}$. Using this fact, the equation for volume can be simplified to

$$V = \tfrac{1}{3}\pi\left(\tfrac{h}{2}\right)^2 h = \tfrac{\pi}{12}h^3.$$

Step 4: Applying the chain rule while differentiating both sides of this equation with respect to time t, we obtain

$$\frac{dV}{dt} = \frac{\pi}{4}h^2\frac{dh}{dt}.$$

Step 5: We want to find $\frac{dh}{dt}$ when $h = \frac{1}{2}$ ft. Since water is leaving at the rate of $0.03\,\text{ft}^3/\text{sec}$, we know that $\frac{dV}{dt} = -0.03\,\text{ft}^3/\text{sec}$. Therefore,

$$-0.03 = \frac{\pi}{4}\left(\frac{1}{2}\right)^2\frac{dh}{dt},$$

which implies

$$-0.03 = \frac{\pi}{16}\frac{dh}{dt}.$$

It follows that

$$\frac{dh}{dt} = -\frac{0.48}{\pi} = -0.153\,\text{ft/sec}.$$

? Exercise 4.1.4

At what rate is the height of the water changing when the height of the water is $\frac{1}{4}$ ft?

Hint

We need to find $\frac{dh}{dt}$ when $h = \frac{1}{4}$.

Answer

-0.61 ft/sec

Key Concepts

- To solve a related rates problem, first draw a picture that illustrates the relationship between the two or more related quantities that are changing with respect to time.
- In terms of the quantities, state the information given and the rate to be found.
- Find an equation relating the quantities.
- Use differentiation, applying the chain rule as necessary, to find an equation that relates the rates.
- Be sure not to substitute a variable quantity for one of the variables until after finding an equation relating the rates.

Glossary

related rates

are rates of change associated with two or more related quantities that are changing over time

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4.1E: EXERCISES FOR SECTION 4.1

In exercises 1 - 3, find the quantities for the given equation.

1) Find $\frac{dy}{dt}$ at $x = 1$ and $y = x^2 + 3$ if $\frac{dx}{dt} = 4$.

Answer

$$\frac{dy}{dt} = 8$$

2) Find $\frac{dx}{dt}$ at $x = -2$ and $y = 2x^2 + 1$ if $\frac{dy}{dt} = -1$.

3) Find $\frac{dz}{dt}$ at $(x, y) = (1, 3)$ and $z^2 = x^2 + y^2$ if $\frac{dx}{dt} = 4$ and $\frac{dy}{dt} = 3$.

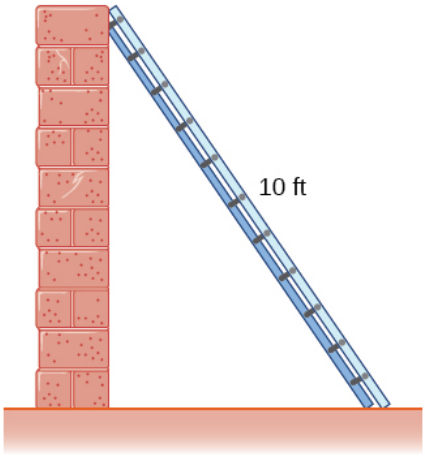
Answer

$$\frac{dz}{dt} = \frac{13}{\sqrt{10}}$$

In exercises 4 - 15, sketch the situation if necessary and used related rates to solve for the quantities.

4) [T] If two electrical resistors are connected in parallel, the total resistance (measured in ohms, denoted by the Greek capital letter omega, Ω) is given by the equation $\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2}$. If R_1 is increasing at a rate of $0.5\Omega/\text{min}$ and R_2 decreases at a rate of $1.1\Omega/\text{min}$, at what rate does the total resistance change when $R_1 = 20\Omega$ and $R_2 = 50\Omega/\text{min}$?

5) A 10-ft ladder is leaning against a wall. If the top of the ladder slides down the wall at a rate of 2 ft/sec, how fast is the bottom moving along the ground when the bottom of the ladder is 5 ft from the wall?

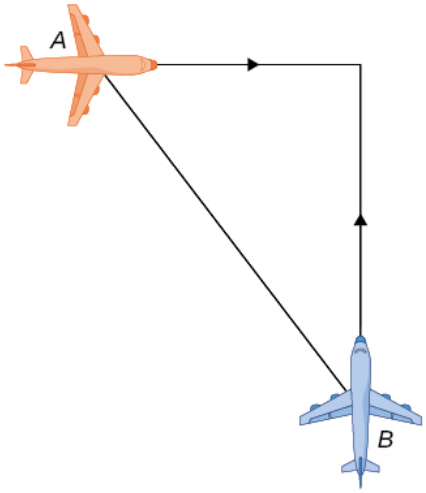


Answer

$2\sqrt{3}$ ft/sec

6) A 25-ft ladder is leaning against a wall. If we push the ladder toward the wall at a rate of 1 ft/sec, and the bottom of the ladder is initially 20 ft away from the wall, how fast does the ladder move up the wall 5 sec after we start pushing?

7) Two airplanes are flying in the air at the same height: airplane A is flying east at 250 mi/h and airplane B is flying north at 300 mi/h. If they are both heading to the same airport, located 30 miles east of airplane A and 40 miles north of airplane B, at what rate is the distance between the airplanes changing?



Answer

The distance is decreasing at 390 mi/h.

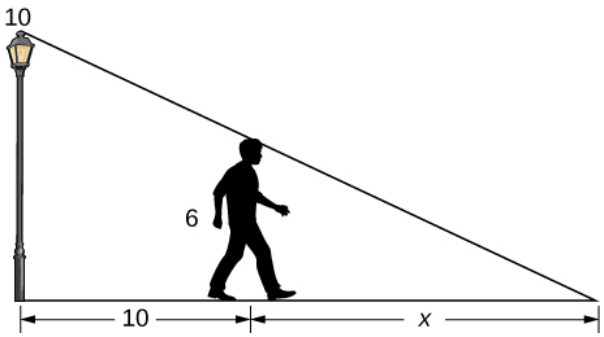
8) You and a friend are riding your bikes to a restaurant that you think is east; your friend thinks the restaurant is north. You both leave from the same point, with you riding at 16 mph east and your friend riding 12 mph north. After you traveled 4 mi, at what rate is the distance between you changing?

9) Two buses are driving along parallel freeways that are 5 mi apart, one heading east and the other heading west. Assuming that each bus drives a constant 55 mph, find the rate at which the distance between the buses is changing when they are 13 mi apart (as the crow flies), heading toward each other.

Answer

The distance between them shrinks at a rate of $\frac{1320}{13} \approx 101.5$ mph.

10) A 6-ft-tall person walks away from a 10-ft lamppost at a constant rate of 3 ft/sec. What is the rate that the tip of the shadow moves away from the pole when the person is 10 ft away from the pole?



11) Using the previous problem, what is the rate at which the tip of the shadow moves away from the person when the person is 10 ft from the pole?

Answer

$\frac{9}{2}$ ft/sec

12) A 5-ft-tall person walks toward a wall at a rate of 2 ft/sec. A spotlight is located on the ground 40 ft from the wall. How fast does the height of the person's shadow on the wall change when the person is 10 ft from the wall?

13) Using the previous problem, what is the rate at which the shadow changes when the person is 10 ft from the wall, if the person is walking away from the wall at a rate of 2 ft/sec?

Answer

It grows at a rate $\frac{4}{9}$ ft/sec

14) A helicopter starting on the ground is rising directly into the air at a rate of 25 ft/sec. You are running on the ground starting directly under the helicopter at a rate of 10 ft/sec. Find the rate of change of the distance between the helicopter and yourself after 5 sec.

15) Using the previous problem, assuming the helicopter is again rising at a rate of 25 ft/sec and you are running on the ground starting directly under the helicopter at a rate of 10 ft/sec, what is the rate at which the distance between you and the helicopter is changing when the helicopter has risen to a height of 60 ft in the air, assuming that, initially, it was 30 ft above you?

Answer

The distance is increasing at $\frac{135\sqrt{26}}{26}$ ft/sec

In exercises 16 - 24, draw and label diagrams to help solve the related-rates problems.

16) The side of a cube increases at a rate of $\frac{1}{2}$ m/sec. Find the rate at which the volume of the cube increases when the side of the cube is 4 m.

17) The volume of a cube decreases at a rate of $10\ \text{m}^3$/sec. Find the rate at which the side of the cube changes when the side of the cube is 2 m.

Answer

$-\frac{5}{6}$ m/sec

18) The radius of a circle increases at a rate of 2 m/sec. Find the rate at which the area of the circle increases when the radius is 5 m.

19) The radius of a sphere decreases at a rate of 3 m/sec. Find the rate at which the surface area decreases when the radius is 10 m.

Answer

$240\pi\ \text{m}^2/\text{sec}$

20) The radius of a sphere increases at a rate of 1 m/sec. Find the rate at which the volume increases when the radius is 20 m.

21) The radius of a sphere is increasing at a rate of 9 cm/sec. Find the radius of the sphere when the volume and the radius of the sphere are increasing at the same numerical rate.

Answer

$\frac{1}{2\sqrt{\pi}}$ cm

22) The base of a triangle is shrinking at a rate of 1 cm/min and the height of the triangle is increasing at a rate of 5 cm/min. Find the rate at which the area of the triangle changes when the height is 22 cm and the base is 10 cm.

23) A triangle has two constant sides of length 3 ft and 5 ft. The angle between these two sides is increasing at a rate of 0.1 rad/sec. Find the rate at which the area of the triangle is changing when the angle between the two sides is $\pi/6$.

Answer

The area is increasing at a rate $\frac{3\sqrt{3}}{8}\ \text{ft}^2/\text{sec}$.

24) A triangle has a height that is increasing at a rate of 2 cm/sec and its area is increasing at a rate of $4\,\text{cm}^2/\text{sec}$. Find the rate at which the base of the triangle is changing when the height of the triangle is 4 cm and the area is $20\,\text{cm}^2$.

In exercises 25 - 27, consider an inverted right cone that is leaking water. (Inverted means the cone's point is facing down, like a funnel.) The dimensions of the conical tank are a height of 16 ft and a radius of 5 ft.

25) How fast does the depth of the water change when the water is 10 ft high if the cone leaks water at a rate of $10\,\text{ft}^3/\text{min}$?

Answer

The depth of the water decreases at $\frac{128}{125\pi}$ ft/min.

26) Find the rate at which the surface area of the water changes when the water is 10 ft high if the cone leaks water at a rate of $10\,\text{ft}^3/\text{min}$.

27) If the water level is decreasing at a rate of 3 in/min when the depth of the water is 8 ft, determine the rate at which water is leaking out of the cone.

Answer

The volume is decreasing at a rate of $\frac{25\pi}{16}\ \text{ft}^3/\text{min}$.

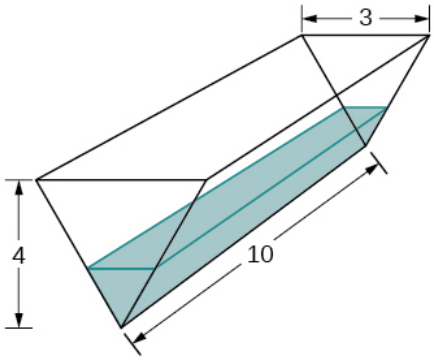
28) A vertical cylinder is leaking water at a rate of $1\,\text{ft}^3/\text{sec}$. If the cylinder has a height of 10 ft and a radius of 1 ft, at what rate is the height of the water changing when the height is 6 ft?

29) A cylinder is leaking water but you are unable to determine at what rate. The cylinder has a height of 2 m and a radius of 2 m. Find the rate at which the water is leaking out of the cylinder if the rate at which the height is decreasing is 10 cm/min when the height is 1 m.

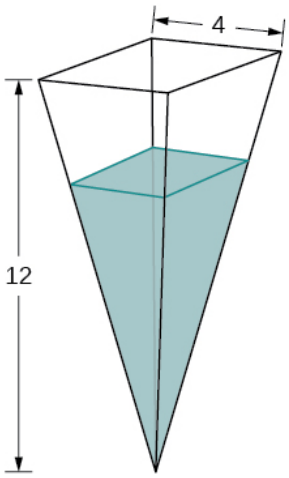
Answer

The water flows out at rate $\frac{2\pi}{5}\,\text{m}^3/\text{min}$.

30) A trough has ends shaped like isosceles triangles, with width 3 m and height 4 m, and the trough is 10 m long. Water is being pumped into the trough at a rate of $5\,\text{m}^3/\text{min}$. At what rate does the height of the water change when the water is 1 m deep?



31) A tank is shaped like an upside-down square pyramid, with base of 4 m by 4 m and a height of 12 m (see the following figure). How fast does the height increase when the water is 2 m deep if water is being pumped in at a rate of $\frac{2}{3}\,\text{m}^3$/sec?



Answer

$\frac{3}{2}$ m/sec

For exercises 32 - 34, consider a pool shaped like the bottom half of a sphere, that is being filled at a rate of $25\,\text{ft}^3/\text{min}$. The radius of the pool is 10 ft.

32) Find the rate at which the depth of the water is changing when the water has a depth of 5 ft.

33) Find the rate at which the depth of the water is changing when the water has a depth of 1 ft.

Answer

$\frac{25}{19\pi}$ ft/min

34) If the height is increasing at a rate of 1 in/sec when the depth of the water is 2 ft, find the rate at which water is being pumped in.

35) Gravel is being unloaded from a truck and falls into a pile shaped like a cone at a rate of $10\,\text{ft}^3/\text{min}$. The radius of the cone base is three times the height of the cone. Find the rate at which the height of the gravel changes when the pile has a height of 5 ft.

https://math.libretexts.org/@go/page/52587

Answer

$\frac{2}{45\pi}$ ft/min

36) Using a similar setup from the preceding problem, find the rate at which the gravel is being unloaded if the pile is 5 ft high and the height is increasing at a rate of 4 in/min.

In exercises 37 - 41, draw the situations and solve the related-rate problems.

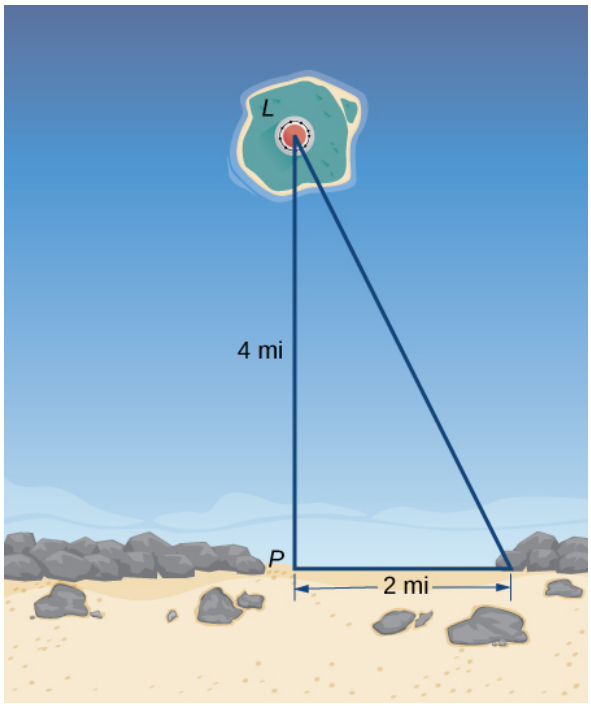
37) You are stationary on the ground and are watching a bird fly horizontally at a rate of 10 m/sec. The bird is located 40 m above your head. How fast does the angle of elevation change when the horizontal distance between you and the bird is 9 m?

Answer

The angle decreases at $\frac{400}{1681}$ rad/sec.

38) You stand 40 ft from a bottle rocket on the ground and watch as it takes off vertically into the air at a rate of 20 ft/sec. Find the rate at which the angle of elevation changes when the rocket is 30 ft in the air.

39) A lighthouse, L, is on an island 4 mi away from the closest point, P, on the beach (see the following image). If the lighthouse light rotates clockwise at a constant rate of 10 revolutions/min, how fast does the beam of light move across the beach 2 mi away from the closest point on the beach?



Answer

100π mi/min

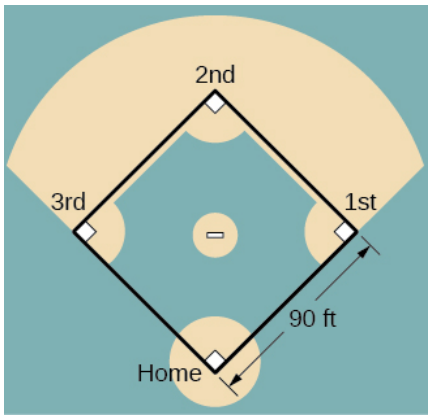
40) Using the same setup as the previous problem, determine at what rate the beam of light moves across the beach 1 mi away from the closest point on the beach.

41) You are walking to a bus stop at a right-angle corner. You move north at a rate of 2 m/sec and are 20 m south of the intersection. The bus travels west at a rate of 10 m/sec away from the intersection – you have missed the bus! What is the rate at which the angle between you and the bus is changing when you are 20 m south of the intersection and the bus is 10 m west of the intersection?

Answer

The angle is changing at a rate of $\frac{11}{25}$ rad/sec.

In exercises 42 - 45, refer to the figure of baseball diamond, which has sides of 90 ft.



42) [T] A batter hits a ball toward third base at **75** ft/sec and runs toward first base at a rate of **24** ft/sec. At what rate does the distance between the ball and the batter change when **2** sec have passed?

43) [T] A batter hits a ball toward second base at **80** ft/sec and runs toward first base at a rate of **30** ft/sec. At what rate does the distance between the ball and the batter change when the runner has covered one-third of the distance to first base? (Hint: Recall the law of cosines.)

Answer

The distance is increasing at a rate of **62.50** ft/sec.

44) [T] A batter hits the ball and runs toward first base at a speed of **22** ft/sec. At what rate does the distance between the runner and second base change when the runner has run **30** ft?

45) [T] Runners start at first and second base. When the baseball is hit, the runner at first base runs at a speed of **18** ft/sec toward second base and the runner at second base runs at a speed of **20** ft/sec toward third base. How fast is the distance between runners changing 1 sec after the ball is hit?

Answer

The distance is decreasing at a rate of **11.99** ft/sec.

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4.2: Linear Approximations and Differentials

Learning Objectives

- Describe the linear approximation to a function at a point.
- Write the linearization of a given function.
- Draw a graph that illustrates the use of differentials to approximate the change in a quantity.
- Calculate the relative error and percentage error in using a differential approximation.

We have just seen how derivatives allow us to compare related quantities that are changing over time. In this section, we examine another application of derivatives: the ability to approximate functions locally by linear functions. Linear functions are the easiest functions with which to work, so they provide a useful tool for approximating function values. In addition, the ideas presented in this section are generalized later in the text when we study how to approximate functions by higher-degree polynomials Introduction to Power Series and Functions.

Linear Approximation of a Function at a Point

Consider a function f that is differentiable at a point $x = a$. Recall that the tangent line to the graph of f at a is given by the equation

$$y = f(a) + f'(a)(x - a).$$

For example, consider the function $f(x) = \frac{1}{x}$ at $a = 2$. Since f is differentiable at $x = 2$ and $f'(x) = -\frac{1}{x^2}$, we see that $f'(2) = -\frac{1}{4}$. Therefore, the tangent line to the graph of f at $a = 2$ is given by the equation

$$y = \frac{1}{2} - \frac{1}{4}(x - 2).$$

Figure $4.2.1a$ shows a graph of $f(x) = \frac{1}{x}$ along with the tangent line to f at $x = 2$. Note that for x near 2, the graph of the tangent line is close to the graph of f. As a result, we can use the equation of the tangent line to approximate $f(x)$ for x near 2. For example, if $x = 2.1$, the y value of the corresponding point on the tangent line is

$$y = \frac{1}{2} - \frac{1}{4}(2.1 - 2) = 0.475.$$

The actual value of $f(2.1)$ is given by

$$f(2.1) = \frac{1}{2.1} \approx 0.47619.$$

Therefore, the tangent line gives us a fairly good approximation of $f(2.1)$ (Figure $4.2.1b$). However, note that for values of x far from 2, the equation of the tangent line does not give us a good approximation. For example, if $x = 10$, the y-value of the corresponding point on the tangent line is

$$y = \frac{1}{2} - \frac{1}{4}(10 - 2) = \frac{1}{2} - 2 = -1.5,$$

whereas the value of the function at $x = 10$ is $f(10) = 0.1$.

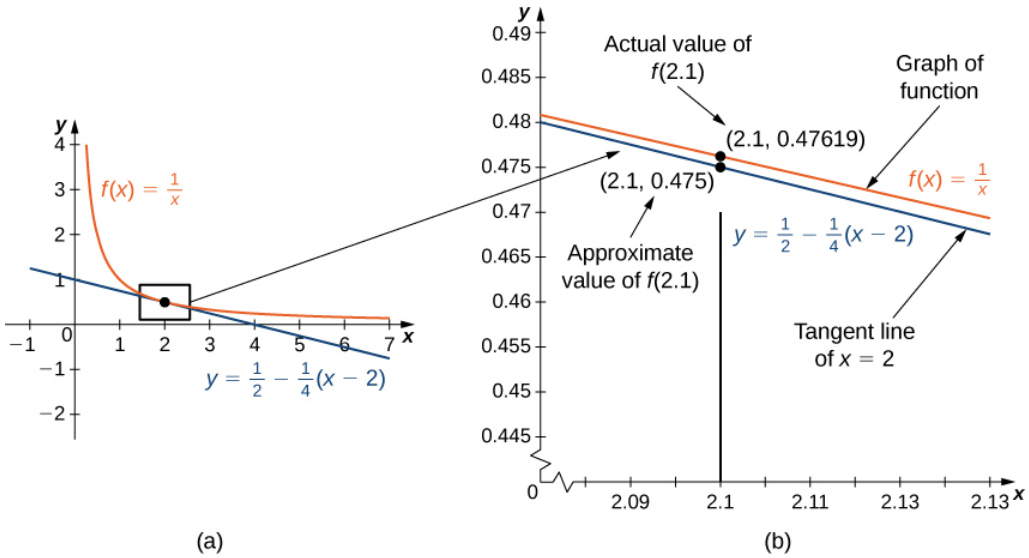


Figure 4.2.1: (a) The tangent line to $f(x) = 1/x$ at $x = 2$ provides a good approximation to f for x near 2. (b) At $x = 2.1$, the value of y on the tangent line to $f(x) = 1/x$ is 0.475. The actual value of $f(2.1)$ is $1/2.1$, which is approximately 0.47619.

In general, for a differentiable function f, the equation of the tangent line to f at $x = a$ can be used to approximate $f(x)$ for x near a. Therefore, we can write

$$f(x) \approx f(a) + f'(a)(x - a) \text{ for } x \text{ near } a.$$

We call the linear function

$$L(x) = f(a) + f'(a)(x - a) \tag{4.2.1}$$

the **linear approximation,** or **tangent line approximation**, of f at $x = a$. This function L is also known as the **linearization** of f at $x = a$.

To show how useful the linear approximation can be, we look at how to find the linear approximation for $f(x) = \sqrt{x}$ at $x = 9$.

Example 4.2.1: Linear Approximation of $\sqrt{x}$

Find the linear approximation of $f(x) = \sqrt{x}$ at $x = 9$ and use the approximation to estimate $\sqrt{9.1}$.

Solution

Since we are looking for the linear approximation at $x = 9$, using Equation 4.2.1 we know the linear approximation is given by

$$L(x) = f(9) + f'(9)(x - 9).$$

We need to find $f(9)$ and $f'(9)$.

$$f(x) = \sqrt{x} \Rightarrow f(9) = \sqrt{9} = 3$$

$$f'(x) = \tfrac{1}{2\sqrt{x}} \Rightarrow f'(9) = \tfrac{1}{2\sqrt{9}} = \tfrac{1}{6}$$

Therefore, the linear approximation is given by Figure 4.2.2.

$$L(x) = 3 + \frac{1}{6}(x - 9)$$

Using the linear approximation, we can estimate $\sqrt{9.1}$ by writing

$$\sqrt{9.1} = f(9.1) \approx L(9.1) = 3 + \frac{1}{6}(9.1 - 9) \approx 3.0167.$$

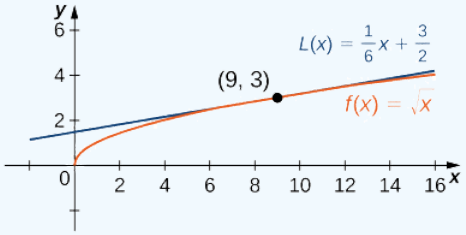


Figure 4.2.2: The local linear approximation to $f(x) = \sqrt{x}$ at $x = 9$ provides an approximation to f for x near 9.

Analysis

Using a calculator, the value of $\sqrt{9.1}$ to four decimal places is 3.0166. The value given by the linear approximation, 3.0167, is very close to the value obtained with a calculator, so it appears that using this linear approximation is a good way to estimate $\sqrt{x}$, at least for x near 9. At the same time, it may seem odd to use a linear approximation when we can just push a few buttons on a calculator to evaluate $\sqrt{9.1}$. However, how does the calculator evaluate $\sqrt{9.1}$? The calculator uses an approximation! In fact, calculators and computers use approximations all the time to evaluate mathematical expressions; they just use higher-degree approximations.

? Exercise 4.2.1

Find the local linear approximation to $f(x) = \sqrt[3]{x}$ at $x = 8$. Use it to approximate $\sqrt[3]{8.1}$ to five decimal places.

Hint

$L(x) = f(a) + f'(a)(x - a)$

Answer

$L(x) = 2 + \frac{1}{12}(x - 8)$; 2.00833

✓ Example 4.2.2: Linear Approximation of $\sin x$

Find the linear approximation of $f(x) = \sin x$ at $x = \frac{\pi}{3}$ and use it to approximate $\sin(62°)$.

Solution

First we note that since $\frac{\pi}{3}$ rad is equivalent to $60°$, using the linear approximation at $x = \pi/3$ seems reasonable. The linear approximation is given by

$$L(x) = f(\tfrac{\pi}{3}) + f'(\tfrac{\pi}{3})(x - \tfrac{\pi}{3}).$$

We see that

$$f(x) = \sin x \Rightarrow f(\tfrac{\pi}{3}) = \sin(\tfrac{\pi}{3}) = \tfrac{\sqrt{3}}{2}$$

$$f'(x) = \cos x \Rightarrow f'(\tfrac{\pi}{3}) = \cos(\tfrac{\pi}{3}) = \tfrac{1}{2}$$

Therefore, the linear approximation of f at $x = \pi/3$ is given by Figure 4.2.3.

$$L(x) = \tfrac{\sqrt{3}}{2} + \tfrac{1}{2}(x - \tfrac{\pi}{3})$$

To estimate $\sin(62°)$ using L, we must first convert $62°$ to radians. We have $62° = \frac{62\pi}{180}$ radians, so the estimate for $\sin(62°)$ is given by

$$\sin(62°) = f(\tfrac{62\pi}{180}) \approx L(\tfrac{62\pi}{180}) = \tfrac{\sqrt{3}}{2} + \tfrac{1}{2}(\tfrac{62\pi}{180} - \tfrac{\pi}{3}) = \tfrac{\sqrt{3}}{2} + \tfrac{1}{2}(\tfrac{2\pi}{180}) = \tfrac{\sqrt{3}}{2} + \tfrac{\pi}{180} \approx 0.88348.$$

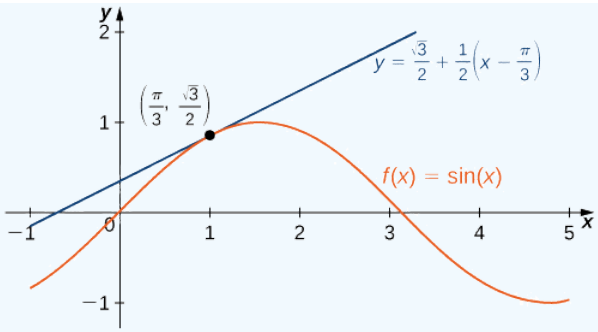


Figure 4.2.3: The linear approximation to $f(x) = \sin x$ at $x = \pi/3$ provides an approximation to $\sin x$ for x near $\pi/3$.

? Exercise 4.2.2

Find the linear approximation for $f(x) = \cos x$ at $x = \frac{\pi}{2}$.

Hint

$L(x) = f(a) + f'(a)(x - a)$

Answer

$L(x) = -x + \frac{\pi}{2}$

Linear approximations may be used in estimating roots and powers. In the next example, we find the linear approximation for $f(x) = (1 + x)^n$ at $x = 0$, which can be used to estimate roots and powers for real numbers near 1. The same idea can be extended to a function of the form $f(x) = (m + x)^n$ to estimate roots and powers near a different number m.

✓ Example 4.2.3: Approximating Roots and Powers

Find the linear approximation of $f(x) = (1 + x)^n$ at $x = 0$. Use this approximation to estimate $(1.01)^3$.

Solution

The linear approximation at $x = 0$ is given by

$$L(x) = f(0) + f'(0)(x - 0).$$

Because

$$f(x) = (1 + x)^n \Rightarrow f(0) = 1$$

$$f'(x) = n(1 + x)^{n-1} \Rightarrow f'(0) = n,$$

the linear approximation is given by Figure 4.2.4*a*.

$$L(x) = 1 + n(x - 0) = 1 + nx$$

We can approximate $(1.01)^3$ by evaluating $L(0.01)$ when $n = 3$. We conclude that

$$(1.01)^3 = f(1.01) \approx L(1.01) = 1 + 3(0.01) = 1.03.$$

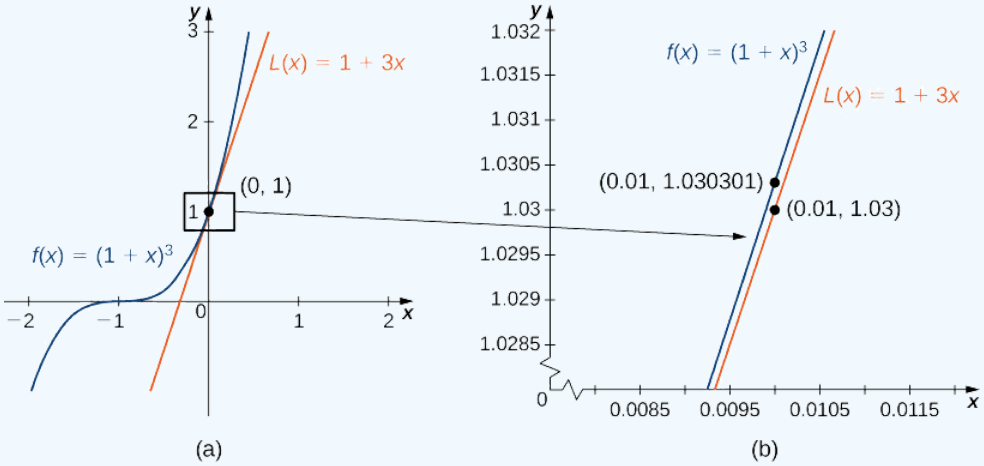


Figure 4.2.4: (a) The linear approximation of $f(x)$ at $x = 0$ is $L(x)$. (b) The actual value of 1.01^3 is 1.030301. The linear approximation of $f(x)$ at $x = 0$ estimates 1.01^3 to be 1.03.

? Exercise 4.2.3

Find the linear approximation of $f(x) = (1 + x)^4$ at $x = 0$ without using the result from the preceding example.

Hint

$f'(x) = 4(1 + x)^3$

Answer

$L(x) = 1 + 4x$

Differentials

We have seen that linear approximations can be used to estimate function values. They can also be used to estimate the amount a function value changes as a result of a small change in the input. To discuss this more formally, we define a related concept: **differentials**. Differentials provide us with a way of estimating the amount a function changes as a result of a small change in input values.

When we first looked at derivatives, we used the Leibniz notation dy/dx to represent the derivative of y with respect to x. Although we used the expressions dy and dx in this notation, they did not have meaning on their own. Here we see a meaning to the expressions dy and dx. Suppose $y = f(x)$ is a differentiable function. Let dx be an independent variable that can be assigned any nonzero real number, and define the dependent variable dy by

$$dy = f'(x)\,dx. \tag{4.2.2}$$

It is important to notice that dy is a function of both x and dx. The expressions dy and dx are called differentials. We can divide both sides of Equation 4.2.2 by dx, which yields

$$\frac{dy}{dx} = f'(x). \tag{4.2.3}$$

This is the familiar expression we have used to denote a derivative. Equation 4.2.3 is known as the **differential form** of Equation 4.2.2.

✓ Example 4.2.4: Computing Differentials

For each of the following functions, find dy and evaluate when $x = 3$ and $dx = 0.1$.

a. $y = x^2 + 2x$
b. $y = \cos x$

Solution

The key step is calculating the derivative. When we have that, we can obtain dy directly.

a. Since $f(x) = x^2 + 2x$, we know $f'(x) = 2x + 2$, and therefore

$$dy = (2x + 2)\,dx.$$

When $x = 3$ and $dx = 0.1$,

$$dy = (2 \cdot 3 + 2)(0.1) = 0.8.$$

b. Since $f(x) = \cos x$, $f'(x) = -\sin(x)$. This gives us

$$dy = -\sin x\,dx.$$

When $x = 3$ and $dx = 0.1$,

$$dy = -\sin(3)(0.1) = -0.1\sin(3).$$

? Exercise 4.2.4

For $y = e^{x^2}$, find dy.

Hint

$dy = f'(x)\,dx$

Answer

$dy = 2xe^{x^2}dx$

We now connect differentials to linear approximations. Differentials can be used to estimate the change in the value of a function resulting from a small change in input values. Consider a function f that is differentiable at point a. Suppose the input x changes by a small amount. We are interested in how much the output y changes. If x changes from a to $a + dx$, then the change in x is dx (also denoted Δx), and the change in y is given by

$$\Delta y = f(a + dx) - f(a).$$

Instead of calculating the exact change in y, however, it is often easier to approximate the change in y by using a linear approximation. For x near a, $f(x)$ can be approximated by the linear approximation (Equation 4.2.1)

$$L(x) = f(a) + f'(a)(x - a).$$

Therefore, if dx is small,

$$f(a + dx) \approx L(a + dx) = f(a) + f'(a)(a + dx - a).$$

That is,

$$f(a + dx) - f(a) \approx L(a + dx) - f(a) = f'(a)\,dx.$$

In other words, the actual change in the function f if x increases from a to $a + dx$ is approximately the difference between $L(a + dx)$ and $f(a)$, where $L(x)$ is the linear approximation of f at a. By definition of $L(x)$, this difference is equal to $f'(a)\,dx$. In summary,

$$\Delta y = f(a + dx) - f(a) \approx L(a + dx) - f(a) = f'(a)\,dx = dy.$$

Therefore, we can use the differential $dy = f'(a)\,dx$ to approximate the change in y if x increases from $x = a$ to $x = a + dx$. We can see this in the following graph.

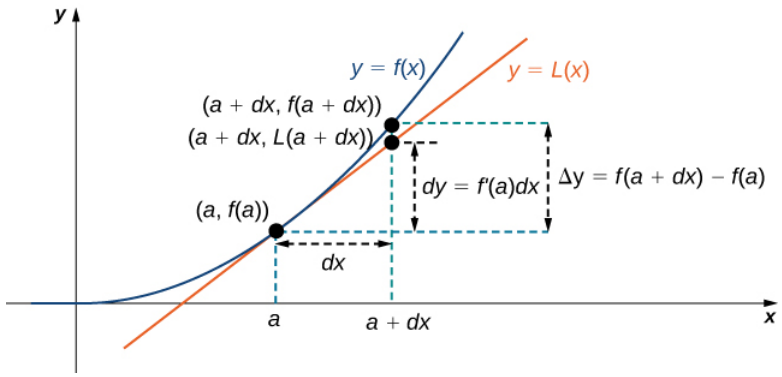


Figure 4.2.5: The differential $dy = f'(a)\,dx$ is used to approximate the actual change in y if x increases from a to $a + dx$.

We now take a look at how to use differentials to approximate the change in the value of the function that results from a small change in the value of the input. Note the calculation with differentials is much simpler than calculating actual values of functions and the result is very close to what we would obtain with the more exact calculation.

Example 4.2.5: Approximating Change with Differentials

Let $y = x^2 + 2x$. Compute Δy and dy at $x = 3$ if $dx = 0.1$.

Solution

The actual change in y if x changes from $x = 3$ to $x = 3.1$ is given by

$$\Delta y = f(3.1) - f(3) = [(3.1)^2 + 2(3.1)] - [3^2 + 2(3)] = 0.81.$$

The approximate change in y is given by $dy = f'(3)\,dx$. Since $f'(x) = 2x + 2$, we have

$$dy = f'(3)\,dx = (2(3) + 2)(0.1) = 0.8.$$

Exercise 4.2.5

For $y = x^2 + 2x$, find Δy and dy at $x = 3$ if $dx = 0.2$.

Hint

$dy = f'(3)\,dx,\ \Delta y = f(3.2) - f(3)$

Answer

$dy = 1.6,\ \Delta y = 1.64$

Calculating the Amount of Error

Any type of measurement is prone to a certain amount of error. In many applications, certain quantities are calculated based on measurements. For example, the area of a circle is calculated by measuring the radius of the circle. An error in the measurement of the radius leads to an error in the computed value of the area. Here we examine this type of error and study how differentials can be used to estimate the error.

Consider a function f with an input that is a measured quantity. Suppose the exact value of the measured quantity is a, but the measured value is $a + dx$. We say the measurement error is dx (or Δx). As a result, an error occurs in the calculated quantity $f(x)$. This type of error is known as a **propagated error** and is given by

$$\Delta y = f(a + dx) - f(a).$$

Since all measurements are prone to some degree of error, we do not know the exact value of a measured quantity, so we cannot calculate the propagated error exactly. However, given an estimate of the accuracy of a measurement, we can use differentials to approximate the propagated error Δy. Specifically, if f is a differentiable function at a,the propagated error is

$$\Delta y \approx dy = f'(a)\,dx.$$

Unfortunately, we do not know the exact value a. However, we can use the measured value $a + dx$, and estimate

$$\Delta y \approx dy \approx f'(a + dx)\,dx.$$

In the next example, we look at how differentials can be used to estimate the error in calculating the volume of a box if we assume the measurement of the side length is made with a certain amount of accuracy.

Example 4.2.6: Volume of a Cube

Suppose the side length of a cube is measured to be 5 cm with an accuracy of 0.1 cm.

a. Use differentials to estimate the error in the computed volume of the cube.
b. Compute the volume of the cube if the side length is (i) 4.9 cm and (ii) 5.1 cm to compare the estimated error with the actual potential error.

Solution

a. The measurement of the side length is accurate to within ± 0.1 cm. Therefore,

$$-0.1 \le dx \le 0.1.$$

The volume of a cube is given by $V = x^3$, which leads to

$$dV = 3x^2 dx.$$

Using the measured side length of 5 cm, we can estimate that

$$-3(5)^2(0.1) \le dV \le 3(5)^2(0.1).$$

Therefore,

$$-7.5 \le dV \le 7.5.$$

b. If the side length is actually 4.9 cm, then the volume of the cube is

$$V(4.9) = (4.9)^3 = 117.649\text{cm}^3.$$

If the side length is actually 5.1 cm, then the volume of the cube is

$$V(5.1) = (5.1)^3 = 132.651\text{cm}^3.$$

Therefore, the actual volume of the cube is between 117.649 and 132.651. Since the side length is measured to be 5 cm, the computed volume is $V(5) = 5^3 = 125$. Therefore, the error in the computed volume is

$$117.649 - 125 \le \Delta V \le 132.651 - 125.$$

That is,

$$-7.351 \le \Delta V \le 7.651.$$

We see the estimated error dV is relatively close to the actual potential error in the computed volume.

Exercise 4.2.6

Estimate the error in the computed volume of a cube if the side length is measured to be 6 cm with an accuracy of 0.2 cm.

Hint

$dV = 3x^2 dx$

Answer

The volume measurement is accurate to within $21.6\,\text{cm}^3$.

The measurement error $dx\ (=\Delta x)$ and the propagated error Δy are absolute errors. We are typically interested in the size of an error relative to the size of the quantity being measured or calculated. Given an absolute error Δq for a particular quantity, we define the **relative error** as $\frac{\Delta q}{q}$, where q is the actual value of the quantity. The **percentage error** is the relative error expressed as a percentage. For example, if we measure the height of a ladder to be 63 in. when the actual height is 62 in., the absolute error is 1 in. but the relative error is $\frac{1}{62} = 0.016$, or 1.6%. By comparison, if we measure the width of a piece of cardboard to be 8.25 in. when the actual width is 8 in., our absolute error is $\frac{1}{4}$ in., whereas the relative error is $\frac{0.25}{8} = \frac{1}{32}$, or 3.1%. Therefore, the percentage error in the measurement of the cardboard is larger, even though 0.25 in. is less than 1 in.

Example 4.2.7: Relative and Percentage Error

An astronaut using a camera measures the radius of Earth as 4000 mi with an error of ± 80 mi. Let's use differentials to estimate the relative and percentage error of using this radius measurement to calculate the volume of Earth, assuming the planet is a perfect sphere.

Solution: If the measurement of the radius is accurate to within ± 80, we have

$$-80 \le dr \le 80.$$

Since the volume of a sphere is given by $V = (\frac{4}{3})\pi r^3$, we have

$$dV = 4\pi r^2 dr.$$

Using the measured radius of 4000 mi, we can estimate

$$-4\pi(4000)^2(80) \le dV \le 4\pi(4000)^2(80).$$

To estimate the relative error, consider $\dfrac{dV}{V}$. Since we do not know the exact value of the volume V, use the measured radius $r = 4000$ mi to estimate V. We obtain $V \approx (\frac{4}{3})\pi(4000)^3$. Therefore the relative error satisfies

$$\frac{-4\pi(4000)^2(80)}{4\pi(4000)^3/3} \le \frac{dV}{V} \le \frac{4\pi(4000)^2(80)}{4\pi(4000)^3/3},$$

which simplifies to

$$-0.06 \le \frac{dV}{V} \le 0.06.$$

The relative error is 0.06 and the percentage error is 6%.

Exercise 4.2.7

Determine the percentage error if the radius of Earth is measured to be 3950 mi with an error of ± 100 mi.

Hint

Use the fact that $dV = 4\pi r^2 dr$ to find dV/V.

Answer

7.6%

Key Concepts

- A differentiable function $y = f(x)$ can be approximated at a by the linear function

$$L(x) = f(a) + f'(a)(x - a).$$

- For a function $y = f(x)$, if x changes from a to $a + dx$, then

$$dy = f'(x)\,dx$$

is an approximation for the change in y. The actual change in y is

$$\Delta y = f(a + dx) - f(a).$$

- A measurement error dx can lead to an error in a calculated quantity $f(x)$. The error in the calculated quantity is known as the propagated error. The propagated error can be estimated by

$$dy \approx f'(x)\,dx.$$

- To estimate the relative error of a particular quantity q, we estimate $\frac{\Delta q}{q}$.

Key Equations

- **Linear approximation**

$L(x) = f(a) + f'(a)(x - a)$

- **A differential**

$$dy = f'(x)\,dx$$

Glossary

differential

the differential dx is an independent variable that can be assigned any nonzero real number; the differential dy is defined to be $dy = f'(x)\,dx$

differential form

given a differentiable function $y = f'(x)$, the equation $dy = f'(x)\,dx$ is the differential form of the derivative of y with respect to x

linear approximation

the linear function $L(x) = f(a) + f'(a)(x - a)$ is the linear approximation of f at $x = a$

percentage error

the relative error expressed as a percentage

propagated error

the error that results in a calculated quantity $f(x)$ resulting from a measurement error dx

relative error

given an absolute error Δq for a particular quantity, $\frac{\Delta q}{q}$ is the relative error.

tangent line approximation (linearization)

since the linear approximation of f at $x = a$ is defined using the equation of the tangent line, the linear approximation of f at $x = a$ is also known as the tangent line approximation to f at $x = a$

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4.2E: EXERCISES FOR SECTION 4.2

1) What is the linear approximation for any generic linear function $y = mx + b$?

2) Determine the necessary conditions such that the linear approximation function is constant. Use a graph to prove your result.

Answer

$f'(a) = 0$

3) Explain why the linear approximation becomes less accurate as you increase the distance between x and a. Use a graph to prove your argument.

4) When is the linear approximation exact?

Answer

The linear approximation exact when $y = f(x)$ is linear or constant.

In exercises 5 - 10, find the linear approximation $L(x)$ to $y = f(x)$ near $x = a$ for the function.

5) [T] $f(x) = x + x^4, \quad a = 0$

6) [T] $f(x) = \dfrac{1}{x}, \quad a = 2$

Answer

$L(x) = \frac{1}{2} - \frac{1}{4}(x - 2)$

7) [T] $f(x) = \tan x, \quad a = \frac{\pi}{4}$

8) [T] $f(x) = \sin x, \quad a = \frac{\pi}{2}$

Answer

$L(x) = 1$

9) [T] $f(x) = x \sin x, \quad a = 2\pi$

10) [T] $f(x) = \sin^2 x, \quad a = 0$

Answer

$L(x) = 0$

In exercises 11 - 16, compute the values given within 0.01 by deciding on the appropriate $f(x)$ and a, and evaluating $L(x) = f(a) + f'(a)(x - a)$. Check your answer using a calculator.

11) [T] $(2.001)^6$

12) [T] $\sin(0.02)$

Answer

$\sin(0.02) \approx 0.02$

13) [T] $\cos(0.03)$

14) [T] $(15.99)^{1/4}$

Answer

$(15.99)^{1/4} \approx 1.9996875$

15) [T] $\dfrac{1}{0.98}$

16) [T] $\sin(3.14)$

Answer

$\sin(3.14) \approx 0.001593$

In exercises 17 - 22, determine the appropriate $f(x)$ and a, and evaluate $L(x) = f(a) + f'(a)(x-a)$. Calculate the numerical error in the linear approximations that follow.

17) $(1.01)^3$

18) $\cos(0.01)$

Answer

$\cos(0.01) \approx L(0.01) = f(0) + f'(0)(0-0.01) = 1$; error, 0.00005

19) $(\sin(0.01))^2$

20) $(1.01)^{-3}$

Answer

$(1.01)^{-3} \approx L(1.01) = f(1) + f'(1)(1.01-1) = 0.97$; error, 0.0006

21) $\left(1+\frac{1}{10}\right)^{10}$

22) $\sqrt{8.99}$

Answer

$\sqrt{8.99} \approx L(8.99) = f(9) + f'(9)(8.99-9) = 3 - \frac{1}{600}$; error, 4.632×10^{-7}

In exercises 23 - 26, find the differential of the function.

23) $y = 3x^4 + x^2 - 2x + 1$

24) $y = x\cos x$

Answer

$dy = (\cos x - x\sin x)\,dx$

25) $y = \sqrt{1+x}$

26) $y = \dfrac{x^2+2}{x-1}$

Answer

$dy = \left(\dfrac{x^2-2x-2}{(x-1)^2}\right)dx$

In exercises 27 - 32, find the differential and evaluate for the given x and dx.

27) $y = 3x^2 - x + 6,\ x = 2,\ dx = 0.1$

28) $y = \dfrac{1}{x+1},\ x = 1,\ dx = 0.25$

Answer

$dy = -\dfrac{1}{(x+1)^2}dx, \quad dy = -\frac{1}{16}$

29) $y = \tan x,\ x = 0,\ dx = \frac{\pi}{10}$

30) $y = \dfrac{3x^2+2}{\sqrt{x+1}},\ x = 0,\ dx = 0.1$

Answer

$dy = \dfrac{9x^2+12x-2}{2(x+1)^{3/2}}dx, \quad dy = -0.1$

31) $y = \dfrac{\sin(2x)}{x},\ x = \pi,\ dx = 0.25$

32) $y = x^3 + 2x + \dfrac{1}{x},\ x = 1,\ dx = 0.05$

Answer

$dy = \left(3x^2 + 2 - \dfrac{1}{x^2}\right)dx, \quad dy = 0.2$

In exercises 33 - 38, find the change in volume dV or in surface area dA.

33) dV if the sides of a cube change from 10 to 10.1.

34) dA if the sides of a cube change from x to $x+dx$.

Answer

$dA = 12x\,dx$

35) dA if the radius of a sphere changes from r by dr.

36) dV if the radius of a sphere changes from r by dr.

Answer

$dV = 4\pi r^2 dr$

37) dV if a circular cylinder with $r = 2$ changes height from 3 cm to $3.05cm$.

38) dV if a circular cylinder of height 3 changes from $r = 2$ to $r = 1.9$ cm.

Answer

$dV = -1.2\pi\,\text{cm}^3$

In exercises 39 - 41, use differentials to estimate the maximum and relative error when computing the surface area or volume.

39) A spherical golf ball is measured to have a radius of 5 mm, with a possible measurement error of 0.1 mm. What is the possible change in volume?

40) A pool has a rectangular base of 10 ft by 20 ft and a depth of 6 ft. What is the change in volume if you only fill it up to 5.5 ft?

Answer

$-100\,\text{ft}^3$

41) An ice cream cone has height 4 in. and radius 1 in. If the cone is 0.1 in. thick, what is the difference between the volume of the cone, including the shell, and the volume of the ice cream you can fit inside the shell?

In exercises 42 - 44, confirm the approximations by using the linear approximation at $x = 0$.

42) $\sqrt{1-x} \approx 1 - \frac{1}{2}x$

43) $\dfrac{1}{\sqrt{1-x^2}} \approx 1$

44) $\sqrt{c^2 + x^2} \approx c$

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4.3: Maxima and Minima

Learning Objectives

- Define absolute extrema.
- Define local extrema.
- Explain how to find the critical points of a function over a closed interval.
- Describe how to use critical points to locate absolute extrema over a closed interval.

Given a particular function, we are often interested in determining the largest and smallest values of the function. This information is important in creating accurate graphs. Finding the maximum and minimum values of a function also has practical significance, because we can use this method to solve optimization problems, such as maximizing profit, minimizing the amount of material used in manufacturing an aluminum can, or finding the maximum height a rocket can reach. In this section, we look at how to use derivatives to find the largest and smallest values for a function.

Absolute Extrema

Consider the function $f(x) = x^2 + 1$ over the interval $(-\infty, \infty)$. As $x \to \pm\infty, f(x) \to \infty$. Therefore, the function does not have a largest value. However, since $x^2 + 1 \geq 1$ for all real numbers x and $x^2 + 1 = 1$ when $x = 0$, the function has a smallest value, 1, when $x = 0$. We say that 1 is the absolute minimum of $f(x) = x^2 + 1$ and it occurs at $x = 0$. We say that $f(x) = x^2 + 1$ does not have an absolute maximum (Figure 4.3.1).

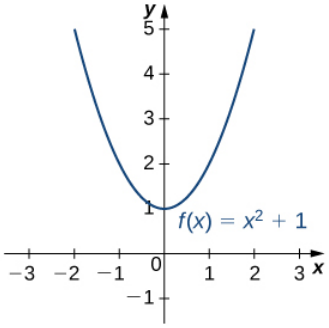


Figure 4.3.1: The given function has an absolute minimum of 1 at $x = 0$. The function does not have an absolute maximum.

Definition: Absolute Extrema

Let f be a function defined over an interval I and let $c \in I$. We say f has an *absolute maximum* on I at c if $f(c) \geq f(x)$ for all $x \in I$. We say f has an *absolute minimum* on I at c if $f(c) \leq f(x)$ for all $x \in I$. If f has an absolute maximum on I at c or an absolute minimum on I at c, we say f has *an absolute extremum* on I at c.

Before proceeding, let's note two important issues regarding this definition. First, the term absolute here does not refer to absolute value. An absolute extremum may be positive, negative, or zero. Second, if a function f has an absolute extremum over an interval I at c, the absolute extremum is $f(c)$. The real number c is a point in the domain at which the absolute extremum occurs. For example, consider the function $f(x) = 1/(x^2 + 1)$ over the interval $(-\infty, \infty)$. Since

$$f(0) = 1 \geq \frac{1}{x^2 + 1} = f(x)$$

for all real numbers x, we say f has an absolute maximum over $(-\infty, \infty)$ at $x = 0$. The absolute maximum is $f(0) = 1$. It occurs at $x = 0$, as shown in Figure 4.3.2(b).

A function may have both an absolute maximum and an absolute minimum, just one extremum, or neither. Figure 4.3.2 shows several functions and some of the different possibilities regarding absolute extrema. However, the following theorem, called the **Extreme Value Theorem**, guarantees that a continuous function f over a closed, bounded interval $[a, b]$ has both an absolute maximum and an absolute minimum.

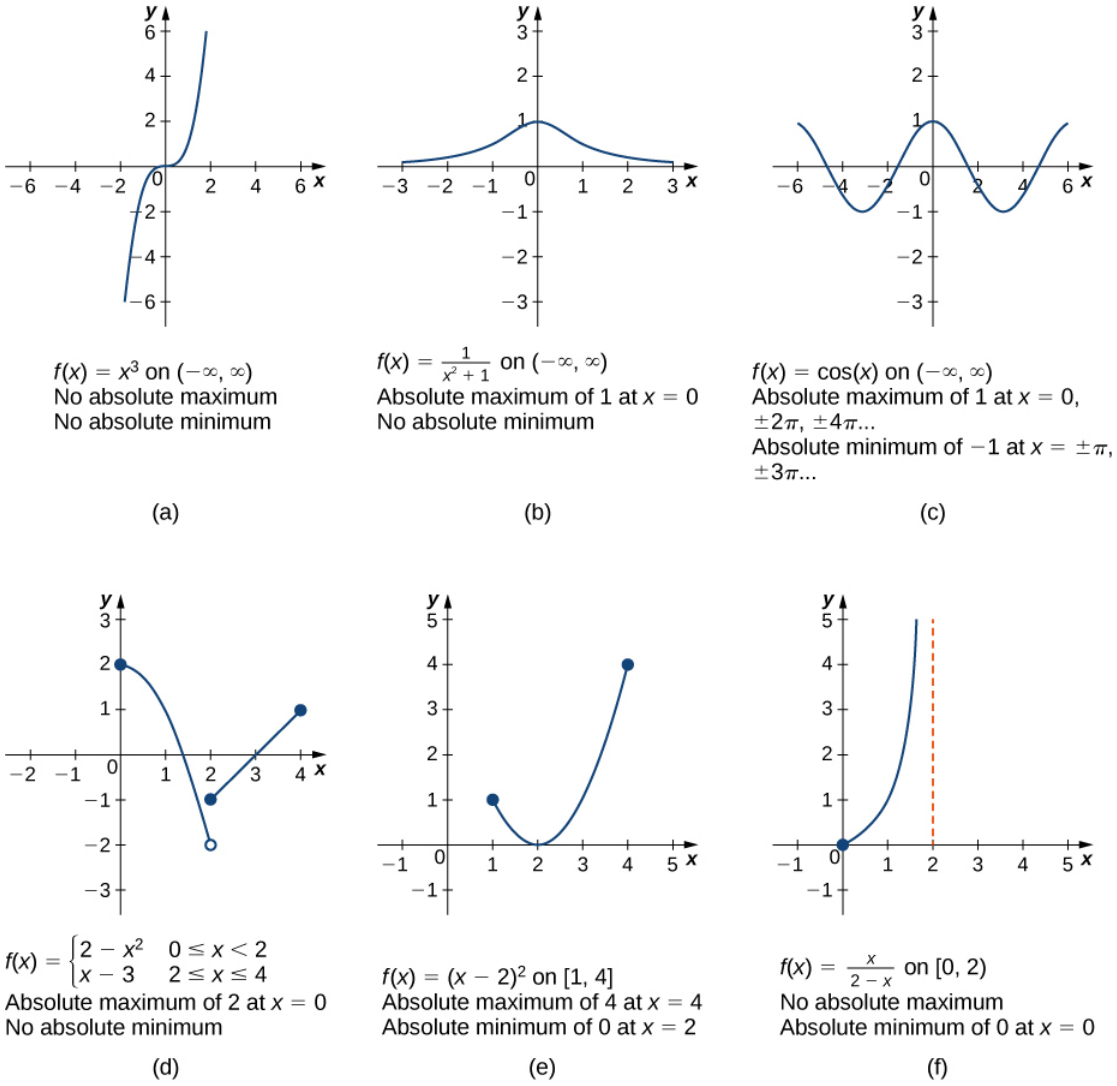


Figure 4.3.2: Graphs (a), (b), and (c) show several possibilities for absolute extrema for functions with a domain of $(-\infty, \infty)$. Graphs (d), (e), and (f) show several possibilities for absolute extrema for functions with a domain that is a bounded interval.

Theorem 4.3.1: Extreme Value Theorem

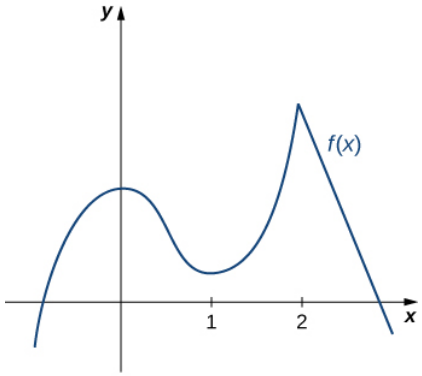
If f is a continuous function over the closed, bounded interval $[a, b]$, then there is a point in $[a, b]$ at which f has an absolute maximum over $[a, b]$ and there is a point in $[a, b]$ at which f has an absolute minimum over $[a, b]$.

The proof of the extreme value theorem is beyond the scope of this text. Typically, it is proved in a course on real analysis. There are a couple of key points to note about the statement of this theorem. For the extreme value theorem to apply, the function must be continuous over a closed, bounded interval. If the interval I is open or the function has even one point of discontinuity, the function may not have an absolute maximum or absolute minimum over I. For example, consider the functions shown in Figure 4.3.2 (d), (e), and (f). All three of these functions are defined over bounded intervals. However, the function in graph (e) is the only one that has both an absolute maximum and an absolute minimum over its domain. The extreme value theorem cannot be applied to the functions in graphs (d) and (f) because neither of these functions is continuous over a closed, bounded interval. Although the function in graph (d) is defined over the closed interval $[0, 4]$, the function is discontinuous at $x = 2$. The function has an absolute maximum over $[0, 4]$ but does not have an absolute minimum. The function in graph (f) is continuous over the half-open interval $[0, 2)$, but is not defined at $x = 2$, and therefore is not continuous over a closed, bounded interval. The function has an absolute minimum over $[0, 2)$, but does not have an absolute maximum over $[0, 2)$. These two graphs illustrate why a function over a bounded interval may fail to have an absolute maximum and/or absolute minimum.

Before looking at how to find absolute extrema, let's examine the related concept of local extrema. This idea is useful in determining where absolute extrema occur.

Local Extrema and Critical Points

Consider the function f shown in Figure 4.3.3. The graph can be described as two mountains with a valley in the middle. The absolute maximum value of the function occurs at the higher peak, at $x = 2$. However, $x = 0$ is also a point of interest. Although $f(0)$ is not the largest value of f, the value $f(0)$ is larger than $f(x)$ for all x near 0. We say f has a local maximum at $x = 0$. Similarly, the function f does not have an absolute minimum, but it does have a local minimum at $x = 1$ because $f(1)$ is less than $f(x)$ for x near 1.



f(*x*) defined on $(-\infty, \infty)$
Local maxima at $x = 0$ and $x = 2$
Local minimum at $x = 1$

Figure 4.3.3: This function f has two local maxima and one local minimum. The local maximum at $x = 2$ is also the absolute maximum.

Definition: Local Extrema

A function f has a *local maximum* at c if there exists an open interval I containing c such that I is contained in the domain of f and $f(c) \geq f(x)$ for all $x \in I$. A function f has a *local minimum* at c if there exists an open interval I containing c such that I is contained in the domain of f and $f(c) \leq f(x)$ for all $x \in I$. A function f has a *local extremum* at c if f has a local maximum at c or f has a local minimum at c.

Note that if f has an absolute extremum at c and f is defined over an interval containing c, then $f(c)$ is also considered a local extremum. If an absolute extremum for a function f occurs at an endpoint, we do not consider that to be a local extremum, but instead refer to that as an endpoint extremum.

Given the graph of a function f, it is sometimes easy to see where a local maximum or local minimum occurs. However, it is not always easy to see, since the interesting features on the graph of a function may not be visible because they occur at a very small scale. Also, we may not have a graph of the function. In these cases, how can we use a formula for a function to determine where these extrema occur?

To answer this question, let's look at Figure 4.3.3 again. The local extrema occur at $x = 0, x = 1$, and $x = 2$. Notice that at $x = 0$ and $x = 1$, the derivative $f'(x) = 0$. At $x = 2$, the derivative $f'(x)$ does not exist, since the function f has a corner there. In fact, if f has a local extremum at a point $x = c$, the derivative $f'(c)$ must satisfy one of the following conditions: either $f'(c) = 0$ or $f'(c)$ is undefined. Such a value c is known as a critical point and it is important in finding extreme values for functions.

Definition: Critical Points

Let c be an interior point in the domain of f. We say that c is a *critical point* of f if $f'(c) = 0$ or $f'(c)$ is undefined.

As mentioned earlier, if f has a local extremum at a point $x = c$, then c must be a critical point of f. This fact is known as **Fermat's theorem.**

Theorem 4.3.2: Fermat's Theorem

If f has a local extremum at c and f is differentiable at c, then $f'(c) = 0$.

Proof

Suppose f has a local extremum at c and f is differentiable at c. We need to show that $f'(c) = 0$. To do this, we will show that $f'(c) \geq 0$ and $f'(c) \leq 0$, and therefore $f'(c) = 0$. Since f has a local extremum at c, f has a local maximum or local minimum at c. Suppose f has a local maximum at c. The case in which f has a local minimum at c can be handled similarly. There then exists an open interval I such that $f(c) \geq f(x)$ for all $x \in I$. Since f is differentiable at c, from the definition of the derivative, we know that

$$f'(c) = \lim_{x \to c} \frac{f(x) - f(c)}{x - c}.$$

Since this limit exists, both one-sided limits also exist and equal $f'(c)$. Therefore,

$$f'(c) = \lim_{x \to c^+} \frac{f(x) - f(c)}{x - c,} \tag{4.3.1}$$

and

$$f'(c) = \lim_{x \to c^-} \frac{f(x) - f(c)}{x - c}.$$

Since $f(c)$ is a local maximum, we see that $f(x) - f(c) \leq 0$ for x near c. Therefore, for x near c, but $x > c$, we have $\frac{f(x)-f(c)}{x-c} \leq 0$. From Equation 4.3.1 we conclude that $f'(c) \leq 0$. Similarly, it can be shown that $f'(c) \geq 0$. Therefore, $f'(c) = 0$.

□

From Fermat's theorem, we conclude that if f has a local extremum at c, then either $f'(c) = 0$ or $f'(c)$ is undefined. In other words, local extrema can only occur at critical points.

Note this theorem does not claim that a function f must have a local extremum at a critical point. Rather, it states that critical points are candidates for local extrema. For example, consider the function $f(x) = x^3$. We have $f'(x) = 3x^2 = 0$ when $x = 0$. Therefore, $x = 0$ is a critical point. However, $f(x) = x^3$ is increasing over $(-\infty, \infty)$, and thus f does not have a local extremum at $x = 0$. In Figure 4.3.4, we see several different possibilities for critical points. In some of these cases, the functions have local extrema at critical points, whereas in other cases the functions do not. Note that these graphs do not show all possibilities for the behavior of a function at a critical point.

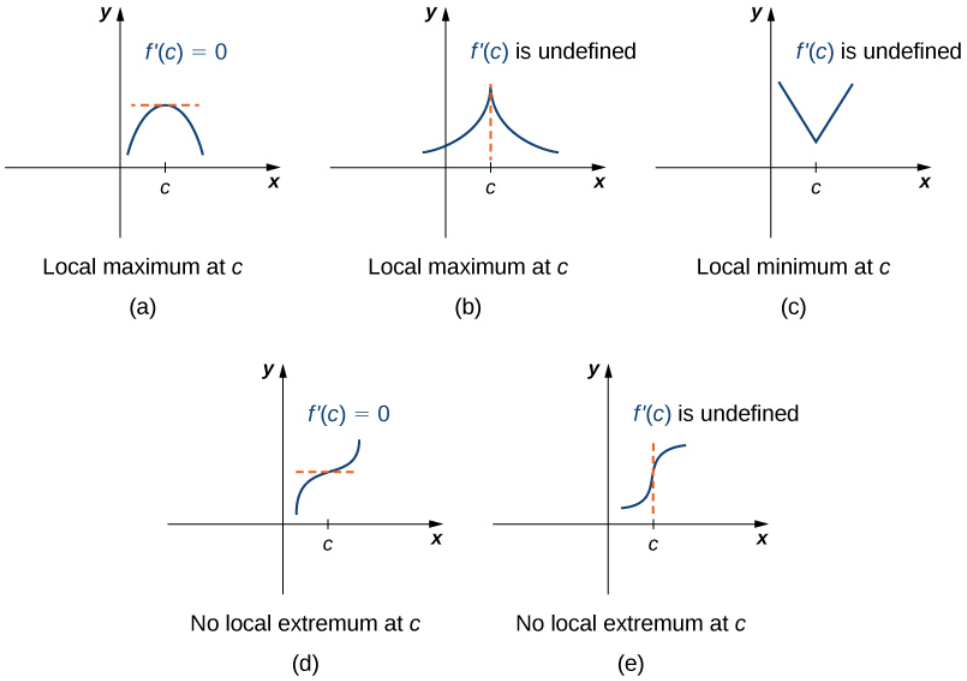


Figure 4.3.4: (a–e) A function f has a critical point at c if $f'(c) = 0$ or $f'(c)$ is undefined. A function may or may not have a local extremum at a critical point.

Later in this chapter we look at analytical methods for determining whether a function actually has a local extremum at a critical point. For now, let's turn our attention to finding critical points. We will use graphical observations to determine whether a critical point is associated with a local extremum.

Example 4.3.1: Locating Critical Points

For each of the following functions, find all critical points. Use a graphing utility to determine whether the function has a local extremum at each of the critical points.

a. $f(x) = \frac{1}{3}x^3 - \frac{5}{2}x^2 + 4x$
b. $f(x) = (x^2 - 1)^3$
c. $f(x) = \frac{4x}{1+x^2}$

Solution

a. The derivative $f'(x) = x^2 - 5x + 4$ is defined for all real numbers x. Therefore, we only need to find the values for x where $f'(x) = 0$. Since $f'(x) = x^2 - 5x + 4 = (x - 4)(x - 1)$, the critical points are $x = 1$ and $x = 4$. From the graph of f in Figure 4.3.5, we see that f has a local maximum at $x = 1$ and a local minimum at $x = 4$.

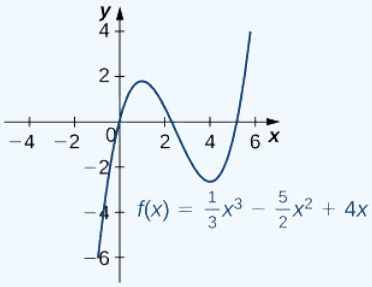


Figure 4.3.5: This function has a local maximum and a local minimum.

b. Using the chain rule, we see the derivative is

$$f'(x) = 3(x^2 - 1)^2(2x) = 6x(x^2 - 1)^2.$$

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Therefore, f has critical points when $x = 0$ and when $x^2 - 1 = 0$. We conclude that the critical points are $x = 0, \pm 1$. From the graph of f in Figure 4.3.6, we see that f has a local (and absolute) minimum at $x = 0$, but does not have a local extremum at $x = 1$ or $x = -1$.

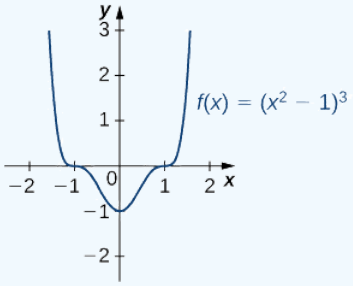


Figure 4.3.6: This function has three critical points: $x = 0$, $x = 1$, and $x = -1$. The function has a local (and absolute) minimum at $x = 0$, but does not have extrema at the other two critical points.

c. By the quotient rule, we see that the derivative is

$$f'(x) = \frac{4(1+x^2) - 4x(2x)}{(1+x^2)^2} = \frac{4-4x^2}{(1+x^2)^2}.$$

The derivative is defined everywhere. Therefore, we only need to find values for x where $f'(x) = 0$. Solving $f'(x) = 0$, we see that $4 - 4x^2 = 0$, which implies $x = \pm 1$. Therefore, the critical points are $x = \pm 1$. From the graph of f in Figure 4.3.7, we see that f has an absolute maximum at $x = 1$ and an absolute minimum at $x = -1$. Hence, f has a local maximum at $x = 1$ and a local minimum at $x = -1$. (Note that if f has an absolute extremum over an interval I at a point c that is not an endpoint of I, then f has a local extremum at c.)

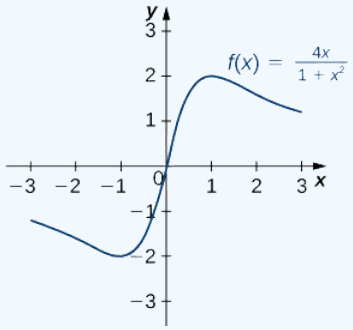


Figure 4.3.7: This function has an absolute maximum and an absolute minimum.

? Exercise 4.3.1

Find all critical points for $f(x) = x^3 - \frac{1}{2}x^2 - 2x + 1$.

Hint

Calculate $f'(x)$.

Answer

$x = \frac{-2}{3}, x = 1$

Locating Absolute Extrema

The extreme value theorem states that a continuous function over a closed, bounded interval has an absolute maximum and an absolute minimum. As shown in Figure 4.3.2, one or both of these absolute extrema could occur at an endpoint. If an absolute extremum does not occur at an endpoint, however, it must occur at an interior point, in which case the absolute extremum is a local extremum. Therefore, by Fermat's Theorem, the point c at which the local extremum occurs must be a critical point. We summarize this result in the following theorem.

Theorem 4.3.3: Location of Absolute Extrema

Let f be a continuous function over a closed, bounded interval I. The absolute maximum of f over I and the absolute minimum of f over I must occur at endpoints of I or at critical points of f in I.

With this idea in mind, let's examine a procedure for locating absolute extrema.

Problem-Solving Strategy: Locating Absolute Extrema over a Closed Interval

Consider a continuous function f defined over the closed interval $[a, b]$.

1. Evaluate f at the endpoints $x = a$ and $x = b$.
2. Find all critical points of f that lie over the interval (a, b) and evaluate f at those critical points.
3. Compare all values found in (1) and (2). From "Location of Absolute Extrema," the absolute extrema must occur at endpoints or critical points. Therefore, the largest of these values is the absolute maximum of f. The smallest of these values is the absolute minimum of f.

Now let's look at how to use this strategy to find the absolute maximum and absolute minimum values for continuous functions.

Example 4.3.2: Locating Absolute Extrema

For each of the following functions, find the absolute maximum and absolute minimum over the specified interval and state where those values occur.

a. $f(x) = -x^2 + 3x - 2$ over $[1, 3]$.
b. $f(x) = x^2 - 3x^{2/3}$ over $[0, 2]$.

Solution

a. Step 1. Evaluate f at the endpoints $x = 1$ and $x = 3$.

$$f(1) = 0 \text{ and } f(3) = -2$$

Step 2. Since $f'(x) = -2x + 3$, f' is defined for all real numbers x. Therefore, there are no critical points where the derivative is undefined. It remains to check where $f'(x) = 0$. Since $f'(x) = -2x + 3 = 0$ at $x = \frac{3}{2}$ and $\frac{3}{2}$ is in the interval $[1, 3]$, $f(\frac{3}{2})$ is a candidate for an absolute extremum of f over $[1, 3]$. We evaluate $f(\frac{3}{2})$ and find

$$f\left(\tfrac{3}{2}\right) = \tfrac{1}{4}.$$

Step 3. We set up the following table to compare the values found in steps 1 and 2.

x	$f(x)$	Conclusion
1	0	
$\frac{3}{2}$	$\frac{1}{4}$	Absolute maximum
3	-2	Absolute minimum

From the table, we find that the absolute maximum of f over the interval [1, 3] is $\frac{1}{4}$, and it occurs at $x = \frac{3}{2}$. The absolute minimum of f over the interval $[1, 3]$ is -2, and it occurs at $x = 3$ as shown in Figure 4.3.8.

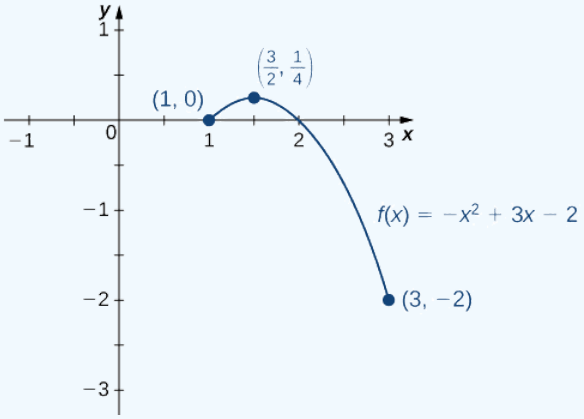


Figure 4.3.8: This function has both an absolute maximum and an absolute minimum.

b. Step 1. Evaluate f at the endpoints $x = 0$ and $x = 2$.

$$f(0) = 0 \text{ and } f(2) = 4 - 3\,(2)^{2/3} \approx -0.762$$

Step 2. The derivative of f is given by

$$f'(x) = 2x - \tfrac{2}{x^{1/3}} = \frac{2x^{4/3} - 2}{x^{1/3}}$$

for $x \neq 0$. The derivative is zero when $2x^{4/3} - 2 = 0$, which implies $x = \pm 1$. The derivative is undefined at $x = 0$. Therefore, the critical points of f are $x = 0, 1, -1$. The point $x = 0$ is an endpoint, so we already evaluated $f(0)$ in step 1. The point $x = -1$ is not in the interval of interest, so we need only evaluate $f(1)$. We find that

$$f(1) = -2.$$

Step 3. We compare the values found in steps 1 and 2, in the following table.

x	$f(x)$	Conclusion
0	0	Absolute maximum
1	-2	Absolute minimum
2	-0.762	

We conclude that the absolute maximum of f over the interval $[0, 2]$ is zero, and it occurs at $x = 0$. The absolute minimum is -2, and it occurs at $x = 1$ as shown in Figure 4.3.9.

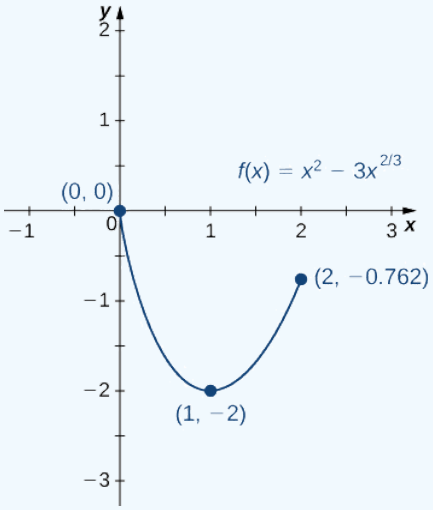


Figure 4.3.9: This function has an absolute maximum at an endpoint of the interval.

Exercise 4.3.2

Find the absolute maximum and absolute minimum of $f(x) = x^2 - 4x + 3$ over the interval $[1, 4]$.

Hint

Look for critical points. Evaluate f at all critical points and at the endpoints.

Answer

The absolute maximum is 3 and it occurs at $x = 4$. The absolute minimum is -1 and it occurs at $x = 2$.

At this point, we know how to locate absolute extrema for continuous functions over closed intervals. We have also defined local extrema and determined that if a function f has a local extremum at a point c, then c must be a critical point of f. However, c being a critical point is not a sufficient condition for f to have a local extremum at c. Later in this chapter, we show how to determine whether a function actually has a local extremum at a critical point. First, however, we need to introduce the Mean Value Theorem, which will help as we analyze the behavior of the graph of a function.

Key Concepts

- A function may have both an absolute maximum and an absolute minimum, have just one absolute extremum, or have no absolute maximum or absolute minimum.
- If a function has a local extremum, the point at which it occurs must be a critical point. However, a function need not have a local extremum at a critical point.
- A continuous function over a closed, bounded interval has an absolute maximum and an absolute minimum. Each extremum occurs at a critical point or an endpoint.

Glossary

absolute extremum

if f has an absolute maximum or absolute minimum at c, we say f has an absolute extremum at c

absolute maximum

if $f(c) \geq f(x)$ for all x in the domain of f, we say f has an absolute maximum at c

absolute minimum

if $f(c) \leq f(x)$ for all x in the domain of f, we say f has an absolute minimum at c

critical point

if $f'(c) = 0$ or $f'(c)$ is undefined, we say that c is a critical point of f

extreme value theorem

if f is a continuous function over a finite, closed interval, then f has an absolute maximum and an absolute minimum

Fermat's theorem

if f has a local extremum at c, then c is a critical point of f

local extremum

if f has a local maximum or local minimum at c, we say f has a local extremum at c

local maximum

if there exists an interval I such that $f(c) \geq f(x)$ for all $x \in I$, we say f has a local maximum at c

local minimum

if there exists an interval I such that $f(c) \leq f(x)$ for all $x \in I$, we say f has a local minimum at c

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4.3E: EXERCISES FOR SECTION 4.3

1) In precalculus, you learned a formula for the position of the maximum or minimum of a quadratic equation $y = ax^2 + bx + c$, which was $m = -\frac{b}{2a}$. Prove this formula using calculus.

2) If you are finding an absolute minimum over an interval $[a, b]$, why do you need to check the endpoints? Draw a graph that supports your hypothesis.

Answer

On a closed interval, the endpoints often lie above or below any local (relative) extrema. Answers may vary for the graph.

3) If you are examining a function over an interval (a, b), for a and b finite, is it possible not to have an absolute maximum or absolute minimum?

4) When you are checking for critical points to locate the extrema of a function f, explain why you also need to determine points where $f'(x)$ is undefined. Draw a graph to support your explanation.

Answer

Points on the graph of f where there is a corner, a cusp, or a jump discontinuity or removable discontinuity can easily be absolute (or local) extrema of the function. Answers may vary for the graph.

5) Can you have a finite absolute maximum for $y = ax^2 + bx + c$ over $(-\infty, \infty)$? Explain why or why not using graphical arguments.

6) Can you have a finite absolute maximum for $y = ax^3 + bx^2 + cx + d$ over $(-\infty, \infty)$ assuming a is non-zero? Explain why or why not using graphical arguments.

Answer

No; answers will vary

7) Let m be the number of local minima and M be the number of local maxima. Can you create a function where $M > m + 2$? Draw a graph to support your explanation.

8) Is it possible to have more than one absolute maximum? Use a graphical argument to prove your hypothesis.

Answer

Since the absolute maximum is the function (output) value rather than the x value, the answer is no; answers will vary

9) Is it possible to have no absolute minimum or maximum for a function? If so, construct such a function. If not, explain why this is not possible.

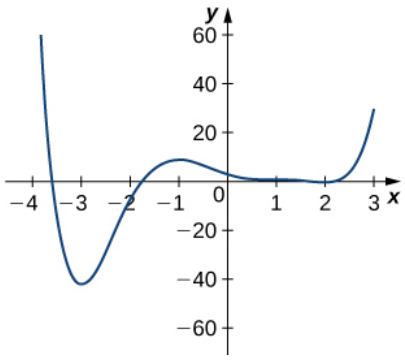
10) [T] Graph the function $y = e^{ax}$. For which values of a, on any infinite domain, will you have an absolute minimum and absolute maximum?

Answer

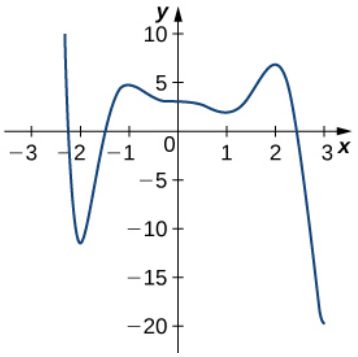
When $a = 0$

In exercises 11 - 14, determine where the local and absolute maxima and minima occur on the graph given. Assume domains are closed intervals unless otherwise specified.

11)



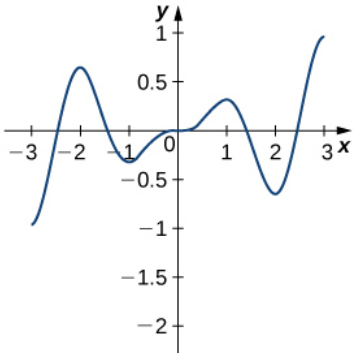
12)



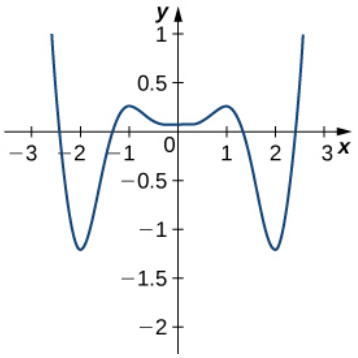
Answer

Absolute minimum at 3; Absolute maximum at −2.2; local minima at −2, 1; local maima at −1, 2

13)



14)



Answer

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Absolute minima at −2, 2; absolute maxima at −2.5, 2.5; local minimum at 0; local maxima at −1, 1

For exercises 15 - 18, draw graphs of $f(x)$, which is continuous, over the interval $[-4,4]$ with the following properties:

15) Absolute maximum at $x=2$ and absolute minima at $x=\pm 3$

16) Absolute minimum at $x=1$ and absolute maximum at $x=2$

Answer

Answers may vary.

17) Absolute maximum at $x=4$, absolute minimum at $x=-1$, local maximum at $x=-2$, and a critical point that is not a maximum or minimum at $x=2$

18) Absolute maxima at $x=2$ and $x=-3$, local minimum at $x=1$, and absolute minimum at $x=4$

Answer

Answers may vary.

In exercises 19 - 28, find the critical points in the domains of the given functions.

19) $y=4x^3-3x$

20) $y=4\sqrt{x}-x^2$

Answer

$x=1$

21) $y=\dfrac{1}{x-1}$

22) $y=\ln(x-2)$

Answer

None

23) $y=\tan(x)$

24) $y=\sqrt{4-x^2}$

Answer

$x=0$

25) $y=x^{3/2}-3x^{5/2}$

26) $y=\dfrac{x^2-1}{x^2+2x-3}$

Answer

None

27) $y=\sin^2(x)$

28) $y=x+\dfrac{1}{x}$

Answer

$x=-1$ and $x=1$

In exercises 29 - 39, find the local and/or absolute maxima for the functions over the specified domain.

29) $f(x)=x^2+3$ over $[-1,4]$

30) $y=x^2+\dfrac{2}{x}$ over $[1,4]$

Answer

Absolute maximum: $x=4, y=\frac{33}{2}$; absolute minimum: $x=1, y=3$

31) $y = (x - x^2)^2$ over $[-1, 1]$

32) $y = \dfrac{1}{x - x^2}$ over $[0, 1]$

Answer

Absolute minimum: $x = \frac{1}{2}, y = 4$

33) $y = \sqrt{9 - x}$ over $[1, 9]$

34) $y = x + \sin(x)$ over $[0, 2\pi]$

Answer

Absolute maximum: $x = 2\pi, y = 2\pi$; absolute minimum: $x = 0, y = 0$

35) $y = \dfrac{x}{1 + x}$ over $[0, 100]$

36) $y = |x + 1| + |x - 1|$ over $[-3, 2]$

Answer

Absolute maximum: $x = -3, y = 6$; absolute minimum: $-1 \le x \le 1, y = 2$

37) $y = \sqrt{x} - \sqrt{x^3}$ over $[0, 4]$

38) $y = \sin x + \cos x$ over $[0, 2\pi]$

Answer

Absolute maximum: $x = \frac{\pi}{4}, y = \sqrt{2}$; absolute minimum: $x = \frac{5\pi}{4}, y = -\sqrt{2}$

39) $y = 4 \sin \theta - 3 \cos \theta$ over $[0, 2\pi]$

In exercises 40 - 45, find the local and absolute minima and maxima for the functions over $(-\infty, \infty)$.

40) $y = x^2 + 4x + 5$

Answer

Absolute minimum: $x = -2, y = 1$

41) $y = x^3 - 12x$

42) $y = 3x^4 + 8x^3 - 18x^2$

Answer

Absolute minimum: $x = -3, y = -135$; local maximum: $x = 0, y = 0$; local minimum: $x = 1, y = -7$

43) $y = x^3(1 - x)^6$

44) $y = \dfrac{x^2 + x + 6}{x - 1}$

Answer

Local maximum: $x = 1 - 2\sqrt{2}, y = 3 - 4\sqrt{2}$; local minimum: $x = 1 + 2\sqrt{2}, y = 3 + 4\sqrt{2}$

45) $y = \dfrac{x^2 - 1}{x - 1}$

In exercises 46 - 50, use a calculator to graph the function and to estimate the absolute and local maxima and minima. Then, solve for them explicitly.

46) [T] $y = 3x\sqrt{1 - x^2}$

Answer

Absolute maximum: $x = \frac{\sqrt{2}}{2}, y = \frac{3}{2}$; absolute minimum: $x = -\frac{\sqrt{2}}{2}, y = -\frac{3}{2}$

47) [T] $y = x + \sin(x)$

48) [T] $y = 12x^5 + 45x^4 + 20x^3 - 90x^2 - 120x + 3$

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Answer

Local maximum: $x = -2, y = 59$; local minimum: $x = 1, y = -130$

49) [T] $y = \dfrac{x^3 + 6x^2 - x - 30}{x - 2}$

50) [T] $y = \dfrac{\sqrt{4 - x^2}}{\sqrt{4 + x^2}}$

Answer

Absolute maximum: $x = 0, y = 1$; absolute minimum: $x = -2, 2, y = 0$

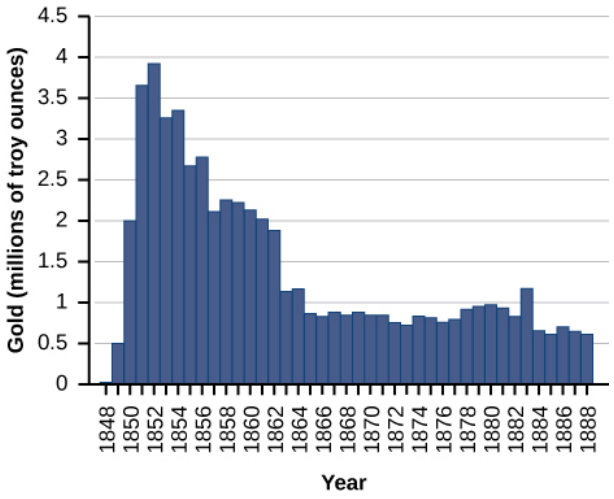
51) A company that produces cell phones has a cost function of $C = x^2 - 1200x + 36,400$, where C is cost in dollars and x is number of cell phones produced (in thousands). How many units of cell phone (in thousands) minimizes this cost function?

52) A ball is thrown into the air and its position is given by $h(t) = -4.9t^2 + 60t + 5m$. Find the height at which the ball stops ascending. How long after it is thrown does this happen?

Answer

$h = \frac{9245}{49}$ m, $t = \frac{300}{49}$ s

For exercises 53-54, consider the production of gold during the California gold rush (1848–1888). The production of gold can be modeled by $G(t) = \dfrac{(25t)}{(t^2 + 16)}$, where t is the number of years since the rush began $(0 \le t \le 40)$ and G is ounces of gold produced (in millions). A summary of the data is shown in the following figure.



53) Find when the maximum (local and global) gold production occurred, and the amount of gold produced during that maximum.

54) Find when the minimum (local and global) gold production occurred. What was the amount of gold produced during this minimum?

Answer

The global minimum was in 1848, when no gold was produced.

In exercises 55 & 56, find the critical points, maxima, and minima for the given piecewise functions.

55) $y = \begin{cases} x^2 - 4x, & \text{if } 0 \le x \le 1 \\ x^2 - 4, & \text{if } 1 < x \le 2 \end{cases}$

56) $y = \begin{cases} x^2 + 1, & \text{if } x \le 1 \\ x^2 - 4x + 5, & \text{if } x > 1 \end{cases}$

Answer

Absolute minima: $x = 0, x = 2, y = 1$; local maximum at $x = 1, y = 2$

In exercises 57 - 58, find the critical points of the following generic functions. Are they maxima, minima, or neither? State the necessary conditions.

57) $y = ax^2 + bx + c$, given that $a > 0$

58) $y = (x-1)^a$, given that $a > 1$

Answer

No maxima/minima if a is odd, minimum at $x = 1$ if a is even

- Gilbert Strang (MIT) and Edwin “Jed” Herman (Harvey Mudd) with many contributing authors. This content by OpenStax is licensed with a CC-BY-SA-NC 4.0 license. Download for free at http://cnx.org.
- Paul Seeburger (Monroe Community College) added answers for exercises 2 and 4.

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4.4: The Mean Value Theorem

Learning Objectives

- Explain the meaning of Rolle's theorem.
- Describe the significance of the Mean Value Theorem.
- State three important consequences of the Mean Value Theorem.

The *Mean Value Theorem* is one of the most important theorems in calculus. We look at some of its implications at the end of this section. First, let's start with a special case of the Mean Value Theorem, called Rolle's theorem.

Rolle's Theorem

Informally, **Rolle's theorem** states that if the outputs of a differentiable function f are equal at the endpoints of an interval, then there must be an interior point c where $f'(c) = 0$. Figure 4.4.1 illustrates this theorem.

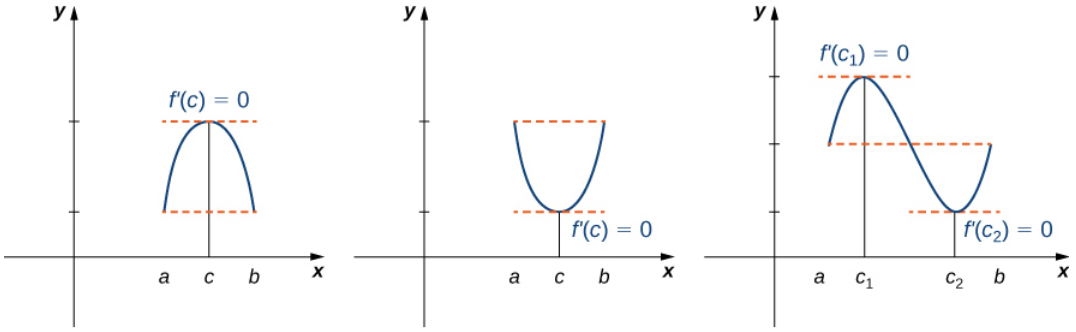


Figure 4.4.1: If a differentiable function f satisfies $f(a) = f(b)$, then its derivative must be zero at some point(s) between a and b.

Rolle's Theorem

Let f be a continuous function over the closed interval $[a, b]$ and differentiable over the open interval (a, b) such that $f(a) = f(b)$. There then exists at least one $c \in (a, b)$ such that $f'(c) = 0$.

Proof

Let $k = f(a) = f(b)$. We consider three cases:

1. $f(x) = k$ for all $x \in (a, b)$.
2. There exists $x \in (a, b)$ such that $f(x) > k$.
3. There exists $x \in (a, b)$ such that $f(x) < k$.

Case 1: If $f(x) = k$ for all $x \in (a, b)$, then $f'(x) = 0$ for all $x \in (a, b)$.

Case 2: Since f is a continuous function over the closed, bounded interval $[a, b]$, by the extreme value theorem, it has an absolute maximum. Also, since there is a point $x \in (a, b)$ such that $f(x) > k$, the absolute maximum is greater than k. Therefore, the absolute maximum does not occur at either endpoint. As a result, the absolute maximum must occur at an interior point $c \in (a, b)$. Because f has a maximum at an interior point c, and f is differentiable at c, by Fermat's theorem, $f'(c) = 0$.

Case 3: The case when there exists a point $x \in (a, b)$ such that $f(x) < k$ is analogous to case 2, with maximum replaced by minimum.

□

An important point about Rolle's theorem is that the differentiability of the function f is critical. If f is not differentiable, even at a single point, the result may not hold. For example, the function $f(x) = |x| - 1$ is continuous over $[-1, 1]$ and $f(-1) = 0 = f(1)$, but $f'(c) \neq 0$ for any $c \in (-1, 1)$ as shown in the following figure.

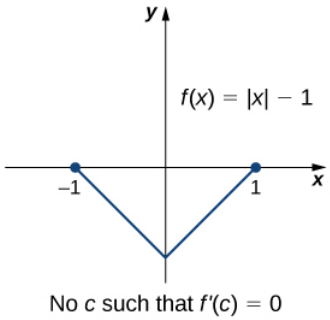


Figure 4.4.2: Since $f(x) = |x| - 1$ is not differentiable at $x = 0$, the conditions of Rolle's theorem are not satisfied. In fact, the conclusion does not hold here; there is no $c \in (-1, 1)$ such that $f'(c) = 0$.

Let's now consider functions that satisfy the conditions of Rolle's theorem and calculate explicitly the points c where $f'(c) = 0$.

Example 4.4.1: Using Rolle's Theorem

For each of the following functions, verify that the function satisfies the criteria stated in Rolle's theorem and find all values c in the given interval where $f'(c) = 0$.

a. $f(x) = x^2 + 2x$ over $[-2, 0]$
b. $f(x) = x^3 - 4x$ over $[-2, 2]$

Solution

a. Since f is a polynomial, it is continuous and differentiable everywhere. In addition, $f(-2) = 0 = f(0)$. Therefore, f satisfies the criteria of Rolle's theorem. We conclude that there exists at least one value $c \in (-2, 0)$ such that $f'(c) = 0$. Since $f'(x) = 2x + 2 = 2(x + 1)$, we see that $f'(c) = 2(c + 1) = 0$ implies $c = -1$ as shown in the following graph.

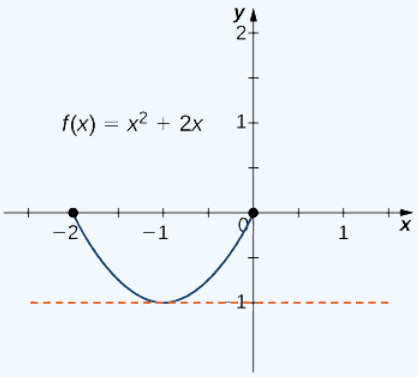


Figure 4.4.3: This function is continuous and differentiable over [−2,0], $f'(c) = 0$ when $c = -1$.

b. As in part a. f is a polynomial and therefore is continuous and differentiable everywhere. Also, $f(-2) = 0 = f(2)$. That said, f satisfies the criteria of Rolle's theorem. Differentiating, we find that $f'(x) = 3x^2 - 4$. Therefore, $f'(c) = 0$ when $x = \pm\frac{2}{\sqrt{3}}$. Both points are in the interval $[-2, 2]$, and, therefore, both points satisfy the conclusion of Rolle's theorem as shown in the following graph.

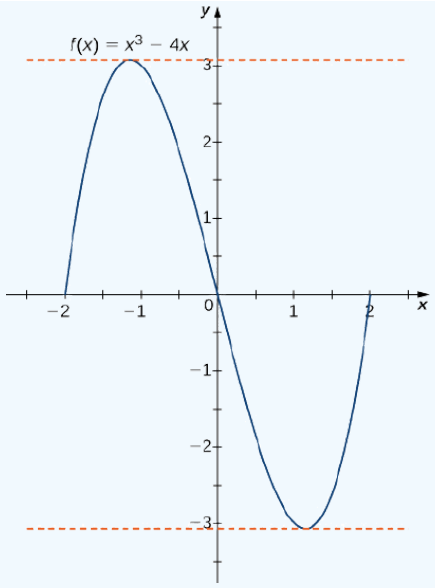


Figure 4.4.4: For this polynomial over $[-2, 2]$, $f'(c) = 0$ at $x = \pm 2/\sqrt{3}$.

Exercise 4.4.1

Verify that the function $f(x) = 2x^2 - 8x + 6$ defined over the interval $[1, 3]$ satisfies the conditions of Rolle's theorem. Find all points c guaranteed by Rolle's theorem.

Hint

Find all values c, where $f'(c) = 0$.

Answer

$c = 2$

The Mean Value Theorem and Its Meaning

Rolle's theorem is a special case of the Mean Value Theorem. In Rolle's theorem, we consider differentiable functions f that are zero at the endpoints. The Mean Value Theorem generalizes Rolle's theorem by considering functions that are not necessarily zero at the endpoints. Consequently, we can view the Mean Value Theorem as a slanted version of Rolle's theorem (Figure 4.4.5). The Mean Value Theorem states that if f is continuous over the closed interval $[a, b]$ and differentiable over the open interval (a, b), then there exists a point $c \in (a, b)$ such that the tangent line to the graph of f at c is parallel to the secant line connecting $(a, f(a))$ and $(b, f(b))$.

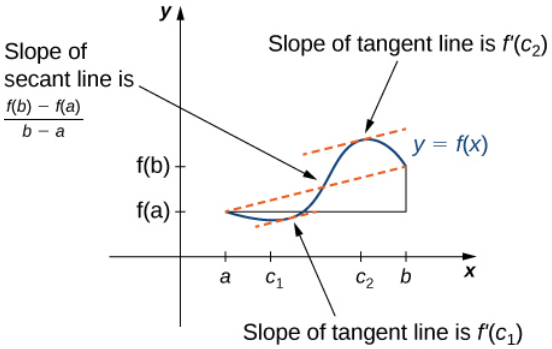


Figure 4.4.5: The Mean Value Theorem says that for a function that meets its conditions, at some point the tangent line has the same slope as the secant line between the ends. For this function, there are two values c_1 and c_2 such that the tangent line to f at c_1 and c_2 has the same slope as the secant line.

Mean Value Theorem

Let f be continuous over the closed interval $[a, b]$ and differentiable over the open interval (a, b). Then, there exists at least one point $c \in (a, b)$ such that

$$f'(c) = \frac{f(b) - f(a)}{b - a}$$

Proof

The proof follows from Rolle's theorem by introducing an appropriate function that satisfies the criteria of Rolle's theorem. Consider the line connecting $(a, f(a))$ and $(b, f(b))$. Since the slope of that line is

$$\frac{f(b) - f(a)}{b - a}$$

and the line passes through the point $(a, f(a))$, the equation of that line can be written as

$$y = \frac{f(b) - f(a)}{b - a}(x - a) + f(a).$$

Let $g(x)$ denote the vertical difference between the point $(x, f(x))$ and the point (x, y) on that line. Therefore,

$$g(x) = f(x) - \left[\frac{f(b) - f(a)}{b - a}(x - a) + f(a)\right].$$

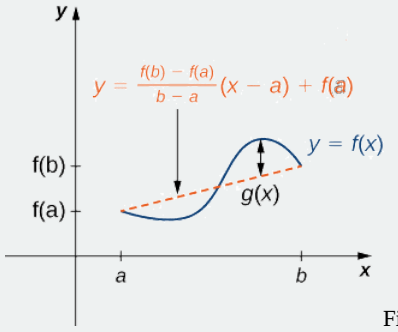


Figure 4.4.6: The value $g(x)$ is the vertical difference between the point $(x, f(x))$ and the point (x, y) on the secant line connecting $(a, f(a))$ and $(b, f(b))$.

Since the graph of f intersects the secant line when $x = a$ and $x = b$, we see that $g(a) = 0 = g(b)$. Since f is a differentiable function over (a, b), g is also a differentiable function over (a, b). Furthermore, since f is continuous over $[a, b]$, g is also continuous over $[a, b]$. Therefore, g satisfies the criteria of Rolle's theorem. Consequently, there exists a point $c \in (a, b)$ such that $g'(c) = 0$. Since

$$g'(x) = f'(x) - \frac{f(b) - f(a)}{b - a},$$

we see that

$$g'(c) = f'(c) - \frac{f(b) - f(a)}{b - a}.$$

Since $g'(c) = 0$, we conclude that

$$f'(c) = \frac{f(b) - f(a)}{b - a}.$$

□

In the next example, we show how the Mean Value Theorem can be applied to the function $f(x) = \sqrt{x}$ over the interval $[0, 9]$. The method is the same for other functions, although sometimes with more interesting consequences.

Example 4.4.2: Verifying that the Mean Value Theorem Applies

For $f(x) = \sqrt{x}$ over the interval $[0, 9]$, show that f satisfies the hypothesis of the Mean Value Theorem, and therefore there exists at least one value $c \in (0, 9)$ such that $f'(c)$ is equal to the slope of the line connecting $(0, f(0))$ and $(9, f(9))$. Find these values c guaranteed by the Mean Value Theorem.

Solution

We know that $f(x) = \sqrt{x}$ is continuous over $[0, 9]$ and differentiable over $(0, 9)$. Therefore, f satisfies the hypotheses of the Mean Value Theorem, and there must exist at least one value $c \in (0, 9)$ such that $f'(c)$ is equal to the slope of the line connecting $(0, f(0))$ and $(9, f(9))$ (Figure 4.4.7). To determine which value(s) of c are guaranteed, first calculate the derivative of f. The derivative $f'(x) = \frac{1}{(2\sqrt{x})}$. The slope of the line connecting $(0, f(0))$ and $(9, f(9))$ is given by

$$\frac{f(9) - f(0)}{9 - 0} = \frac{\sqrt{9} - \sqrt{0}}{9 - 0} = \frac{3}{9} = \frac{1}{3}.$$

We want to find c such that $f'(c) = \frac{1}{3}$. That is, we want to find c such that

$$\frac{1}{2\sqrt{c}} = \frac{1}{3}.$$

Solving this equation for c, we obtain $c = \frac{9}{4}$. At this point, the slope of the tangent line equals the slope of the line joining the endpoints.

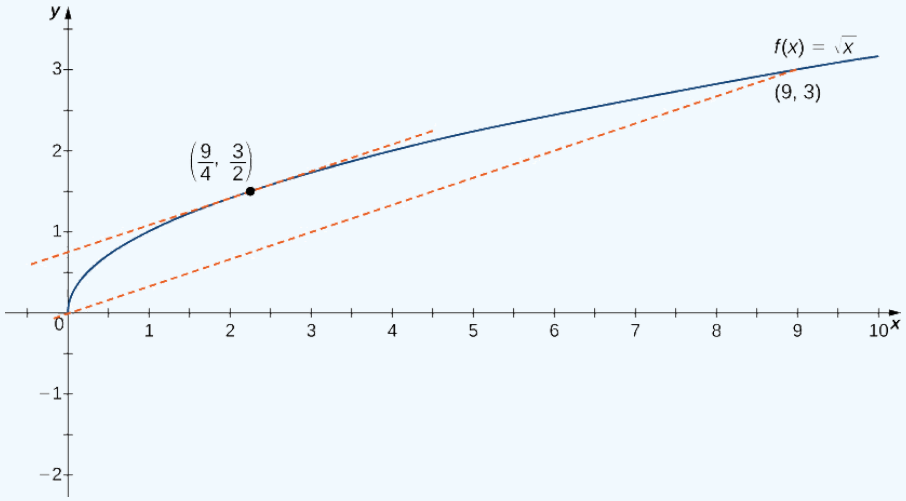


Figure 4.4.7: The slope of the tangent line at $c = 9/4$ is the same as the slope of the line segment connecting (0,0) and (9,3).

One application that helps illustrate the Mean Value Theorem involves velocity. For example, suppose we drive a car for 1 h down a straight road with an average velocity of 45 mph. Let $s(t)$ and $v(t)$ denote the position and velocity of the car, respectively, for $0 \le t \le 1$ h. Assuming that the position function $s(t)$ is differentiable, we can apply the Mean Value Theorem to conclude that, at some time $c \in (0, 1)$, the speed of the car was exactly

$$v(c) = s'(c) = \frac{s(1) - s(0)}{1 - 0} = 45 \text{ mph}.$$

Example 4.4.3: Mean Value Theorem and Velocity

If a rock is dropped from a height of 100 ft, its position t seconds after it is dropped until it hits the ground is given by the function $s(t) = -16t^2 + 100$.

a. Determine how long it takes before the rock hits the ground.
b. Find the average velocity v_{avg} of the rock for when the rock is released and the rock hits the ground.
c. Find the time t guaranteed by the Mean Value Theorem when the instantaneous velocity of the rock is v_{avg}.

Solution

a. When the rock hits the ground, its position is $s(t) = 0$. Solving the equation $-16t^2 + 100 = 0$ for t, we find that $t = \pm\frac{5}{2} sec$. Since we are only considering $t \geq 0$, the ball will hit the ground $\frac{5}{2}$ sec after it is dropped.

b. The average velocity is given by

$$v_{avg} = \frac{s(5/2) - s(0)}{5/2 - 0} = \frac{0 - 100}{5/2} = -40 \text{ ft/sec}.$$

c. The instantaneous velocity is given by the derivative of the position function. Therefore, we need to find a time t such that $v(t) = s'(t) = v_{avg} = -40$ ft/sec. Since $s(t)$ is continuous over the interval $[0, 5/2]$ and differentiable over the interval $(0, 5/2)$, by the Mean Value Theorem, there is guaranteed to be a point $c \in (0, 5/2)$ such that

$$s'(c) = \frac{s(5/2) - s(0)}{5/2 - 0} = -40.$$

Taking the derivative of the position function $s(t)$, we find that $s'(t) = -32t$. Therefore, the equation reduces to $s'(c) = -32c = -40$. Solving this equation for c, we have $c = \frac{5}{4}$. Therefore, $\frac{5}{4}$ sec after the rock is dropped, the instantaneous velocity equals the average velocity of the rock during its free fall: -40 ft/sec.

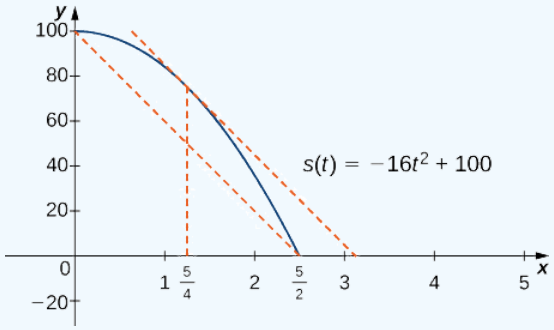


Figure 4.4.8: At time $t = 5/4$ sec, the velocity of the rock is equal to its average velocity from the time it is dropped until it hits the ground.

Exercise 4.4.2

Suppose a ball is dropped from a height of 200 ft. Its position at time t is $s(t) = -16t^2 + 200$. Find the time t when the instantaneous velocity of the ball equals its average velocity.

Hint

First, determine how long it takes for the ball to hit the ground. Then, find the average velocity of the ball from the time it is dropped until it hits the ground.

Answer

$\frac{5}{2\sqrt{2}}$ sec

Corollaries of the Mean Value Theorem

Let's now look at three corollaries of the Mean Value Theorem. These results have important consequences, which we use in upcoming sections.

At this point, we know the derivative of any constant function is zero. The Mean Value Theorem allows us to conclude that the converse is also true. In particular, if $f'(x) = 0$ for all x in some interval I, then $f(x)$ is constant over that interval. This result may seem intuitively obvious, but it has important implications that are not obvious, and we discuss them shortly.

Corollary 1: Functions with a Derivative of Zero

Let f be differentiable over an interval I. If $f'(x) = 0$ for all $x \in I$, then $f(x) =$ constant for all $x \in I$.

Proof

Since f is differentiable over I, f must be continuous over I. Suppose $f(x)$ is not constant for all x in I. Then there exist $a, b \in I$, where $a \neq b$ and $f(a) \neq f(b)$. Choose the notation so that $a < b$. Therefore,

$$\frac{f(b) - f(a)}{b - a} \neq 0.$$

Since f is a differentiable function, by the Mean Value Theorem, there exists $c \in (a, b)$ such that

$$f'(c) = \frac{f(b) - f(a)}{b - a}.$$

Therefore, there exists $c \in I$ such that $f'(c) \neq 0$, which contradicts the assumption that $f'(x) = 0$ for all $x \in I$.

□

From "Corollary 1: Functions with a Derivative of Zero," it follows that if two functions have the same derivative, they differ by, at most, a constant.

Corollary 2: Constant Difference Theorem

If f and g are differentiable over an interval I and $f'(x) = g'(x)$ for all $x \in I$, then $f(x) = g(x) + C$ for some constant C.

Proof

Let $h(x) = f(x) - g(x)$. Then, $h'(x) = f'(x) - g'(x) = 0$ for all $x \in I$. By Corollary 1, there is a constant C such that $h(x) = C$ for all $x \in I$. Therefore, $f(x) = g(x) + C$ for all $x \in I$.

□

The third corollary of the Mean Value Theorem discusses when a function is increasing and when it is decreasing. Recall that a function f is increasing over I if $f(x_1) < f(x_2)$ whenever $x_1 < x_2$, whereas f is decreasing over I if $f(x_1) > f(x_2)$ whenever $x_1 < x_2$. Using the Mean Value Theorem, we can show that if the derivative of a function is positive, then the function is increasing; if the derivative is negative, then the function is decreasing (Figure 4.4.9). We make use of this fact in the next section, where we show how to use the derivative of a function to locate local maximum and minimum values of the function, and how to determine the shape of the graph.

This fact is important because it means that for a given function f, if there exists a function F such that $F'(x) = f(x)$; then, the only other functions that have a derivative equal to f are $F(x) + C$ for some constant C. We discuss this result in more detail later in the chapter.

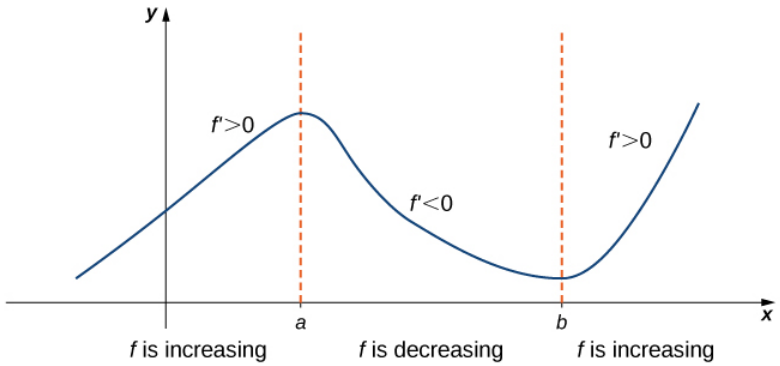


Figure 4.4.9: If a function has a positive derivative over some interval I, then the function increases over that interval I; if the derivative is negative over some interval I, then the function decreases over that interval I.

Corollary 3: Increasing and Decreasing Functions

Let f be continuous over the closed interval $[a, b]$ and differentiable over the open interval (a, b).

i. If $f'(x) > 0$ for all $x \in (a, b)$, then f is an increasing function over $[a, b]$.
ii. If $f'(x) < 0$ for all $x \in (a, b)$, then f is a decreasing function over $[a, b]$.

Proof

We will prove i.; the proof of ii. is similar. Suppose f is not an increasing function on I. Then there exist a and b in I such that $a < b$, but $f(a) \geq f(b)$. Since f is a differentiable function over I, by the Mean Value Theorem there exists $c \in (a, b)$ such that

$$f'(c) = \frac{f(b) - f(a)}{b - a}.$$

Since $f(a) \geq f(b)$, we know that $f(b) - f(a) \leq 0$. Also, $a < b$ tells us that $b - a > 0$. We conclude that

$$f'(c) = \frac{f(b) - f(a)}{b - a} \leq 0.$$

However, $f'(x) > 0$ for all $x \in I$. This is a contradiction, and therefore f must be an increasing function over I.

□

Key Concepts

- If f is continuous over $[a, b]$ and differentiable over (a, b) and $f(a) = f(b)$, then there exists a point $c \in (a, b)$ such that $f'(c) = 0$. This is Rolle's theorem.
- If f is continuous over $[a, b]$ and differentiable over (a, b), then there exists a point $c \in (a, b)$ such that

$$f'(c) = \frac{f(b) - f(a)}{b - a}.$$

  This is the Mean Value Theorem.
- If $f'(x) = 0$ over an interval I, then f is constant over I.
- If two differentiable functions f and g satisfy $f'(x) = g'(x)$ over I, then $f(x) = g(x) + C$ for some constant C.
- If $f'(x) > 0$ over an interval I, then f is increasing over I. If $f'(x) < 0$ over I, then f is decreasing over I.

Glossary

mean value theorem

if f is continuous over $[a, b]$ and differentiable over (a, b), then there exists $c \in (a, b)$ such that $f'(c) = \frac{f(b)-f(a)}{b-a}$

rolle's theorem

if f is continuous over $[a, b]$ and differentiable over (a, b), and if $f(a) = f(b)$, then there exists $c \in (a, b)$ such that $f'(c) = 0$

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4.4E: EXERCISES FOR SECTION 4.4

1)Why do you need continuity to apply the Mean Value Theorem? Construct a counterexample.

2) Why do you need differentiability to apply the Mean Value Theorem? Find a counterexample.

Answer

One example is $f(x) = |x| + 3, -2 \leq x \leq 2$

3) When are Rolle's theorem and the Mean Value Theorem equivalent?

4) If you have a function with a discontinuity, is it still possible to have $f'(c)(b-a) = f(b) - f(a)$? Draw such an example or prove why not.

Answer

Yes, but the Mean Value Theorem still does not apply

In exercises 5 - 9, determine over what intervals (if any) the Mean Value Theorem applies. Justify your answer.

5) $y = \sin(\pi x)$

6) $y = \dfrac{1}{x^3}$

Answer

$(-\infty, 0), (0, \infty)$

7) $y = \sqrt{4 - x^2}$

8) $y = \sqrt{x^2 - 4}$

Answer

$(-\infty, -2), (2, \infty)$

9) $y = \ln(3x - 5)$

In exercises 10 - 13, graph the functions on a calculator and draw the secant line that connects the endpoints. Estimate the number of points c such that $f'(c)(b-a) = f(b) - f(a)$.

10) [T] $y = 3x^3 + 2x + 1$ over $[-1, 1]$

Answer

2 points

11) [T] $y = \tan\left(\frac{\pi}{4}x\right)$ over $\left[-\frac{3}{2}, \frac{3}{2}\right]$

12) [T] $y = x^2 \cos(\pi x)$ over $[-2, 2]$

Answer

5 points

13) [T] $y = x^6 - \frac{3}{4}x^5 - \frac{9}{8}x^4 + \frac{15}{16}x^3 + \frac{3}{32}x^2 + \frac{3}{16}x + \frac{1}{32}$ over $[-1, 1]$

In exercises 14 - 19, use the Mean Value Theorem and find all points $0 < c < 2$ such that $f(2) - f(0) = f'(c)(2-0)$.

14) $f(x) = x^3$

Answer

$c = \frac{2\sqrt{3}}{3}$

15) $f(x) = \sin(\pi x)$

16) $f(x) = \cos(2\pi x)$

Answer

$c = \frac{1}{2}, 1, \frac{3}{2}$

17) $f(x) = 1 + x + x^2$

18) $f(x) = (x-1)^{10}$

Answer

$c = 1$

19) $f(x) = (x-1)^9$

In exercises 20 - 23, show there is no c such that $f(1) - f(-1) = f'(c)(2)$. Explain why the Mean Value Theorem does not apply over the interval $[-1, 1]$.

20) $f(x) = \left|x - \frac{1}{2}\right|$

Answer

Not differentiable

21) $f(x) = \dfrac{1}{x^2}$

22) $f(x) = \sqrt{|x|}$

Answer

Not differentiable

23) $f(x) = \lfloor x \rfloor$ (Hint: This is called the floor function and it is defined so that $f(x)$ is the largest integer less than or equal to x.)

In exercises 24 - 34, determine whether the Mean Value Theorem applies for the functions over the given interval $[a, b]$. Justify your answer.

24) $y = e^x$ over $[0, 1]$

Answer

Yes

25) $y = \ln(2x + 3)$ over $[-\frac{3}{2}, 0]$

26) $f(x) = \tan(2\pi x)$ over $[0, 2]$

Answer

The Mean Value Theorem does not apply since the function is discontinuous at $x = \frac{1}{4}, \frac{3}{4}, \frac{5}{4}, \frac{7}{4}$.

27) $y = \sqrt{9 - x^2}$ over $[-3, 3]$

28) $y = \dfrac{1}{|x+1|}$ over $[0, 3]$

Answer

Yes

29) $y = x^3 + 2x + 1$ over $[0, 6]$

30) $y = \dfrac{x^2 + 3x + 2}{x}$ over $[-1, 1]$

Answer

The Mean Value Theorem does not apply; discontinuous at $x = 0$.

31) $y = \dfrac{x}{\sin(\pi x) + 1}$ over $[0, 1]$

32) $y = \ln(x + 1)$ over $[0, e - 1]$

Answer

Yes

33) $y = x \sin(\pi x)$ over $[0, 2]$

34) $y = 5 + |x|$ over $[-1, 1]$

Answer

The Mean Value Theorem does not apply; not differentiable at $x = 0$.

For exercises 35 - 37, consider the roots of each equation.

35) Show that the equation $y = x^3 + 3x^2 + 16$ has exactly one real root. What is it?

36) Find the conditions for exactly one root (double root) for the equation $y = x^2 + bx + c$

Answer

$b = \pm 2\sqrt{c}$

37) Find the conditions for $y = e^x - b$ to have one root. Is it possible to have more than one root?

In exercises 38 - 42, use a calculator to graph the function over the interval $[a, b]$ and graph the secant line from a to b. Use the calculator to estimate all values of c as guaranteed by the Mean Value Theorem. Then, find the exact value of c, if possible, or write the final equation and use a calculator to estimate to four digits.

38) [T] $y = \tan(\pi x)$ over $\left[-\frac{1}{4}, \frac{1}{4}\right]$

Answer

$c \approx \pm 0.1533$

$c = \pm \frac{1}{\pi} \cos^{-1}\left(\frac{\sqrt{\pi}}{2}\right)$

39) [T] $y = \dfrac{1}{\sqrt{x+1}}$ over $[0, 3]$

40) [T] $y = |x^2 + 2x - 4|$ over $[-4, 0]$

Answer

The Mean Value Theorem does not apply.

41) [T] $y = x + \dfrac{1}{x}$ over $\left[\frac{1}{2}, 4\right]$

42) [T] $y = \sqrt{x+1} + \dfrac{1}{x^2}$ over $[3, 8]$

Answer

$$\frac{1}{2\sqrt{c+1}} - \frac{2}{c^3} = \frac{521}{2880}$$

$c \approx 3.133, 5.867$

43) At 10:17 a.m., you pass a police car at 55 mph that is stopped on the freeway. You pass a second police car at 55 mph at 10:53 a.m., which is located 39 mi from the first police car. If the speed limit is 60 mph, can the police cite you for speeding?

44) Two cars drive from one stoplight to the next, leaving at the same time and arriving at the same time. Is there ever a time when they are going the same speed? Prove or disprove.

Answer

Yes

45) Show that $y = \sec^2 x$ and $y = \tan^2 x$ have the same derivative. What can you say about $y = \sec^2 x - \tan^2 x$?

46) Show that $y = \csc^2 x$ and $y = \cot^2 x$ have the same derivative. What can you say about $y = \csc^2 x - \cot^2 x$?

Answer

It is constant.

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4.5: Derivatives and the Shape of a Graph

Learning Objectives

- Explain how the sign of the first derivative affects the shape of a function's graph.
- State the first derivative test for critical points.
- Use concavity and inflection points to explain how the sign of the second derivative affects the shape of a function's graph.
- Explain the concavity test for a function over an open interval.
- Explain the relationship between a function and its first and second derivatives.
- State the second derivative test for local extrema.

Earlier in this chapter we stated that if a function f has a local extremum at a point c, then c must be a critical point of f. However, a function is not guaranteed to have a local extremum at a critical point. For example, $f(x) = x^3$ has a critical point at $x = 0$ since $f'(x) = 3x^2$ is zero at $x = 0$, but f does not have a local extremum at $x = 0$. Using the results from the previous section, we are now able to determine whether a critical point of a function actually corresponds to a local extreme value. In this section, we also see how the second derivative provides information about the shape of a graph by describing whether the graph of a function curves upward or curves downward.

The First Derivative Test

Corollary 3 of the Mean Value Theorem showed that if the derivative of a function is positive over an interval I then the function is increasing over I. On the other hand, if the derivative of the function is negative over an interval I, then the function is decreasing over I as shown in the following figure.

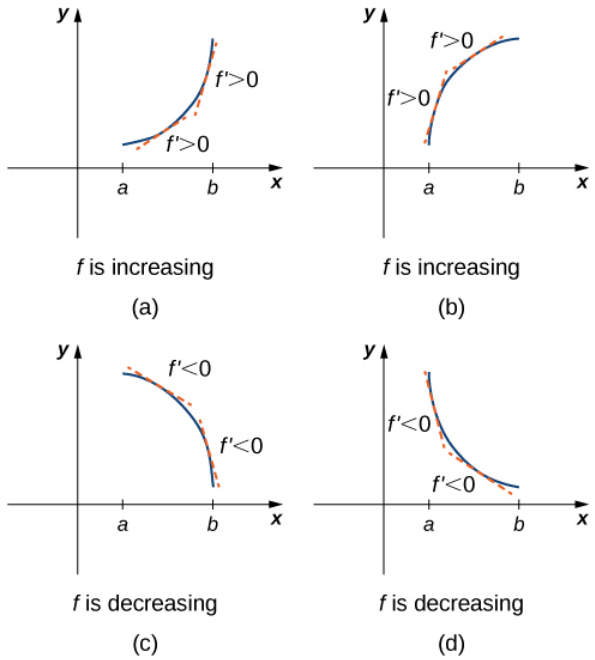


Figure 4.5.1: Both functions are increasing over the interval (a, b). At each point x, the derivative $f'(x) > 0$. Both functions are decreasing over the interval (a, b). At each point x, the derivative $f'(x) < 0$.

A continuous function f has a local maximum at point c if and only if f switches from increasing to decreasing at point c. Similarly, f has a local minimum at c if and only if f switches from decreasing to increasing at c. If f is a continuous function over an interval I containing c and differentiable over I, except possibly at c, the only way f can switch from increasing to decreasing (or vice versa) at point c is if f' changes sign as x increases through c. If f is differentiable at c, the only way that f' can change sign as x increases through c is if $f'(c) = 0$. Therefore, for a function f that is continuous over an interval I containing c and differentiable over I, except possibly at c, the only way f can switch from increasing to decreasing (or vice versa) is if $f'(c) = 0$ or $f'(c)$ is